Linear Algebra with Applications

A Textbook for MTH2021

Monash University

Built from the MTH2021 seminar notes,
in the spirit of Anton & Rorres

Semester 1, 2026

CONTENTS

CHAPTER 1

LINEAR SYSTEMS AND MATRICES

What this chapter is about. Linear algebra began as the art of solving systems of linear equations, and that is where we begin too. We develop a single mechanical procedure — Gaussian elimination — that decides whether *any* linear system has no solution, one solution, or infinitely many, and produces those solutions when they exist. We then study *matrices* as objects in their own right: how they add, multiply, transpose, and invert. The chapter culminates in a list of equivalent conditions for a square matrix to be invertible (Theorem 1.5.7), a theorem we will extend repeatedly throughout the unit. By the end you should row-reduce fluently, justify every step, and read the structure of a solution set directly off an echelon form.

Reference. A&R §§1.1–1.6, with the graph-theory application drawn from A&R §10.5.

1.1 Systems of Linear Equations

A great deal of applied mathematics reduces, eventually, to solving equations in which the unknowns appear only to the first power and are never multiplied together. Such equations are called *linear*; they are the simplest equations that remain genuinely useful.

Definition 1.1.1 (Linear equation, linear system)
A *linear equation* in the variables $x_1, x_2, \ldots, x_n$ is an equation

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b,$$

in which the *coefficients* $a_1, \ldots, a_n$ and the *constant term* b are fixed real numbers. A finite collection of linear equations in the same variables is a *system of linear equations*, or simply a *linear system*.

Geometry gives the definition meaning. In two variables, $a_1x_1 + a_2x_2 = b$ is the equation of a straight line (provided a_1, a_2 are not both zero); in three variables, $a_1x_1 + a_2x_2 + a_3x_3 = b$ is a plane; and in general $a_1x_1 + \cdots + a_nx_n = b$ describes a *hyperplane* in $\mathbb{R}^n$. To solve a system is to find the points lying simultaneously on *all* of these hyperplanes — their common intersection.

A general system of m equations in n variables,

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1, \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2, \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m, \end{aligned} \tag{1.1}$$

is unwieldy written out in full, so we package the coefficients into a rectangular array. Define the $m \times n$

coefficient matrix A, the n-vector of unknowns $\mathbf{x}$, and the m-vector of constants $\mathbf{b}$:

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}, \qquad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \qquad \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}.$$

System (1.1) then has the compact *matrix form* $A\mathbf{x} = \mathbf{b}$. For the purpose of solving, all relevant data is captured by the *augmented matrix*, obtained by adjoining the column $\mathbf{b}$:

$$[\,A \mid \mathbf{b}\,] = \left[\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{array}\right].$$

A *solution* of (1.1) is an n-vector $\mathbf{x}$ satisfying every equation simultaneously. The system is *consistent* if it has at least one solution and *inconsistent* if it has none.

Example 1.1.2 (Matrix form)

Write the system $x - y = 1,\ x + 2y = 4$ in matrix form, and describe it geometrically.

Solution. Here $A = \left[\begin{smallmatrix} 1 & -1 \\ 1 & 2 \end{smallmatrix}\right]$, $\mathbf{x} = \left[\begin{smallmatrix} x \\ y \end{smallmatrix}\right]$, $\mathbf{b} = \left[\begin{smallmatrix} 1 \\ 4 \end{smallmatrix}\right]$, so the system is $A\mathbf{x} = \mathbf{b}$. Geometrically it asks for the intersection of two lines in the plane; their slopes differ, so they meet in the single point $(x, y) = (2, 1)$.

How many solutions can a linear system have? Geometric intuition in two and three variables suggests three outcomes: no intersection, a single point, or a whole line or plane of common points. The next theorem confirms these are the *only* possibilities — a linear system can never have, say, exactly two solutions.

Theorem 1.1.3 (The infinitely-many-solutions theorem)
If a linear system has more than one solution, then it has infinitely many solutions.

Proof.
Since the system has more than one solution, we can choose two *distinct* solutions $\mathbf{x}_1 \neq \mathbf{x}_2$, so $A\mathbf{x}_1 = \mathbf{b}$ and $A\mathbf{x}_2 = \mathbf{b}$. For each scalar $\alpha \in \mathbb{R}$, define

$$\mathbf{x}_\alpha = \mathbf{x}_1 + \alpha(\mathbf{x}_1 - \mathbf{x}_2).$$

We check that every $\mathbf{x}_\alpha$ is a solution. Using the distributive and scalar properties of matrix multiplication (Theorem 1.4.1),

$$A\mathbf{x}_\alpha = A\mathbf{x}_1 + \alpha\, A(\mathbf{x}_1 - \mathbf{x}_2) = \mathbf{b} + \alpha(\mathbf{b} - \mathbf{b}) = \mathbf{b}. \qquad (A\mathbf{x}_1 = A\mathbf{x}_2 = \mathbf{b})$$

So $\mathbf{x}_\alpha$ solves $A\mathbf{x} = \mathbf{b}$ for every α. Finally, these solutions are all distinct: if $\mathbf{x}_\alpha = \mathbf{x}_\beta$ then $(\alpha - \beta)(\mathbf{x}_1 - \mathbf{x}_2) = \mathbf{0}$, and since $\mathbf{x}_1 - \mathbf{x}_2 \neq \mathbf{0}$ (because $\mathbf{x}_1 \neq \mathbf{x}_2$), this forces $\alpha = \beta$. Hence $\{\mathbf{x}_\alpha : \alpha \in \mathbb{R}\}$ is an infinite set of distinct solutions. ■

1.2 Gaussian Elimination

The strategy for solving $A\mathbf{x} = \mathbf{b}$ is to replace it by a simpler system with the *same* solutions — simple enough to solve by inspection. The replacements are made by three *elementary row operations* on the augmented matrix:

(a) multiply one row by a nonzero scalar: $R_i \mapsto cR_i \ (c \neq 0)$;

(b) interchange two rows: $R_i \leftrightarrow R_j$;

(c) add a multiple of one row to another: $R_i \mapsto R_i + cR_j$.

Each operation is *reversible*: $R_i \mapsto cR_i$ is undone by $R_i \mapsto \frac{1}{c}R_i$; a swap is undone by repeating it; and $R_i \mapsto R_i + cR_j$ is undone by $R_i \mapsto R_i - cR_j$. This reversibility is precisely what keeps the solution set intact.

Definition 1.2.1 (Row equivalence)
Two matrices A and B are *row equivalent*, written $A \sim B$, if either can be obtained from the other by a sequence of elementary row operations.

Remark 1.2.2. If two linear systems have row-equivalent augmented matrices, then they have exactly the same set of solutions. This is the entire justification of the method: row reduction neither gains nor loses solutions.

Definition 1.2.3 (Echelon and reduced echelon form)
A matrix is in *(row) echelon form* if:

1. all zero rows, if any, lie at the bottom; and
2. the first nonzero entry of each nonzero row — its *pivot* — lies strictly to the right of the pivot in the row above.

It is in *reduced (row) echelon form* if, in addition,

3. each pivot equals 1; and
4. each pivot is the only nonzero entry in its column.

Remark 1.2.4. Anton & Rorres require every pivot to equal 1 even in unreduced echelon form, and call pivots *leading* 1*'s*. We adopt the more common convention above, in which a pivot is merely the first nonzero entry of its row. The distinction is cosmetic and never affects the number or position of pivots.

Algorithm 1.2.5 (Gaussian elimination)
To bring a matrix to echelon form:

(i) Find the leftmost column with a nonzero entry a; by a row swap, bring a (the pivot) to the top of the current block of rows.

(ii) Subtract multiples of the pivot row from the rows below so that every entry beneath the pivot becomes 0. (This finishes the pivot row; all further work is on the rows below it.)

(iii) Repeat (i)–(ii) on the remaining rows.

Algorithm 1.2.6 (Gauss–Jordan elimination)
To reach reduced echelon form: perform Gaussian elimination; scale each nonzero row by the reciprocal of its pivot; then, starting from the last nonzero row and working upward, use type-(c) operations to clear the entries *above* each pivot.

Remark 1.2.7. Echelon forms are *not* unique: different choices of operations yield different echelon forms. However, the *number* of pivots and their *column positions* are the same for every echelon form of a given matrix. Consequently every matrix has a *unique* reduced echelon form.

1.2.1 Reading off the solutions

Let $[A \mid \mathbf{b}] \sim [U \mid \mathbf{c}]$ with $[U \mid \mathbf{c}]$ in echelon form; the solutions of $U\mathbf{x} = \mathbf{c}$ coincide with those of $A\mathbf{x} = \mathbf{b}$ and are found by back-substitution. Variables whose columns in U contain a pivot are *leading variables*; for a consistent system, the variables whose columns contain no pivot are *free variables*. Free variables may take arbitrary values; the leading variables are then expressed in terms of them. Three mutually exclusive cases arise:

(i) **No solution** (inconsistent): some row of $[U \mid \mathbf{c}]$ reads $[0\ 0\ \cdots\ 0 \mid c]$ with $c \neq 0$, asserting the impossibility $0 = c$.

(ii) **Infinitely many solutions**: the system is consistent and has at least one free variable.

(iii) **Exactly one solution**: the system is consistent with no free variable.

Theorem 1.2.8 (Counting solutions)
Consider a system of m equations in n variables whose echelon form has r pivots.

(a) If $r = n$, there is no solution or exactly one solution.

(b) If $r < n$, there is no solution or infinitely many solutions.

(c) If $n > m$, there is no solution or infinitely many solutions.

Proof.
A pivot column corresponds to a leading variable, so r is the number of leading variables and $n - r$ is the number of free variables.

(a) If $r = n$, there are no free variables. A consistent system with no free variables has its solution completely determined, so there is exactly one solution; an inconsistent system has none.

(b) If $r < n$, there is at least one free variable. If the system is consistent, that free variable can take any value, yielding infinitely many solutions; otherwise there are none.

(c) The number of pivots cannot exceed the number of rows, so $r \leq m$. Combined with $n > m$ this gives $r \leq m < n$, i.e. $r < n$, and case (b) applies. ■

Example 1.2.9 (A system depending on a parameter)

Solve, for each real k,

$$\begin{aligned} x_1 + 3x_2 + 3x_3 + 2x_4 &= 1, \\ 2x_1 + 6x_2 + 9x_3 + 5x_4 &= 1, \\ -x_1 - 3x_2 + 3x_3 &= k. \end{aligned}$$

Solution. Row-reduce the augmented matrix:

$$\left[\begin{array}{cccc|c} 1 & 3 & 3 & 2 & 1 \\ 2 & 6 & 9 & 5 & 1 \\ -1 & -3 & 3 & 0 & k \end{array}\right] \xrightarrow[R_3 \mapsto R_3 + R_1]{R_2 \mapsto R_2 - 2R_1} \left[\begin{array}{cccc|c} 1 & 3 & 3 & 2 & 1 \\ 0 & 0 & 3 & 1 & -1 \\ 0 & 0 & 6 & 2 & k+1 \end{array}\right] \xrightarrow{R_3 \mapsto R_3 - 2R_2} \left[\begin{array}{cccc|c} 1 & 3 & 3 & 2 & 1 \\ 0 & 0 & 3 & 1 & -1 \\ 0 & 0 & 0 & 0 & k+3 \end{array}\right].$$

Pivots sit in columns 1 and 3, so x_1, x_3 are leading and x_2, x_4 are free. The last row reads $0 = k + 3$.

- **(i)** $k \neq -3$**:** the last row is inconsistent — *no solution.*
- **(ii)** $k = -3$**:** the last row vanishes; two free variables remain — *infinitely many solutions.* Setting $x_2 = s$, $x_4 = t$, back-substitution gives $x_3 = \frac{-1-t}{3}$ and $x_1 = 2 - 3s - t$, so

$$(x_1, x_2, x_3, x_4) = \big(2 - 3s - t,\ s,\ \tfrac{-1-t}{3},\ t\big), \qquad s, t \in \mathbb{R}.$$

- **(iii)** No value of k gives a unique solution, since x_2, x_4 are always free.

1.2.2 Homogeneous systems

Definition 1.2.10 (Homogeneous system)
A linear system is *homogeneous* if it has the form $A\mathbf{x} = \mathbf{0}$. Geometrically the hyperplanes all pass through the origin.

Theorem 1.2.11

For any homogeneous system $A\mathbf{x} = \mathbf{0}$:

(a) the system is consistent; and

(b) if there are more variables than equations, it has infinitely many solutions.

Proof.

(a) The zero vector satisfies $A\mathbf{0} = \mathbf{0}$ — the *trivial solution* — so the system is consistent.

(b) With $n > m$, Theorem 1.2.8(c) allows only "no solution" or "infinitely many"; part (a) rules out the former, leaving infinitely many. ■

The trivial solution is always present; the genuine question is whether *nontrivial* solutions exist, which happens precisely when a free variable is present.

Theorem 1.2.12 (Structure of the general solution)

Suppose $A\mathbf{x} = \mathbf{b}$ is consistent with a particular solution $\mathbf{p}$. Then every solution can be written $\mathbf{x} = \mathbf{p}+\mathbf{z}$, where $\mathbf{z}$ solves the associated homogeneous system $A\mathbf{z} = \mathbf{0}$.

Proof.

If $\mathbf{x}$ is any solution then $A(\mathbf{x} - \mathbf{p}) = \mathbf{b} - \mathbf{b} = \mathbf{0}$, so $\mathbf{z} := \mathbf{x} - \mathbf{p}$ solves the homogeneous system and $\mathbf{x} = \mathbf{p} + \mathbf{z}$. Conversely, if $A\mathbf{z} = \mathbf{0}$ then $A(\mathbf{p} + \mathbf{z}) = \mathbf{b} + \mathbf{0} = \mathbf{b}$, so $\mathbf{p} + \mathbf{z}$ is a solution. ■

This is a recurring theme: *the solution set of $A\mathbf{x} = \mathbf{b}$ is one particular solution shifted by the entire solution set of $A\mathbf{x} = \mathbf{0}$.*

1.3 Matrix Operations

We now study matrices in their own right. An $m \times n$ matrix is a rectangular array of numbers with m rows and n columns; we write $(A)_{ij}$ for the entry in row i, column j ($1 \le i \le m$, $1 \le j \le n$). Two matrices of the same size are *equal* when they agree entrywise.

Definition 1.3.1–1.3.3 (Sum, scalar multiple, product)

Let the sizes be compatible as stated.

- *Sum* (same size): $(A + B)_{ij} = (A)_{ij} + (B)_{ij}$.
- *Scalar multiple*: $(kA)_{ij} = k\,(A)_{ij}$.
- *Product* (A is $m \times r$, B is $r \times n$, so AB is $m \times n$): $(AB)_{ij} = \displaystyle\sum_{k=1}^{r} (A)_{ik}(B)_{kj}$.

The product is defined only when the number of columns of A equals the number of rows of B; the entry $(AB)_{ij}$ is the dot product of row i of A with column j of B.

Definition 1.3.4–1.3.6 (Transpose, symmetric matrix, trace)

The *transpose* of an $m \times n$ matrix A is the $n \times m$ matrix A^T with $(A^\mathsf{T})_{ji} = (A)_{ij}$. A square matrix is *symmetric* if $A^\mathsf{T} = A$. The *trace* of a square matrix is the sum of its diagonal entries, $\operatorname{tr}(A) = \sum_{i=1}^{n} (A)_{ii}$.

Example 1.3.8 (Multiplication is not commutative)

With $A = \left[\begin{smallmatrix}1&2\\3&1\end{smallmatrix}\right]$ and $B = \left[\begin{smallmatrix}1&2\\3&0\end{smallmatrix}\right]$ we get $AB = \left[\begin{smallmatrix}7&2\\6&6\end{smallmatrix}\right]$ but $BA = \left[\begin{smallmatrix}7&4\\3&6\end{smallmatrix}\right]$. So in general $AB \neq BA$; one product may even be defined while the other is not.

Some named matrices recur throughout: the $n \times n$ *identity* $I = I_n$ (ones on the diagonal, zeros elsewhere), satisfying $AI = IA = A$; the *zero matrix* O, satisfying $A + O = A$; and the *upper-* and *lower-triangular* matrices U (with $(U)_{ij} = 0$ for $i > j$) and L (with $(L)_{ij} = 0$ for $i < j$). Triangular shape survives multiplication.

Theorem 1.3.9 (Products of triangular matrices)
The product of two lower-triangular matrices is lower-triangular, and the product of two upper-triangular matrices is upper-triangular.

Proof.
Let A, B be lower-triangular and fix $i < j$; we show $(AB)_{ij} = 0$. Split the defining sum at $k = i$:

$$(AB)_{ij} = \sum_{k=1}^{n} (A)_{ik}(B)_{kj} = \underbrace{\sum_{k=1}^{i} (A)_{ik}(B)_{kj}}_{\text{(I)}} + \underbrace{\sum_{k=i+1}^{n} (A)_{ik}(B)_{kj}}_{\text{(II)}} . \qquad \text{(split at } k = i\text{)}$$

- In (I), $k \leq i < j$, so $(B)_{kj} = 0$ since B is lower-triangular.
- In (II), $k > i$, so $(A)_{ik} = 0$ since A is lower-triangular.

Both sums vanish, so $(AB)_{ij} = 0$ for all $i < j$, i.e. AB is lower-triangular. The upper-triangular case is symmetric. ■

The single most useful way to read $A\mathbf{x}$ is as a combination of the columns of A — the bridge, in Chapter 3, between solving $A\mathbf{x} = \mathbf{b}$ and asking whether $\mathbf{b}$ lies in the span of those columns.

Theorem 1.3.10 (Column interpretation of $A\mathbf{x}$)
Let $A = [\,\mathbf{a}_1\ \mathbf{a}_2\ \cdots\ \mathbf{a}_n\,]$ be an $m \times n$ matrix with columns $\mathbf{a}_j$. For $\mathbf{x} = (x_1, \ldots, x_n)^{\mathsf{T}}$,

$$A\mathbf{x} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n.$$

Proof.
The ith entry of $A\mathbf{x}$ is $\sum_{k=1}^{n} (A)_{ik}x_k = \sum_{k=1}^{n} x_k(A)_{ik}$, which is the ith entry of $\sum_k x_k\mathbf{a}_k$ by the definitions of scalar multiplication and matrix addition. ■

Corollary 1.3.11
The ith column of A is $A\mathbf{e}_i$, where $\mathbf{e}_i$ is the ith standard basis vector.

Theorem 1.3.12
Let A be $m \times n$. If $A\mathbf{x} = \mathbf{0}$ for all $\mathbf{x} \in \mathbb{R}^n$, then $A = O$.

Proof.
Taking $\mathbf{x} = \mathbf{e}_i$ and applying Corollary 1.3.11, the ith column of A is $A\mathbf{e}_i = \mathbf{0}$. As this holds for each i, every column is zero, i.e. $A = O$. ■

A product AB can also be expanded as a sum of *outer products* — column-times-row matrices — which is occasionally the most convenient viewpoint (and reappears in the singular value expansion of Chapter 7).

Theorem 1.3.13 (Outer-product expansion)
Let A have columns $\mathbf{a}_1, \dots, \mathbf{a}_n$ and let B have rows $\mathbf{b}_1^\mathsf{T}, \dots, \mathbf{b}_n^\mathsf{T}$. Then

$$AB = \sum_{l=1}^{n} \mathbf{a}_l \, \mathbf{b}_l^\mathsf{T}.$$

Remark 1.3.14. The $n \times n$ matrices $\mathbf{a}_l \mathbf{b}_l^\mathsf{T}$ appearing above are called *outer products.*

Matrices can be multiplied in compatible blocks exactly as if the blocks were scalars — a fact that simplifies many computations and proofs.

Theorem 1.3.15 (Block multiplication)
Partition A and B conformably into blocks,

$$A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}, \qquad B = \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix},$$

where the block sizes match so that each product below is defined. Then

$$AB = \begin{bmatrix} A_{11}B_{11} + A_{12}B_{21} & A_{11}B_{12} + A_{12}B_{22} \\ A_{21}B_{11} + A_{22}B_{21} & A_{21}B_{12} + A_{22}B_{22} \end{bmatrix}.$$

1.4 Algebraic Properties and Inverses

Matrix arithmetic obeys most of the familiar laws of ordinary algebra — with the single, crucial exception that multiplication is not commutative.

Theorem 1.4.1 (Matrix arithmetic)
For all matrices A, B, C of compatible sizes and all scalars a, b:

(a) $A + B = B + A$;
(b) $A + (B + C) = (A + B) + C$;
(c) $A(BC) = (AB)C$;
(d) $A(B + C) = AB + AC$;
(e) $(B + C)A = BA + CA$;
(f) $a(B + C) = aB + aC$;
(g) $(a + b)C = aC + bC$;
(h) $a(bC) = (ab)C$;
(i) $a(BC) = (aB)C = B(aC)$.

Proof of (e), as a representative.
Let B, C be $m \times n$ and A be $n \times p$. Fix entry (i, j) with $1 \le i \le m$, $1 \le j \le p$:

$$\big((B + C)A\big)_{ij} = \sum_{k=1}^{n} (B + C)_{ik}(A)_{kj} \qquad \text{(definition of product)}$$

$$= \sum_{k=1}^{n} \big((B)_{ik} + (C)_{ik}\big)(A)_{kj} \qquad \text{(definition of matrix sum)}$$

$$= \sum_{k=1}^{n} (B)_{ik}(A)_{kj} + \sum_{k=1}^{n} (C)_{ik}(A)_{kj} \qquad \text{(distributivity in } \mathbb{R})$$

$$= (BA)_{ij} + (CA)_{ij}. \qquad \text{(definition of product)}$$

The remaining parts are proved similarly, entrywise. The associativity of multiplication (c) is the one that genuinely uses the summation interchange, and is the reason the order of factors — though not their grouping — matters. ∎

Definition 1.4.2 (Invertible matrix)
A square matrix A is *invertible* (or *nonsingular*) if there is a matrix B with $AB = BA = I$; we write $B = A^{-1}$. A square matrix with no inverse is *singular*.

Theorem 1.4.3 (Properties of the inverse)

(a) If B and C are both inverses of A, then $B = C$ (the inverse is unique).
(b) If A, B are invertible of the same size, then AB is invertible and $(AB)^{-1} = B^{-1}A^{-1}$.
(c) If A is invertible, then so is A^{-1}, and $(A^{-1})^{-1} = A$.
(d) If A is invertible, then so is A^{T}, and $(A^{\mathsf{T}})^{-1} = (A^{-1})^{\mathsf{T}}$.

Proof.
(a) If both B and C invert A:
$B = BI = B(AC) = (BA)C = IC = C.$ (associativity; $AC = I$, $BA = I$)
(b) Check $B^{-1}A^{-1}$ inverts AB on both sides:
$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AIA^{-1} = I,$ ($BB^{-1} = I$)
$(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}IB = I.$ ($A^{-1}A = I$)
By uniqueness (part (a)), $(AB)^{-1} = B^{-1}A^{-1}$.
(c) Immediate from $A^{-1}A = AA^{-1} = I$, which says A inverts A^{-1}; so $(A^{-1})^{-1} = A$.
(d) Transpose $AA^{-1} = I$ using Lemma 1.4.4(d):
$(A^{-1})^{\mathsf{T}}A^{\mathsf{T}} = (AA^{-1})^{\mathsf{T}} = I^{\mathsf{T}} = I,$ (Lemma 1.4.4(d))
and symmetrically $A^{\mathsf{T}}(A^{-1})^{\mathsf{T}} = I$. Hence $(A^{\mathsf{T}})^{-1} = (A^{-1})^{\mathsf{T}}$. ∎

Lemma 1.4.4 (Properties of the transpose)
Whenever the operations are defined,

$$(A^{\mathsf{T}})^{\mathsf{T}} = A, \qquad (A+B)^{\mathsf{T}} = A^{\mathsf{T}} + B^{\mathsf{T}}, \qquad (kA)^{\mathsf{T}} = kA^{\mathsf{T}}, \qquad (AB)^{\mathsf{T}} = B^{\mathsf{T}}A^{\mathsf{T}}.$$

Proof.
The first three follow entrywise from the definition. For the reversal rule, with A of size $m \times n$ and B of size $n \times p$, fix entry (i, j):

$$\begin{aligned}
\big((AB)^{\mathsf{T}}\big)_{ij} &= (AB)_{ji} && \text{(definition of transpose)}\\
&= \sum_k (A)_{jk}(B)_{ki} && \text{(definition of product)}\\
&= \sum_k (B^{\mathsf{T}})_{ik}(A^{\mathsf{T}})_{kj} && \text{(rewrite each factor as a transpose entry)}\\
&= (B^{\mathsf{T}}A^{\mathsf{T}})_{ij}. && \text{(definition of product)}
\end{aligned}$$

Since the entries agree in every position, $(AB)^{\mathsf{T}} = B^{\mathsf{T}}A^{\mathsf{T}}$. ∎

Both the transpose and the inverse *reverse* the order of a product: $(AB)^{\mathsf{T}} = B^{\mathsf{T}}A^{\mathsf{T}}$ and $(AB)^{-1} = B^{-1}A^{-1}$. Writing $A^{\mathsf{T}}B^{\mathsf{T}}$ or $A^{-1}B^{-1}$ is among the most common errors in the unit.

Example 1.4.5 (The Sherman–Morrison formula)

Let $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ be columns with $\mathbf{v}^\mathsf{T}\mathbf{u} \neq -1$. Show $I + \mathbf{u}\mathbf{v}^\mathsf{T}$ is invertible with $(I + \mathbf{u}\mathbf{v}^\mathsf{T})^{-1} = I - \dfrac{\mathbf{u}\mathbf{v}^\mathsf{T}}{1 + \mathbf{v}^\mathsf{T}\mathbf{u}}$.

Solution. Write $\beta = 1 + \mathbf{v}^\mathsf{T}\mathbf{u}$ (a scalar). Treating $\mathbf{v}^\mathsf{T}\mathbf{u}$ as the scalar it is,

$$(I + \mathbf{u}\mathbf{v}^\mathsf{T})\Big(I - \tfrac{\mathbf{u}\mathbf{v}^\mathsf{T}}{\beta}\Big) = I + \mathbf{u}\mathbf{v}^\mathsf{T} - \frac{\mathbf{u}\mathbf{v}^\mathsf{T}}{\beta} - \frac{\mathbf{u}(\mathbf{v}^\mathsf{T}\mathbf{u})\mathbf{v}^\mathsf{T}}{\beta} = I + \mathbf{u}\mathbf{v}^\mathsf{T}\Big(1 - \tfrac{1}{\beta} - \tfrac{\mathbf{v}^\mathsf{T}\mathbf{u}}{\beta}\Big).$$

The bracket is $\dfrac{\beta - 1 - \mathbf{v}^\mathsf{T}\mathbf{u}}{\beta} = 0$, so the product is I. By Theorem 1.5.11 (a right inverse of a square matrix is the inverse), the formula holds.

1.5 Elementary Matrices

We now tie row operations to matrix multiplication. The link is a special family of matrices.

Definition 1.5.1 (Elementary matrix)
An *elementary matrix* is a matrix obtained from the identity by performing a single elementary row operation.

For instance, applying $R_3 \mapsto R_3 + 2R_1$ to I_3 gives $E_3 = \left[\begin{smallmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{smallmatrix}\right]$.

Theorem 1.5.4 (Row operations are left multiplications)
Let μ be an elementary row operation on m-row matrices and $E = \mu(I_m)$. Then for any $m \times n$ matrix A,

$$\mu(A) = EA.$$

Theorem 1.5.5 (Elementary matrices are invertible)
Every elementary matrix is invertible, and its inverse is the elementary matrix of the reverse row operation.

Proof.
Let μ have reverse operation μ', so $\mu'(\mu(I)) = I$ and $\mu(\mu'(I)) = I$. With $E = \mu(I)$ and $F = \mu'(I)$, Theorem 1.5.4 gives $FE = \mu'(\mu(I)) = I$ and $EF = \mu(\mu'(I)) = I$. Hence $E^{-1} = F$, itself elementary. ■

These pieces assemble into the central theorem of the chapter: a list of conditions on a square matrix, all equivalent. We extend this list in Theorem 2.2.12 and again in Chapter 3.

Theorem 1.5.7 (Equivalent statements (invertibility))
For an $n \times n$ matrix A, the following are equivalent:

(a) A is invertible;

(b) $A\mathbf{x} = \mathbf{0}$ has only the trivial solution $\mathbf{x} = \mathbf{0}$;

(c) the reduced echelon form of A is I_n;

(d) A is a product of elementary matrices.

Proof.

We prove the cycle (a)⇒(b)⇒(c)⇒(d)⇒(a).

(a)⇒(b) If A is invertible and $A\mathbf{x} = \mathbf{0}$, then

$$\mathbf{x} = I\mathbf{x} = A^{-1}A\mathbf{x} = A^{-1}\mathbf{0} = \mathbf{0}, \qquad \text{(multiply by } A^{-1}\text{)}$$

so only the trivial solution exists.

(b)⇒(c) Only the trivial solution means no free variables, so every column has a pivot; for a square matrix this forces $\mathrm{RREF}(A) = I_n$.

(c)⇒(d) If $\mathrm{RREF}(A) = I_n$ then $E_k \cdots E_1 A = I_n$ for elementary E_i (Theorem 1.5.4). Each E_i is invertible (Theorem 1.5.5), so $A = E_1^{-1} \cdots E_k^{-1}$, a product of elementary matrices.

(d)⇒(a) A product of elementary matrices is a product of invertible matrices (Theorem 1.5.5), hence invertible (Theorem 1.4.3(b)). ■

Algorithm 1.5.8 (Inversion algorithm)

For a square matrix A, form $[\,A \mid I\,]$ and row-reduce until the left block is in reduced echelon form.

- If the left block becomes I_n, the right block is A^{-1}: $[\,A \mid I\,] \sim [\,I \mid A^{-1}\,]$.
- If a zero row appears on the left, A is singular and the algorithm stops (Remark 1.5.9).

The method works because the product $E_k \cdots E_1$ that reduces A to I simultaneously turns I into $E_k \cdots E_1 = A^{-1}$.

Remark 1.5.9. If A is not invertible, step (ii) reveals it: a row of zeros appears on the left of the partition, and the algorithm may be halted.

Example 1.5.10 (Inverting a 3×3 matrix)

Show $A = \left[\begin{smallmatrix} 1 & 1 & 3 \\ 0 & 2 & 0 \\ 1 & 4 & 4 \end{smallmatrix}\right]$ is invertible and find A^{-1}.

Solution. Reduce $[\,A \mid I\,]$:

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 3 & 1 & 0 & 0 \\ 0 & 2 & 0 & 0 & 1 & 0 \\ 1 & 4 & 4 & 0 & 0 & 1 \end{array}\right] \xrightarrow{R_3 \mapsto R_3 - R_1} \left[\begin{array}{ccc|ccc} 1 & 1 & 3 & 1 & 0 & 0 \\ 0 & 2 & 0 & 0 & 1 & 0 \\ 0 & 3 & 1 & -1 & 0 & 1 \end{array}\right] \xrightarrow{\text{reduce}} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -4 & \frac{4}{2} & \frac{3}{2} \\ 0 & 1 & 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 1 & 1 & -\frac{3}{2} & -\frac{1}{2} \end{array}\right].$$

The left block reached I_3, so A is invertible (Theorem 1.5.7) and $A^{-1} = \left[\begin{smallmatrix} -4 & 2 & \frac{3}{2} \\ 0 & \frac{1}{2} & 0 \\ 1 & -\frac{3}{2} & -\frac{1}{2} \end{smallmatrix}\right]$. (One verifies $AA^{-1} = I$.)

For square matrices, a *one-sided* inverse is automatically two-sided — a convenient and frequently-used shortcut.

Theorem 1.5.11 (One-sided inverses suffice)

If a square matrix A has a left inverse ($BA = I$) or a right inverse ($AB = I$), then A is invertible and its inverse is B.

Proof.

Left inverse Suppose $BA = I$. If $A\mathbf{x} = \mathbf{0}$, then multiplying on the left by B,

$$\mathbf{x} = I\mathbf{x} = (BA)\mathbf{x} = B(A\mathbf{x}) = B\mathbf{0} = \mathbf{0}.$$

So $A\mathbf{x} = \mathbf{0}$ has only the trivial solution, and A is invertible by Theorem 1.5.7 (b)⇒(a). Multiplying

$BA = I$ on the right by A^{-1} then gives $B = A^{-1}$.

Right inverse Suppose instead $AB = I$. Then A is a left inverse of B, so by the case just proved B is invertible with $B^{-1} = A$. Hence $A = B^{-1}$ is invertible, and $A^{-1} = (B^{-1})^{-1} = B$ (Theorem 1.4.3(c)). ■

Lemma 1.5.12

If either A or B is not invertible, then AB is not invertible.

Proof.

Contrapositive. If AB is invertible with inverse C, then $A(BC) = (AB)C = I$ shows A has a right inverse, and $(CA)B = C(AB) = I$ shows B has a left inverse; by Theorem 1.5.11 both A and B are invertible. Hence if either factor is singular, AB cannot be invertible. ■

The list of equivalences extends naturally to statements about $A\mathbf{x} = \mathbf{b}$.

Theorem 1.5.13 (Equivalent statements (consistency))

For an $n \times n$ matrix A, the following are equivalent:

(a) A is invertible;

(b) $A\mathbf{x} = \mathbf{b}$ is consistent for every $\mathbf{b}$;

(c) $A\mathbf{x} = \mathbf{b}$ has exactly one solution for every $\mathbf{b}$.

Proof.

We prove (a)⇒(c)⇒(b)⇒(a).

(a)⇒(c) If A is invertible, then $\mathbf{x} = A^{-1}\mathbf{b}$ solves $A\mathbf{x} = \mathbf{b}$, and any solution satisfies $\mathbf{x} = A^{-1}A\mathbf{x} = A^{-1}\mathbf{b}$ — so the solution exists and is unique.

(c)⇒(b) Exactly one solution is, in particular, at least one.

(b)⇒(a) If $A\mathbf{x} = \mathbf{b}$ is consistent for every $\mathbf{b}$, then $A\mathbf{x} = \mathbf{e}_i$ has a solution $\mathbf{c}_i$ for each i. Assembling these columns,

$$AC = A[\,\mathbf{c}_1 \ \cdots \ \mathbf{c}_n\,] = [\,\mathbf{e}_1 \ \cdots \ \mathbf{e}_n\,] = I, \qquad \text{(column } i \text{ is } A\mathbf{c}_i = \mathbf{e}_i\text{)}$$

so A has a right inverse and is invertible by Theorem 1.5.11. ■

1.6 Application: Walks in Graphs

Matrix powers count paths. We follow A&R §10.5.

Definition 1.6.1 (Graph, digraph)

A *graph* $G = (V, E)$ consists of a set V of *vertices* together with a multiset E of unordered pairs of vertices, called *edges*. A *digraph* is defined the same way except that each edge is an *ordered* pair.

Definition 1.6.4 (Adjacency matrix)

If G has vertices $v_1, \ldots, v_n$, its *adjacency matrix* $M = M(G)$ is the $n \times n$ matrix whose (i, j) entry is the number of edges (or directed edges) from v_i to v_j. For an undirected graph M is symmetric.

Definition 1.6.6 (Walk)

A *walk of length* m from v to w is a sequence of vertices $v_0, v_1, \ldots, v_m$ with $v_0 = v$, $v_m = w$, and an edge from v_i to v_{i+1} for each $i = 0, \ldots, m-1$. Vertices and edges may repeat.

Theorem 1.6.7 (Walk-counting)
The number of walks of length m from v_i to v_j in a graph or digraph G is the (i, j) entry of M^m.

Proof.
We argue by induction on m.
Base case ($m = 1$): by definition of the adjacency matrix, $(M^1)_{ij} = (M)_{ij}$ is the number of edges (length-1 walks) from v_i to v_j.
Inductive step: assume $(M^m)_{ik}$ counts the length-m walks from v_i to v_k, for every k. Any length-$(m+1)$ walk from v_i to v_j consists of a length-m walk from v_i to some intermediate vertex v_k, followed by a single edge $v_k \to v_j$. Summing over all choices of v_k,

$$\#\{\text{walks } i \to j \text{ of length } m{+}1\} = \sum_{k=1}^{n} (M^m)_{ik}(M)_{kj} = (M^{m+1})_{ij},$$

where the last equality is the definition of the matrix product. This completes the induction. ■

Example 1.6.8 (Counting length-3 walks)

Let G have vertices a, b, c, d and edges ab, ac, ad (twice), bd, cc. Then (ordering a, b, c, d)

$$M = \begin{bmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 2 & 1 & 0 & 0 \end{bmatrix}, \qquad M^3 = \begin{bmatrix} * & * & 7 & * \\ * & 4 & * & * \\ 7 & * & * & * \\ * & * & * & * \end{bmatrix}.$$

Reading off entries of M^3: the number of length-3 walks from b to b is $(M^3)_{bb} = 4$, and the number from a to c is $(M^3)_{ac} = 7$.

Remark 1.6.10. In examinations you are typically *given* a power M^m and asked to read off one entry; the only work is identifying which entry answers the question.

Chapter 1 Summary

- A linear system $A\mathbf{x} = \mathbf{b}$ has 0, 1, or infinitely many solutions (Theorem 1.1.3); which one is decided by row-reducing $[A \mid \mathbf{b}]$ and counting pivots (Theorem 1.2.8). Row operations preserve the solution set (Remark 1.2.2); the reduced echelon form is unique (Remark 1.2.7).
- The general solution of a consistent system is one particular solution plus the homogeneous solution set (Theorem 1.2.12).
- Transpose and inverse both reverse products: $(AB)^\mathsf{T} = B^\mathsf{T} A^\mathsf{T}$ (Lemma 1.4.4), $(AB)^{-1} = B^{-1}A^{-1}$ (Theorem 1.4.3).
- **Invertibility (Theorem 1.5.7, Theorem 1.5.13):** for square A, the statements "A invertible", "$A\mathbf{x} = \mathbf{0}$ only trivially", "$\mathrm{RREF}(A) = I$", "A a product of elementary matrices", "$A\mathbf{x} = \mathbf{b}$ always consistent", "$A\mathbf{x} = \mathbf{b}$ always uniquely solvable" are all equivalent. A one-sided inverse is a true inverse (Theorem 1.5.11); a product with a singular factor is singular (Lemma 1.5.12).
- Compute A^{-1} by reducing $[A \mid I] \sim [I \mid A^{-1}]$ (Algorithm 1.5.8).
- Walks of length m from v_i to v_j number $(M^m)_{ij}$ (Theorem 1.6.7).

Chapter 1 Exercises

1.1 For which value(s) of a does $x + 2y - 3z = 4$, $3x - y + 5z = 2$, $4x + y + (a^2 - 14)z = a + 2$ have (i) no solution, (ii) a unique solution, (iii) infinitely many solutions?

1.2 Let $A = \left[\begin{smallmatrix}1&1\\0&1\end{smallmatrix}\right]$, $B = \left[\begin{smallmatrix}1&0\\1&1\end{smallmatrix}\right]$. Compute AB and BA, confirm they differ, and verify $\operatorname{tr}(AB) = \operatorname{tr}(BA)$.

1.3 Prove that if A, B are symmetric $n \times n$ matrices, then AB is symmetric if and only if $AB = BA$.

1.4 Use Algorithm 1.5.8 to find the inverse of $A = \left[\begin{smallmatrix}2&1&1\\1&2&1\\1&1&2\end{smallmatrix}\right]$, or show it is singular.

1.5 Using the Sherman–Morrison formula (Example 1.4.5) with $\mathbf{u} = \left[\begin{smallmatrix}1\\2\end{smallmatrix}\right]$, $\mathbf{v} = \left[\begin{smallmatrix}1\\1\end{smallmatrix}\right]$, compute $(I + \mathbf{u}\mathbf{v}^\top)^{-1}$ and verify your answer.

1.6 A graph has adjacency matrix $M = \left[\begin{smallmatrix}0&1&1\\1&0&1\\1&1&0\end{smallmatrix}\right]$. Compute M^2 and state the number of length-2 walks from v_1 to v_1 and from v_1 to v_2.

1.7 Prove Lemma 1.4.4(b): $(A + B)^\top = A^\top + B^\top$, directly from the definition of the transpose.

Solutions to Chapter 1 Exercises

1.1 Row-reduce $[A \mid \mathbf{b}]$:

$$\left[\begin{array}{ccc|c}1&2&-3&4\\3&-1&5&2\\4&1&a^2-14&a+2\end{array}\right] \xrightarrow[R_3\mapsto R_3-4R_1]{R_2\mapsto R_2-3R_1} \left[\begin{array}{ccc|c}1&2&-3&4\\0&-7&14&-10\\0&-7&a^2-2&a-14\end{array}\right] \xrightarrow{R_3\mapsto R_3-R_2} \left[\begin{array}{ccc|c}1&2&-3&4\\0&-7&14&-10\\0&0&a^2-16&a-4\end{array}\right].$$

The bottom row reads $(a-4)(a+4)z = a-4$.

- **(ii) Unique:** $a \neq \pm 4$ — then z, hence y, x, are determined.
- **(iii) Infinitely many:** $a = 4$ — the row becomes $0 = 0$, leaving z free.
- **(i) No solution:** $a = -4$ — the row becomes $0 = -8$, inconsistent.

1.2 $AB = \left[\begin{smallmatrix}2&1\\1&1\end{smallmatrix}\right]$, $BA = \left[\begin{smallmatrix}1&1\\1&2\end{smallmatrix}\right]$; these differ (the $(1,1)$ entries are 2 and 1). Yet $\operatorname{tr}(AB) = 3 = \operatorname{tr}(BA)$, consistent with $\operatorname{tr}(AB) = \operatorname{tr}(BA)$ (Example 1.3.16).

1.3 Since A, B are symmetric, $(AB)^\top = B^\top A^\top = BA$ by Lemma 1.4.4. Thus AB is symmetric $\iff (AB)^\top = AB \iff BA = AB$. ■

1.4 Reduce $[A \mid I]$ (details omitted) to obtain

$$A^{-1} = \frac{1}{4}\begin{bmatrix}3&-1&-1\\-1&3&-1\\-1&-1&3\end{bmatrix}.$$

A is invertible since the left block reaches I_3 (Theorem 1.5.7).

1.5 Here $\mathbf{v}^\top\mathbf{u} = 3 \neq -1$ and $\mathbf{u}\mathbf{v}^\top = \left[\begin{smallmatrix}1&1\\2&2\end{smallmatrix}\right]$, so

$$(I + \mathbf{u}\mathbf{v}^\top)^{-1} = I - \tfrac{1}{4}\begin{bmatrix}1&1\\2&2\end{bmatrix} = \begin{bmatrix}\frac{3}{4}&-\frac{1}{4}\\-\frac{1}{2}&\frac{1}{2}\end{bmatrix}.$$

Check: $(I + \mathbf{u}\mathbf{v}^\top) = \left[\begin{smallmatrix}2&1\\2&3\end{smallmatrix}\right]$ and $\left[\begin{smallmatrix}2&1\\2&3\end{smallmatrix}\right]\left[\begin{smallmatrix}3/4&-1/4\\-1/2&1/2\end{smallmatrix}\right] = I$. ✓

1.6 $M^2 = \left[\begin{smallmatrix}2&1&1\\1&2&1\\1&1&2\end{smallmatrix}\right]$. By Theorem 1.6.7, the number of length-2 walks from v_1 to v_1 is $(M^2)_{11} = 2$ (namely $v_1v_2v_1$ and $v_1v_3v_1$), and from v_1 to v_2 is $(M^2)_{12} = 1$ (namely $v_1v_3v_2$).

1.7 For all i, j: $\left((A+B)^\top\right)_{ji} = (A+B)_{ij} = (A)_{ij} + (B)_{ij} = (A^\top)_{ji} + (B^\top)_{ji} = (A^\top + B^\top)_{ji}$, using Definition 1.3.4 and Definition 1.3.1. Since the entries agree in every position, $(A+B)^\top = A^\top + B^\top$. ■

CHAPTER 2

DETERMINANTS

What this chapter is about. To every square matrix we attach a single number, its *determinant*, which detects invertibility: a square matrix is invertible exactly when its determinant is nonzero. We define the determinant recursively through cofactor expansion, learn to compute it efficiently by row reduction, and establish its two structural pillars — that it is multiplicative, $\det(AB) = \det(A)\det(B)$, and that it vanishes precisely for singular matrices. These facts let us extend the invertibility equivalences of Chapter 1 with one more condition, $\det(A) \neq 0$, and they prepare the characteristic polynomial of Chapter 6.

Reference. A&R §§2.1–2.3.
A note on scope. Two foundational results — that $\det(A) = \det(A^{\mathsf{T}})$ and the precise effect of row operations on the determinant — are stated here without proof. Their proofs require the permutation (Leibniz) definition of the determinant developed in MTH2025 and are not examinable in MTH2021. Everything built on top of them *is* proved here.

2.1 Definition

The determinant is defined by recursion on the size of the matrix: the $n \times n$ case is reduced to several $(n-1) \times (n-1)$ determinants.

Definition 2.1.1 (Determinant, minor, cofactor)
The *determinant* function det assigns to each $n \times n$ matrix A a real number $\det(A)$, defined recursively:

- $n = 1$: if $A = [a]$ then $\det(A) = a$.
- $n = 2$: if $A = \left[\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right]$ then $\det(A) = ad - bc$.
- $n \geq 2$: the (i, j)-*minor* $M_{ij}(A)$ is the determinant of the $(n-1) \times (n-1)$ matrix obtained by deleting row i and column j of A; the (i, j)-*cofactor* is

$$C_{ij}(A) = (-1)^{i+j} M_{ij}(A).$$

The determinant is then given by *cofactor (Laplace) expansion along the first row*:

$$\det(A) = (A)_{11}C_{11}(A) + (A)_{12}C_{12}(A) + \cdots + (A)_{1n}C_{1n}(A).$$

Remark 2.1.2. The cofactor sign $(-1)^{i+j}$ produces the checkerboard pattern $\det(A) = +(A)_{11}M_{11} - (A)_{12}M_{12} + (A)_{13}M_{13} - \cdots$, so the signs alternate across the expansion row.

Example 2.1.3 (A 3×3 determinant by first-row expansion)

For $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 0 \\ 4 & 1 & 1 \end{bmatrix}$,

$$\det(A) = 1\det\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} - 2\det\begin{bmatrix} 2 & 0 \\ 4 & 1 \end{bmatrix} + 3\det\begin{bmatrix} 2 & 1 \\ 4 & 1 \end{bmatrix} = 1(1) - 2(2) + 3(-2) = 1 - 4 - 6 = -9.$$

A great convenience is that the expansion may be taken along *any* row or column, not just the first; one chooses the row or column with the most zeros to minimise work.

Theorem 2.1.4 (Cofactor expansion along any line)
For any $n \times n$ matrix A and any fixed i or j,

$$\det(A) = \sum_{k=1}^{n} (A)_{ik} C_{ik}(A) \quad (\text{row } i), \qquad \det(A) = \sum_{k=1}^{n} (A)_{kj} C_{kj}(A) \quad (\text{column } j).$$

Remark 2.1.5. If A has a row of zeros or a column of zeros, then $\det(A) = 0$ — expand along that line, and every term carries a factor of 0.

Example 2.1.6 (Choosing a convenient line)

For the matrix A of Example 2.1.3, column 3 has entries $3, 0, 1$. Expanding along it,

$$\det(A) = 3\,C_{13} + 0 \cdot C_{23} + 1 \cdot C_{33} = 3(+1)\det\begin{bmatrix} 2 & 1 \\ 4 & 1 \end{bmatrix} + 1(+1)\det\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} = 3(-2) + (-3) = -9,$$

agreeing with Example 2.1.3, but with one fewer 2×2 determinant to evaluate.

2.2 Properties

Cofactor expansion is impractical for large matrices: an $n \times n$ determinant expands into $n!$ terms. The efficient route is row reduction, which rests on knowing how each elementary operation affects the determinant.

Lemma 2.2.1 (Determinant of a triangular matrix)
If A is upper-triangular or lower-triangular, then $\det(A)$ is the product of its diagonal entries:

$$\det(A) = (A)_{11}(A)_{22} \cdots (A)_{nn}.$$

Proof.
We argue by induction on n, treating the upper-triangular case (the lower-triangular case is symmetric, expanding along the first row instead).
Base case ($n = 1$): $A = [(A)_{11}]$, so $\det(A) = (A)_{11}$.
Inductive step: let A be $n \times n$ upper-triangular. Its first column is $((A)_{11}, 0, \ldots, 0)^{\mathsf{T}}$, so cofactor expansion down column 1 leaves only the first term,

$$\det(A) = (A)_{11} C_{11} = (A)_{11} M_{11}.$$

The minor M_{11} is the determinant of the $(n-1) \times (n-1)$ upper-triangular matrix obtained by deleting row 1 and column 1, whose diagonal is $(A)_{22}, \ldots, (A)_{nn}$. By the inductive hypothesis $M_{11} = (A)_{22} \cdots (A)_{nn}$, so $\det(A) = (A)_{11}(A)_{22} \cdots (A)_{nn}$. ■

Theorem 2.2.2 (Transpose invariance)
$\det(A) = \det(A^{\mathsf{T}})$.

Proof omitted (MTH2025). A consequence worth noting immediately: every statement about the rows of a determinant has an identical counterpart for columns, since transposing swaps rows and columns without changing the value.

Theorem 2.2.3 (Effect of row and column operations)
Let A be square.

(a) If B is obtained from A by multiplying a single row (or column) by a scalar k, then $\det(B) = k\det(A)$.

(b) If B is obtained by interchanging two rows (or two columns), then $\det(B) = -\det(A)$.

(c) If B is obtained by adding a scalar multiple of one row to another row (or one column to another column), then $\det(B) = \det(A)$.

Proof omitted (MTH2025). The 2×2 case is an easy direct check (Exercise 2.1).

Example 2.2.5 (Determinant by reduction to echelon form)

Find $\det(A)$ for $A = \begin{bmatrix} 0 & 3 & 3 \\ 1 & 1 & 2 \\ 3 & 2 & 4 \end{bmatrix}$ by row-reducing and tracking the determinant.

Solution. Swap $R_1 \leftrightarrow R_2$ (this negates the determinant by Theorem 2.2.3(b)):

$$\det(A) = -\det \begin{bmatrix} 1 & 1 & 2 \\ 0 & 3 & 3 \\ 3 & 2 & 4 \end{bmatrix}.$$

Now $R_3 \mapsto R_3 - 3R_1$ leaves the determinant unchanged (Theorem 2.2.3(c)):

$$= -\det \begin{bmatrix} 1 & 1 & 2 \\ 0 & 3 & 3 \\ 0 & -1 & -2 \end{bmatrix}.$$

Then $R_3 \mapsto R_3 + \frac{1}{3}R_2$ (again unchanged):

$$= -\det \begin{bmatrix} 1 & 1 & 2 \\ 0 & 3 & 3 \\ 0 & 0 & -1 \end{bmatrix} = -\big(1 \cdot 3 \cdot (-1)\big) = 3,$$

the last step by Lemma 2.2.1 (triangular determinant). Hence $\det(A) = 3$.

The same operations applied to the identity give the determinants of the elementary matrices.

Theorem 2.2.6 (Determinants of elementary matrices)
Let E be an $n \times n$ elementary matrix.

(a) If E scales a row of I_n by $k \neq 0$, then $\det(E) = k$.

(b) If E interchanges two rows of I_n, then $\det(E) = -1$.

(c) If E adds a multiple of one row of I_n to another, then $\det(E) = 1$.

Proof.

Each elementary matrix E is obtained from I_n by its named row operation, and $\det(I_n) = 1$ by Lemma 2.2.1. Applying the effect of that operation on determinants (Theorem 2.2.3) to I_n:

(a) Scaling a row by k multiplies the determinant by k, so $\det(E) = k\det(I_n) = k$.

(b) Interchanging two rows negates the determinant, so $\det(E) = -\det(I_n) = -1$.

(c) Adding a multiple of one row to another leaves the determinant unchanged, so $\det(E) = \det(I_n) = 1$. ∎

Theorem 2.2.7 (Scaling the whole matrix)
If A is $n \times n$ and k is a scalar, then $\det(kA) = k^n \det(A)$.

Proof.

Multiplying A by k multiplies each of its n rows by k. Applying Theorem 2.2.3(a) once per row pulls out one factor of k each time, for n factors in total: $\det(kA) = k^n \det(A)$. ∎

Theorem 2.2.8 (Determinant of EA)
If A is $n \times n$ and E is an $n \times n$ elementary matrix, then $\det(EA) = \det(E)\det(A)$.

Proof.

By Theorem 1.5.4, EA is the matrix obtained by applying E's row operation to A. There are three types of elementary matrix, one per row operation; we check each, using Theorem 2.2.3 for the effect on $\det(EA)$ and Theorem 2.2.6 for $\det(E)$.

Scaling If E scales row i by k, then $\det(EA) = k\det(A)$ and $\det(E) = k$, so $\det(EA) = k\det(A) = \det(E)\det(A)$.

Interchange If E interchanges two rows, then $\det(EA) = -\det(A)$ and $\det(E) = -1$, so $\det(EA) = -\det(A) = \det(E)\det(A)$.

Row addition If E adds a multiple of one row to another, then $\det(EA) = \det(A)$ and $\det(E) = 1$, so $\det(EA) = \det(A) = \det(E)\det(A)$.

In every case $\det(EA) = \det(E)\det(A)$. ∎

We can now characterise invertibility by the determinant — the result that gives the chapter its purpose.

Theorem 2.2.9 (Invertible $\iff$ nonzero determinant)
A square matrix A is invertible if and only if $\det(A) \neq 0$.

Proof.

Let R be the reduced echelon form of A, so $R = E_k \cdots E_1 A$ for elementary matrices E_i (Theorem 1.5.4). Applying Theorem 2.2.8 repeatedly,

$$\det(R) = \det(E_k)\cdots\det(E_1)\det(A). \qquad \text{(Theorem 2.2.8, } k \text{ times)}$$

By Theorem 2.2.6, each $\det(E_i) \neq 0$, so $\det(R) \neq 0 \iff \det(A) \neq 0$. It remains to relate $\det(R)$ to invertibility.

(A invertible) Then $R = I_n$ (Theorem 1.5.7), so $\det(R) = 1 \neq 0$, forcing $\det(A) \neq 0$.

(A singular) Then $R \neq I_n$; being a square reduced echelon form, R has a zero row, so $\det(R) = 0$ (Remark 2.1.5), forcing $\det(A) = 0$.

Taking contrapositives, $\det(A) \neq 0$ if and only if A is invertible. ∎

Theorem 2.2.10 (Multiplicativity of the determinant)
If A, B are $n \times n$ matrices, then $\det(AB) = \det(A)\det(B)$.

Proof.

Case 1: A invertible Then $A = E_k \cdots E_1$ is a product of elementary matrices (Theorem 1.5.7). Applying Theorem 2.2.8 k times,

$$\det(AB) = \det(E_k \cdots E_1 B) = \det(E_k) \cdots \det(E_1)\det(B). \qquad \text{(Theorem 2.2.8, } k \text{ times)}$$

Taking $B = I$ in this identity gives $\det(A) = \det(E_k) \cdots \det(E_1)$, so $\det(AB) = \det(A)\det(B)$.

Case 2: A singular Then $\det(A) = 0$ (Theorem 2.2.9), so the right side is 0. By Lemma 1.5.12, AB is also singular, so $\det(AB) = 0$ (Theorem 2.2.9). Both sides vanish. ■

Theorem 2.2.11 (Determinant of an inverse)
If A is invertible, then $\det(A^{-1}) = \dfrac{1}{\det(A)}$.

Proof.
From $AA^{-1} = I$ and Theorem 2.2.10, $\det(A)\det(A^{-1}) = \det(I) = 1$. Since A is invertible, $\det(A) \neq 0$ (Theorem 2.2.9), so we may divide to get $\det(A^{-1}) = 1/\det(A)$. ■

These results extend the invertibility equivalences of Chapter 1 with a determinant criterion.

Theorem 2.2.12 (Equivalent statements (with determinant))
For an $n \times n$ matrix A, the following are equivalent:

(a) A is invertible;
(b) $A\mathbf{x} = \mathbf{0}$ has only the trivial solution;
(c) the reduced echelon form of A is I_n;
(d) A is a product of elementary matrices;
(e) $A\mathbf{x} = \mathbf{b}$ is consistent for every $\mathbf{b}$;
(f) $A\mathbf{x} = \mathbf{b}$ has exactly one solution for every $\mathbf{b}$;
(g) $\det(A) \neq 0$.

Proof.
Conditions (a)–(f) are equivalent by Theorem 1.5.7 and Theorem 1.5.13. The new condition (g) is equivalent to (a) by Theorem 2.2.9. ■

Chapter 2 Summary

- The determinant is defined recursively by cofactor expansion (Definition 2.1.1); it may be expanded along any row or column (Theorem 2.1.4), so choose the line with the most zeros.
- In practice, reduce to triangular form and track the operations: a swap negates (Theorem 2.2.3(b)), scaling a row by k multiplies by k (Theorem 2.2.3(a)), and adding a multiple of one row to another leaves it unchanged (Theorem 2.2.3(c)); a triangular determinant is the product of the diagonal (Lemma 2.2.1).
- Structural laws: $\det(kA) = k^n \det(A)$ (Theorem 2.2.7), $\det(AB) = \det(A)\det(B)$ (Theorem 2.2.10), $\det(A^{-1}) = 1/\det(A)$ (Theorem 2.2.11), and $\det(A) = \det(A^\mathsf{T})$ (Theorem 2.2.2).
- **The key fact:** A is invertible $\iff$ $\det(A) \neq 0$ (Theorem 2.2.9), adding condition (g) to the master equivalence list (Theorem 2.2.12).
- Note det is *not* additive: in general $\det(A + B) \neq \det(A) + \det(B)$.

Chapter 2 Exercises

2.1 Prove Theorem 2.2.3 for 2×2 matrices: verify directly that scaling a row by k multiplies the determinant by k, a swap negates it, and adding a multiple of one row to the other leaves it unchanged.

2.2 Evaluate $\det\begin{bmatrix} 2 & 0 & 1 \\ 3 & 1 & 2 \\ 1 & 0 & 3 \end{bmatrix}$ by expanding along the column with the most zeros.

2.3 Compute $\det\begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 7 \\ 3 & 7 & 11 \end{bmatrix}$ by reduction to triangular form, recording the effect of each operation.

2.4 Let A be a 4×4 matrix with $\det(A) = 5$. Find $\det(2A)$, $\det(A^\mathsf{T})$, $\det(A^{-1})$, and $\det(A^3)$.

2.5 Suppose A, B are $n \times n$ with A invertible. Prove that $\det(A^{-1}BA) = \det(B)$.

2.6 Show that if A is an $n \times n$ matrix with $A^2 = A$ (idempotent) and $A \neq I$, then $\det(A) = 0$.

Solutions to Chapter 2 Exercises

2.1 Write $A = \left[\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right]$, so $\det(A) = ad - bc$. *Scaling:* replacing row 1 by (ka, kb) gives $\det = kad - kbc = k(ad - bc) = k\det(A)$. *Swap:* $\left[\begin{smallmatrix} c & d \\ a & b \end{smallmatrix}\right]$ has determinant $cb - da = -(ad - bc) = -\det(A)$. *Add:* replacing row 2 by $(c + ka,\ d + kb)$ gives $a(d + kb) - b(c + ka) = ad + kab - bc - kab = ad - bc = \det(A)$. ■

2.2 Column 2 has entries $0, 1, 0$. Expanding along it: $\det = 0 \cdot C_{12} + 1 \cdot C_{22} + 0 \cdot C_{32} = (+1)\det\left[\begin{smallmatrix} 2 & 1 \\ 1 & 3 \end{smallmatrix}\right] = 6 - 1 = 5$.

2.3 $R_2 \mapsto R_2 - 2R_1$, $R_3 \mapsto R_3 - 3R_1$ (determinant unchanged):

$$\det\begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 7 \\ 3 & 7 & 11 \end{bmatrix} = \det\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 0 & 1 & 2 \end{bmatrix} \xrightarrow{R_3 \mapsto R_3 - R_2} \det\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} = 1 \cdot 1 \cdot 1 = 1.$$

None of these operations changes the value, so the determinant is 1.

2.4 Using Theorem 2.2.7, Theorem 2.2.2, Theorem 2.2.11, Theorem 2.2.10: $\det(2A) = 2^4 \cdot 5 = 80$; $\det(A^\mathsf{T}) = 5$; $\det(A^{-1}) = 1/5$; $\det(A^3) = 5^3 = 125$.

2.5 By multiplicativity (Theorem 2.2.10) and Theorem 2.2.11,

$$\det(A^{-1}BA) = \det(A^{-1})\det(B)\det(A) = \tfrac{1}{\det(A)}\det(B)\det(A) = \det(B). \quad ■$$

(Determinant is invariant under similarity — a fact used heavily in Chapter 6.)

2.6 From $A^2 = A$ and Theorem 2.2.10, $\det(A)^2 = \det(A)$, so $\det(A) \in \{0, 1\}$. Suppose $\det(A) = 1$; then A is invertible (Theorem 2.2.9), and multiplying $A^2 = A$ on the left by A^{-1} gives $A = I$, contradicting $A \neq I$. Hence $\det(A) = 0$. ■

CHAPTER 3

VECTOR SPACES

What this chapter is about. We now abstract. The familiar arithmetic of vectors in $\mathbb{R}^n$ — adding them, scaling them — works identically for polynomials, matrices, and functions. A *vector space* is any set on which addition and scalar multiplication obey ten axioms (A1–A10), and *every* theorem proved from those axioms holds at once for all such sets. We develop the core structural vocabulary: *subspace*, *span*, *linear independence*, *basis*, and *dimension*; the way coordinates let any n-dimensional space stand in for $\mathbb{R}^n$; and the four subspaces attached to a matrix. We close by allowing the scalars to come from any *field*, and apply $\mathbb{Z}_2$-vector spaces to error-correcting codes.

Reference. A&R §§4.1–4.7, with §3.7 (fields) and §3.8 (coding theory) developed from the seminar notes.

Proof convention. Throughout, every step is justified by an axiom (A1–A10) or a previously-proved result. A handful of routine verifications are marked as exercises, matching the seminar notes; one result (Theorem 3.6.6) is stated without proof as it belongs to MTH2025.

3.1 Vector Spaces and Subspaces

Definition 3.1.1 (Vector space)

Let V be a non-empty set equipped with two operations: *addition*, assigning to each $\mathbf{u}, \mathbf{v} \in V$ an element $\mathbf{u} + \mathbf{v}$, and *scalar multiplication*, assigning to each scalar k and each $\mathbf{v} \in V$ an element $k\mathbf{v}$. Then V is a *vector space* if the following ten axioms hold for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$ and all scalars k, m:

A1 (closure, $+$): $\mathbf{u} + \mathbf{v} \in V$.
A2 (commutativity): $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$.
A3 (associativity): $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$.
A4 (zero): there is $\mathbf{0} \in V$ with $\mathbf{u} + \mathbf{0} = \mathbf{u}$.
A5 (negatives): each $\mathbf{u}$ has $-\mathbf{u} \in V$ with $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$.
A6 (closure, scalar): $k\mathbf{u} \in V$.
A7 (distributivity): $k(\mathbf{u} + \mathbf{v}) = k\mathbf{u} + k\mathbf{v}$.
A8 (distributivity): $(k + m)\mathbf{u} = k\mathbf{u} + m\mathbf{u}$.
A9 (associativity, scalar): $k(m\mathbf{u}) = (km)\mathbf{u}$.
A10 (scalar identity): $1\mathbf{u} = \mathbf{u}$.

Remark 3.1.2. Elements of V are called *vectors*. For most of MTH2021 the scalars are real numbers, and V is a *real vector space*. Other scalar fields matter too — the complex numbers $\mathbb{C}$, or the integers modulo 2, $\mathbb{Z}_2$ — as we shall see in §3.7.

The axioms are deliberately spare; even facts that seem obvious must be derived from them. The next two theorems do this groundwork.

Theorem 3.1.3

Let V be a vector space and $\mathbf{v}, \mathbf{w} \in V$. Then (a) $\mathbf{0} + \mathbf{v} = \mathbf{v}$; (b) $-\mathbf{v} + \mathbf{v} = \mathbf{0}$; (c) if $\mathbf{v} + \mathbf{w} = \mathbf{v}$ then $\mathbf{w} = \mathbf{0}$; (d) if $\mathbf{v} + \mathbf{w} = \mathbf{0}$ then $\mathbf{w} = -\mathbf{v}$.

Proof.

(a) By A2 then A4, $\mathbf{0} + \mathbf{v} = \mathbf{v} + \mathbf{0} = \mathbf{v}$.

(b) By A2 then A5, $-\mathbf{v} + \mathbf{v} = \mathbf{v} + (-\mathbf{v}) = \mathbf{0}$.

(c) Suppose $\mathbf{v} + \mathbf{w} = \mathbf{v}$. Add $-\mathbf{v}$ on the left and use A3, (b), then (a):

$$\mathbf{w} = \mathbf{0} + \mathbf{w} = (-\mathbf{v} + \mathbf{v}) + \mathbf{w} = -\mathbf{v} + (\mathbf{v} + \mathbf{w}) = -\mathbf{v} + \mathbf{v} = \mathbf{0}.$$

(d) Suppose $\mathbf{v} + \mathbf{w} = \mathbf{0}$. Then

$$\begin{aligned} \mathbf{w} &= \mathbf{0} + \mathbf{w} = (-\mathbf{v} + \mathbf{v}) + \mathbf{w} = -\mathbf{v} + (\mathbf{v} + \mathbf{w}) && \text{((a), (b), A3)} \\ &= -\mathbf{v} + \mathbf{0} = -\mathbf{v}. && \text{(hypothesis, A4)} \end{aligned}$$

■

Remark 3.1.4. Part (c) shows the zero vector is *unique*: any element acting as a zero must equal $\mathbf{0}$. Part (d) shows each $\mathbf{v}$ has a *unique* additive inverse.

Example 3.1.5 (Standard vector spaces)

Each of the following is a vector space: the zero space $\{\mathbf{0}\}$; n-space $\mathbb{R}^n$; the space $\mathbb{R}^\infty$ of infinite real sequences; the space $\mathbb{R}^{m\times n}$ of $m \times n$ matrices; the space P_n of polynomials of degree at most n; the space P_∞ of all polynomials; and the space $C[0,1]$ of continuous functions $f : [0,1] \to \mathbb{R}$.

Theorem 3.1.6 (Familiar properties)

Let V be a vector space, $\mathbf{u} \in V$, and k a scalar. Then (a) $0\mathbf{u} = \mathbf{0}$; (b) $k\mathbf{0} = \mathbf{0}$; (c) $(-1)\mathbf{u} = -\mathbf{u}$; (d) if $k\mathbf{u} = \mathbf{0}$ then $k = 0$ or $\mathbf{u} = \mathbf{0}$.

Proof.

(a) Starting from A8 and cancelling:

$$\begin{aligned} &0\mathbf{u} = (0+0)\mathbf{u} = 0\mathbf{u} + 0\mathbf{u} && \text{(A8)} \\ &0\mathbf{u} + (-0\mathbf{u}) = (0\mathbf{u} + 0\mathbf{u}) + (-0\mathbf{u}) = 0\mathbf{u} + \big(0\mathbf{u} + (-0\mathbf{u})\big) && \text{(A3)} \\ &\mathbf{0} = 0\mathbf{u} + \mathbf{0} = 0\mathbf{u} && \text{(A5, A4)} \end{aligned}$$

(b) The same cancellation, starting from A7:

$$\begin{aligned} &k\mathbf{0} = k(\mathbf{0} + \mathbf{0}) = k\mathbf{0} + k\mathbf{0} && \text{(A7)} \\ &k\mathbf{0} + (-k\mathbf{0}) = (k\mathbf{0} + k\mathbf{0}) + (-k\mathbf{0}) = k\mathbf{0} + \big(k\mathbf{0} + (-k\mathbf{0})\big) && \text{(A3)} \\ &\mathbf{0} = k\mathbf{0} + \mathbf{0} = k\mathbf{0} && \text{(A5, A4)} \end{aligned}$$

(c) Compute $\mathbf{u} + (-1)\mathbf{u}$:

$$\mathbf{u} + (-1)\mathbf{u} = 1\mathbf{u} + (-1)\mathbf{u} = (1 + (-1))\mathbf{u} = 0\mathbf{u} = \mathbf{0} \qquad \text{(A10, A8, part (a))}$$

So $(-1)\mathbf{u}$ is the additive inverse of $\mathbf{u}$; by Theorem 3.1.3(d), $(-1)\mathbf{u} = -\mathbf{u}$.

(d) Suppose $k\mathbf{u} = \mathbf{0}$. If $k = 0$ we are done. Otherwise $k \neq 0$, so k^{-1} exists, and

$$\mathbf{u} = 1\mathbf{u} = (k^{-1}k)\mathbf{u} = k^{-1}(k\mathbf{u}) = k^{-1}\mathbf{0} = \mathbf{0} \qquad \text{(A10, A9, part (b))}$$

So either $k = 0$ or $\mathbf{u} = \mathbf{0}$. ■

3.1.1 Subspaces

Often a vector space sits inside another. A line through the origin lives inside $\mathbb{R}^2$; the symmetric matrices live inside $\mathbb{R}^{2\times 2}$. Such a subset is a vector space in its own right, and we need only check closure to confirm it.

Definition 3.1.7 (Subspace)
A subset W of a vector space V is a *subspace* of V if W is itself a vector space under the addition and scalar multiplication inherited from V.

Theorem 3.1.8 (Subspace Test)
A non-empty subset W of a vector space V is a subspace if and only if

(i) W is closed under addition: $\mathbf{u}, \mathbf{v} \in W \Rightarrow \mathbf{u} + \mathbf{v} \in W$; and

(ii) W is closed under scalar multiplication: $\mathbf{u} \in W$, k scalar $\Rightarrow k\mathbf{u} \in W$.

Proof.

($\Rightarrow$) If W is a subspace then it is a vector space in its own right, so A1 and A6 hold — and these are exactly conditions (i) and (ii).

($\Leftarrow$) Suppose (i) and (ii) hold. The axioms A2, A3, A7, A8, A9, A10 are *identities*: they hold for all elements of V, hence in particular for all elements of W. So we need only supply A4 (a zero vector) and A5 (negatives).

Since $W \neq \varnothing$, choose some $\mathbf{u} \in W$.

- Applying (ii) with $k = 0$ gives $0\mathbf{u} \in W$. But $0\mathbf{u} = \mathbf{0}$ (Theorem 3.1.6(a)), so $\mathbf{0} \in W$ and A4 holds.
- Applying (ii) with $k = -1$ gives $(-1)\mathbf{u} \in W$ for each $\mathbf{u} \in W$. But $(-1)\mathbf{u} = -\mathbf{u}$ (Theorem 3.1.6(c)), so each element of W has its negative in W and A5 holds.

Thus W satisfies all ten axioms, so W is a subspace. ∎

Theorem 3.1.9
If W is a subspace of V, then W contains the zero vector of V.

Proof.
W is non-empty, so pick $\mathbf{u} \in W$; by closure under scalar multiplication, $0\mathbf{u} = \mathbf{0} \in W$ (Theorem 3.1.6(a)). ∎

Example 3.1.10 (Subspaces)

The following are subspaces: the zero subspace $\{\mathbf{0}\}$; any line through the origin in $\mathbb{R}^d$; any plane through the origin in $\mathbb{R}^d$ ($d \geq 2$); the symmetric matrices in $\mathbb{R}^{2\times 2}$; and P_n inside P_∞.

For the symmetric matrices $W = \{A \in \mathbb{R}^{2\times 2} : A^\mathsf{T} = A\}$: it is non-empty ($O \in W$). If $A, B \in W$ then $(A+B)^\mathsf{T} = A^\mathsf{T} + B^\mathsf{T} = A + B$, so $A + B \in W$; and $(kA)^\mathsf{T} = kA^\mathsf{T} = kA$, so $kA \in W$. By the Subspace Test, W is a subspace.

Example 3.1.11 (Non-subspaces)

The half-plane $H = \{(x, y) \in \mathbb{R}^2 : x \geq 0\}$ is not a subspace: it is not closed under scalar multiplication, since $(1, 0) \in H$ but $(-1)(1, 0) = (-1, 0) \notin H$. The plane $3x + y + z = 5$ is not a subspace: it does not contain the origin (Theorem 3.1.9 fails).

3.2 Span and Linear Independence

Definition 3.2.1 (Linear combination)

A vector $\mathbf{v}$ is a *linear combination* of $\mathbf{w}_1, \ldots, \mathbf{w}_r$ if $\mathbf{v} = \alpha_1\mathbf{w}_1 + \cdots + \alpha_r\mathbf{w}_r$ for some scalars $\alpha_1, \ldots, \alpha_r$, the *coefficients*.

Theorem 3.2.2

Let $S = \{\mathbf{w}_1, \ldots, \mathbf{w}_r\} \subseteq V$. For any scalars α_i, β_i: (a) $\sum_{i=1}^r \alpha_i\mathbf{w}_i \in V$; (b) $\big(\sum \alpha_i\mathbf{w}_i\big) + \big(\sum \beta_i\mathbf{w}_i\big) = \sum(\alpha_i + \beta_i)\mathbf{w}_i$; (c) $k\big(\sum \alpha_i\mathbf{w}_i\big) = \sum(k\alpha_i)\mathbf{w}_i$.

Proof.

(a) We argue by induction on r.
Base case ($r = 1$): $\alpha_1\mathbf{w}_1 \in V$ by A6 (closure under scalar multiplication).
Inductive step: assume $\sum_{i=1}^{r-1} \alpha_i\mathbf{w}_i \in V$. Then

$$\sum_{i=1}^{r} \alpha_i\mathbf{w}_i = \Big(\sum_{i=1}^{r-1} \alpha_i\mathbf{w}_i\Big) + \alpha_r\mathbf{w}_r,$$

which is a sum of two elements of V: the first by the inductive hypothesis, the second by A6. By A1 (closure under addition) the sum lies in V. So the result follows by induction.

(b) Group the two sums termwise and apply A8 (the distributive law $(\alpha_i+\beta_i)\mathbf{w}_i = \alpha_i\mathbf{w}_i+\beta_i\mathbf{w}_i$), using A2 and A3 to regroup:

$$\Big(\sum_i \alpha_i\mathbf{w}_i\Big) + \Big(\sum_i \beta_i\mathbf{w}_i\Big) = \sum_i(\alpha_i\mathbf{w}_i + \beta_i\mathbf{w}_i) = \sum_i(\alpha_i + \beta_i)\mathbf{w}_i.$$

(c) Distribute k across the sum by A7, then apply A9 to each term:

$$k\Big(\sum_i \alpha_i\mathbf{w}_i\Big) = \sum_i k(\alpha_i\mathbf{w}_i) = \sum_i(k\alpha_i)\mathbf{w}_i.$$

■

Theorem 3.2.3 (Span is the smallest containing subspace)

Let $S = \{\mathbf{w}_1, \ldots, \mathbf{w}_r\} \subseteq V$, and let W be the set of all linear combinations of vectors in S. Then (a) W is a subspace of V; and (b) W is the smallest subspace of V containing S.

Proof.

(a) We verify $W = \operatorname{span}(S)$ passes the Subspace Test.

Nonempty: $\mathbf{w}_1 = 1\mathbf{w}_1 \in W$, so $W \neq \varnothing$.

Closure under addition: If $\mathbf{x} = \sum_i \alpha_i\mathbf{w}_i$ and $\mathbf{y} = \sum_i \beta_i\mathbf{w}_i$ lie in W, then $\mathbf{x}+\mathbf{y} = \sum_i(\alpha_i+\beta_i)\mathbf{w}_i \in W$ by Theorem 3.2.2(b).

Closure under scalar multiplication: For a scalar k, $k\mathbf{x} = \sum_i(k\alpha_i)\mathbf{w}_i \in W$ by Theorem 3.2.2(c).

Therefore W is a subspace of V.

(b) Let W' be any subspace containing S. Being closed under addition and scalar multiplication, W' contains every linear combination of the $\mathbf{w}_i$, that is, $W \subseteq W'$. So W is contained in every subspace containing S — it is the smallest such. ■

Definition 3.2.4 (Span)
The subspace W of all linear combinations of $S = \{\mathbf{w}_1, \ldots, \mathbf{w}_r\}$ is the *span* of S, written $W = \operatorname{span}(S) = \operatorname{span}\{\mathbf{w}_1, \ldots, \mathbf{w}_r\}$. We say S *spans* W.

Example 3.2.5 (Spanning sets)

$\{\mathbf{e}_1, \mathbf{e}_2\}$ spans $\mathbb{R}^2$; so does $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2\}$ (the extra vector is redundant). The set $\{1, x, x^2\}$ spans P_2, as does $\{1, x, x^2, x + x^2\}$.

Definition 3.2.6 (Linear independence)
A set $\{\mathbf{v}_1, \ldots, \mathbf{v}_r\}$ is *linearly independent* if the only solution of

$$k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + \cdots + k_r\mathbf{v}_r = \mathbf{0}$$

is the trivial one $k_1 = \cdots = k_r = 0$. Otherwise the set is *linearly dependent*.

Example 3.2.7 (Independence and dependence)

In $\mathbb{R}^2$, $\{\mathbf{e}_1, \mathbf{e}_2\}$ is independent but $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2\}$ is dependent (since $\mathbf{e}_1 + \mathbf{e}_2 - (\mathbf{e}_1 + \mathbf{e}_2) = \mathbf{0}$ is a nontrivial relation). In P_2, $\{1, x, x^2\}$ is independent — a polynomial $k_0 + k_1x + k_2x^2$ is the zero function only if $k_0 = k_1 = k_2 = 0$ — while $\{1, x, x^2, x + x^2\}$ is dependent.

Theorem 3.2.8 (Dependence means redundancy)
A set S with two or more vectors is (a) linearly independent if and only if no vector in S is a linear combination of the others; (b) linearly dependent if and only if at least one vector in S is a linear combination of the others.

Proof.
Statements (a) and (b) are contrapositives of one another, so it suffices to prove (b).

($\Rightarrow$) Suppose $S = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ is dependent. By definition there is a relation

$$k_1\mathbf{v}_1 + \cdots + k_n\mathbf{v}_n = \mathbf{0}$$

in which some coefficient $k_i \neq 0$. Isolating that term and dividing by k_i,

$$\mathbf{v}_i = \sum_{j \neq i} \left(-\tfrac{k_j}{k_i}\right)\mathbf{v}_j,$$

so $\mathbf{v}_i$ is a linear combination of the other vectors.

($\Leftarrow$) Conversely, suppose some $\mathbf{v}_i = \sum_{j \neq i} c_j\mathbf{v}_j$. Moving everything to one side,

$$(-1)\mathbf{v}_i + \sum_{j \neq i} c_j\mathbf{v}_j = \mathbf{0},$$

which is a nontrivial relation (the coefficient of $\mathbf{v}_i$ is $-1 \neq 0$). Hence S is dependent. ∎

Example 3.2.9 (Spanning and independence via a matrix)

Do $(2, 2, 2), (0, 0, 3), (0, 1, 1)$ span $\mathbb{R}^3$? Are they independent?

Solution. Writing $\mathbf{x} = x_1(2,2,2) + x_2(0,0,3) + x_3(0,1,1)$ entrywise gives $A\mathbf{x} = \mathbf{x}$ with $A = \left[\begin{smallmatrix} 2&0&0\\ 2&0&1\\ 2&3&1 \end{smallmatrix}\right]$ (columns are the three vectors). Since $\det(A) = 2\cdot(0\cdot1-1\cdot3) = -6 \neq 0$, A is invertible (Theorem 2.2.9). By Theorem 1.5.13, $A\mathbf{x} = \mathbf{b}$ is consistent for every $\mathbf{b} \in \mathbb{R}^3$, so the vectors span $\mathbb{R}^3$; and by Theorem 1.5.7, $A\mathbf{x} = \mathbf{0}$ has only the trivial solution, so they are linearly independent.

3.3 Basis and Dimension

A basis is a spanning set with no redundancy — just enough vectors to build everything, with none to spare.

Definition 3.3.1 (Basis)

A set $S = \{\mathbf{v}_1, \ldots, \mathbf{v}_r\}$ in a vector space V is a *basis* for V if S is linearly independent and spans V.

Example 3.3.2

$\{\mathbf{e}_1, \mathbf{e}_2\}$ is a basis for $\mathbb{R}^2$, but $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2\}$ is not (not independent). $\{1, x, x^2\}$ is a basis for P_2, but $\{1, x, x^2, x + x^2\}$ is not. The set $\{(2,2,2), (0,0,3), (0,1,1)\}$ of Example 3.2.9 is a basis for $\mathbb{R}^3$.

Example 3.3.3–3.3.5 (Standard bases)

The *standard basis* for $\mathbb{R}^n$ is $\{\mathbf{e}_1, \ldots, \mathbf{e}_n\}$ with $\mathbf{e}_i$ the vector having 1 in position i and 0 elsewhere. The standard basis for P_n is the set of monomials $\{1, x, x^2, \ldots, x^n\}$. The standard basis for $\mathbb{R}^{2\times2}$ is the four matrices $\left[\begin{smallmatrix} 1&0\\ 0&0 \end{smallmatrix}\right], \left[\begin{smallmatrix} 0&1\\ 0&0 \end{smallmatrix}\right], \left[\begin{smallmatrix} 0&0\\ 1&0 \end{smallmatrix}\right], \left[\begin{smallmatrix} 0&0\\ 0&1 \end{smallmatrix}\right]$.

Example 3.3.6 (An infinite-dimensional space)

P_∞ has no finite spanning set. If $S = \{p_1, \ldots, p_m\}$ were one, let n be the highest degree among the p_i; then every combination has degree at most n, so $x^{n+1} \notin \operatorname{span}(S)$, contradicting $\operatorname{span}(S) = P_\infty$.

Definition 3.3.7 (Finite-dimensional)

A vector space is *finite-dimensional* if it has a finite spanning set.

The crucial fact — on which the very notion of dimension rests — is that in a space spanned by n vectors, no independent set can be larger than n.

Theorem 3.3.8

Let V be finite-dimensional with a basis $B = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$. (a) Any subset S with more than n vectors is linearly dependent. (b) Any subset S with fewer than n vectors does not span V.

Proof.

(a) Let $S = \{\mathbf{w}_1, \ldots, \mathbf{w}_m\}$ with $m > n$. Since B is a basis, each $\mathbf{w}_j$ is a combination of the $\mathbf{v}_i$, say $\mathbf{w}_j = \sum_{i=1}^{n} a_{ij}\mathbf{v}_i$. Consider any relation $\sum_{j=1}^{m} k_j\mathbf{w}_j = \mathbf{0}$ and substitute:

$$\sum_{j=1}^{m} k_j \sum_{i=1}^{n} a_{ij}\mathbf{v}_i = \sum_{i=1}^{n}\Big(\sum_{j=1}^{m} a_{ij}k_j\Big)\mathbf{v}_i = \mathbf{0}.$$

Because B is independent, each coefficient must vanish:

$$\sum_{j=1}^{m} a_{ij}k_j = 0 \qquad (i = 1, \ldots, n).$$

This is a homogeneous linear system of n equations in the m unknowns $k_1, \ldots, k_m$. Since $m > n$ (more unknowns than equations), Theorem 1.2.11 guarantees a *nontrivial* solution $(k_1, \ldots, k_m)$. That nontrivial solution is a nontrivial dependence relation, so S is linearly dependent.

(b) Let $S = \{\mathbf{w}_1, \ldots, \mathbf{w}_m\}$ with $m < n$, and suppose for contradiction that S spans V. Then each basis vector $\mathbf{v}_i$ is a combination of the $\mathbf{w}_j$. Reversing the roles in part (a): expressing the n vectors of B in terms of the $m < n$ vectors of S produces a homogeneous system with more unknowns than equations, hence a nontrivial relation among $\mathbf{v}_1, \ldots, \mathbf{v}_n$. This contradicts the independence of B. So S cannot span V. ∎

Theorem 3.3.9 (Dimension Theorem)
Every basis of a finite-dimensional vector space V has the same number of vectors.

Proof.
Let B have n vectors and B' have n' vectors; both are bases. Viewing B as the reference basis, B' is an independent set, so $n' \leq n$ by Theorem 3.3.8(a). Viewing B' as the reference basis, B is independent, so $n \leq n'$. Hence $n = n'$. ∎

Definition 3.3.10 (Dimension)
The *dimension* $\dim(V)$ of a finite-dimensional space V is the number of vectors in any basis. By convention $\dim(\{\mathbf{0}\}) = 0$.

Example 3.3.11

$\dim(\mathbb{R}^n) = n, \quad \dim(P_n) = n + 1, \quad \dim(\mathbb{R}^{2\times 2}) = 4.$

Theorem 3.3.12 (Extending and trimming)
Let S be a nonempty set of vectors in V. (a) If S is independent and $\mathbf{v} \notin \operatorname{span}(S)$, then $S \cup \{\mathbf{v}\}$ is independent. (b) If $\mathbf{v} \in S$ is a linear combination of the others, then $\operatorname{span}(S \setminus \{\mathbf{v}\}) = \operatorname{span}(S)$.

Proof.

(a) Let $S = \{\mathbf{v}_1, \ldots, \mathbf{v}_m\}$ be independent with $\mathbf{v} \notin \operatorname{span}(S)$. Consider a relation

$$k\mathbf{v} + k_1\mathbf{v}_1 + \cdots + k_m\mathbf{v}_m = \mathbf{0}.$$

If $k \neq 0$ we could solve for $\mathbf{v}$, getting $\mathbf{v} = \sum_i (-k_i/k)\mathbf{v}_i \in \operatorname{span}(S)$ — contradicting $\mathbf{v} \notin \operatorname{span}(S)$. So $k = 0$. The relation then reduces to $\sum_i k_i\mathbf{v}_i = \mathbf{0}$, and since S is independent, every $k_i = 0$ as well. Thus the only relation is trivial, so $S \cup \{\mathbf{v}\}$ is independent.

(b) Suppose $\mathbf{v} \in S$ is a combination of the others, and write $S' = S \setminus \{\mathbf{v}\}$ and $\mathbf{v} = \sum_{\mathbf{w}\in S'} c_{\mathbf{w}}\mathbf{w}$. Since $S' \subseteq S$, clearly $\operatorname{span}(S') \subseteq \operatorname{span}(S)$. For the reverse inclusion, take any $\mathbf{u} \in \operatorname{span}(S)$, say $\mathbf{u} = a\mathbf{v} + \sum_{\mathbf{w}\in S'} b_{\mathbf{w}}\mathbf{w}$. Substituting the expression for $\mathbf{v}$,

$$\mathbf{u} = a\sum_{\mathbf{w}\in S'} c_{\mathbf{w}}\mathbf{w} + \sum_{\mathbf{w}\in S'} b_{\mathbf{w}}\mathbf{w} = \sum_{\mathbf{w}\in S'} (ac_{\mathbf{w}} + b_{\mathbf{w}})\mathbf{w} \in \operatorname{span}(S').$$

So $\operatorname{span}(S) \subseteq \operatorname{span}(S')$, and therefore $\operatorname{span}(S \setminus \{\mathbf{v}\}) = \operatorname{span}(S)$. ∎

Theorem 3.3.13 (n vectors in an n-space)
Let $\dim(V) = n$ and let $S \subseteq V$ contain exactly n vectors. (a) S is a basis $\iff$ S is linearly independent.

(b) S is a basis $\iff$ S spans V.

Proof.

(a) ($\Rightarrow$) If S is a basis it is in particular independent. ($\Leftarrow$) Suppose S is independent. If S were not a basis it would fail to span, so some $\mathbf{v} \notin \text{span}(S)$; then $S \cup \{\mathbf{v}\}$ is independent (Theorem 3.3.12(a)) with $n+1$ vectors, contradicting Theorem 3.3.8(a). Hence S spans, and being independent it is a basis.

(b) ($\Rightarrow$) If S is a basis it in particular spans V. ($\Leftarrow$) Suppose S spans V. If S were not a basis it would fail to be independent, so by Theorem 3.2.8(b) some $\mathbf{v} \in S$ is a combination of the others; deleting it gives $S' = S \setminus \{\mathbf{v}\}$ with $\text{span}(S') = \text{span}(S) = V$ (Theorem 3.3.12(b)) and only $n-1$ vectors. But a set of fewer than $n = \dim(V)$ vectors cannot span V (Theorem 3.3.8(b)), a contradiction. Hence S is independent, and being spanning it is a basis. ■

Theorem 3.3.14 (Dimension of a subspace)

If W is a subspace of a finite-dimensional space V, then (a) $\dim(W) \le \dim(V)$; and (b) $W = V$ if and only if $\dim(W) = \dim(V)$.

Proof.

(a) V is finite-dimensional, so it has a finite spanning set; hence any independent subset of V has at most $\dim(V)$ vectors (Theorem 3.3.8(a)). In particular a basis of W is such an independent subset, so $\dim(W) \le \dim(V)$.

(b) ($\Rightarrow$) If $W = V$ then trivially $\dim(W) = \dim(V)$. ($\Leftarrow$) Suppose $\dim(W) = \dim(V) = n$, and let S be a basis of W. Then S is an independent set of n vectors in V, hence a basis of V by Theorem 3.3.13(a). Therefore $V = \text{span}(S) = W$. ■

Theorem 3.3.15 (Bases from spanning/independent sets)

Let S be a finite set in a finite-dimensional space V. (a) If S spans V, then S contains a subset that is a basis. (b) If S is independent, then S extends to a basis.

Proof.

(a) Suppose S spans V. Then $|S| \ge \dim(V)$ (Theorem 3.3.8(b)), so write $|S| = \dim(V) + k$ and induct on k.

Base case ($k = 0$): S spans V and has exactly $\dim(V)$ vectors, so S is a basis by Theorem 3.3.13(b), and contains a basis (itself).

Inductive step: suppose the claim holds for k, and let $|S| = \dim(V) + k + 1$. Since $|S| > \dim(V)$, S is dependent (Theorem 3.3.8(a)), so by Theorem 3.2.8(b) some $\mathbf{v} \in S$ is a combination of the others. Deleting it, $S' = S \setminus \{\mathbf{v}\}$ still spans V (Theorem 3.3.12(b)) and has $\dim(V)+k$ vectors. By the inductive hypothesis S' contains a basis, and since $S' \subset S$, so does S.

(b) Suppose S is independent. If S already spans V, then S is a basis containing S, and we are done. Otherwise there is some $\mathbf{v}_1 \notin \text{span}(S)$; then $S \cup \{\mathbf{v}_1\}$ is independent (Theorem 3.3.12(a)). If this still does not span, pick $\mathbf{v}_2 \notin \text{span}(S \cup \{\mathbf{v}_1\})$, and again $S \cup \{\mathbf{v}_1, \mathbf{v}_2\}$ is independent. Repeat. Because V is finite-dimensional, an independent set cannot grow past $\dim(V)$ vectors (Theorem 3.3.8(a)), so this process must stop — and it stops exactly when the set spans V. At that point $B = S \cup \{\mathbf{v}_1, \ldots, \mathbf{v}_m\}$ is independent and spanning, hence a basis of V containing S. ■

3.4 Direct Sums

Definition 3.4.1 (Sum and direct sum)
Let V, W be subspaces of a vector space U. Their *sum* is

$$V + W = \{\mathbf{v} + \mathbf{w} : \mathbf{v} \in V,\ \mathbf{w} \in W\}.$$

If $V \cap W = \{\mathbf{0}\}$, the sum is *direct*, written $V \oplus W$.

Theorem 3.4.2
If V, W are subspaces of U, then $V + W$ is a subspace of U.

Proof.
We verify $V + W$ passes the Subspace Test.

Nonempty: $\mathbf{0} = \mathbf{0} + \mathbf{0} \in V + W$.

Closure under addition: If $\mathbf{u}_1 = \mathbf{v}_1 + \mathbf{w}_1$ and $\mathbf{u}_2 = \mathbf{v}_2 + \mathbf{w}_2$ (with $\mathbf{v}_i \in V$, $\mathbf{w}_i \in W$), then $\mathbf{u}_1 + \mathbf{u}_2 = (\mathbf{v}_1 + \mathbf{v}_2) + (\mathbf{w}_1 + \mathbf{w}_2) \in V + W$, since V and W are each closed under addition.

Closure under scalar multiplication: For a scalar k, $k\mathbf{u}_1 = k\mathbf{v}_1 + k\mathbf{w}_1 \in V + W$, since V and W are each closed under scalar multiplication.

Therefore $V + W$ is a subspace. ■

Theorem 3.4.3 (Uniqueness characterises directness)
The sum $V + W$ is direct if and only if every $\mathbf{u} \in V + W$ has a *unique* expression $\mathbf{u} = \mathbf{v} + \mathbf{w}$ with $\mathbf{v} \in V$, $\mathbf{w} \in W$.

Proof.

($\Rightarrow$) Suppose $V \cap W = \{\mathbf{0}\}$, and let $\mathbf{u} = \mathbf{v} + \mathbf{w} = \mathbf{v}' + \mathbf{w}'$ be two decompositions with $\mathbf{v}, \mathbf{v}' \in V$ and $\mathbf{w}, \mathbf{w}' \in W$. Rearranging, $\mathbf{v} - \mathbf{v}' = \mathbf{w}' - \mathbf{w}$. The left side lies in V and the right side in W, so this common vector lies in $V \cap W = \{\mathbf{0}\}$. Hence $\mathbf{v} - \mathbf{v}' = \mathbf{0}$ and $\mathbf{w}' - \mathbf{w} = \mathbf{0}$, i.e. $\mathbf{v} = \mathbf{v}'$ and $\mathbf{w} = \mathbf{w}'$. The decomposition is unique.

($\Leftarrow$) Suppose every element has a unique decomposition. Let $\mathbf{x} \in V \cap W$. Then $\mathbf{x}$ has two decompositions,

$$\mathbf{x} = \underbrace{\mathbf{x}}_{\in V} + \underbrace{\mathbf{0}}_{\in W} = \underbrace{\mathbf{0}}_{\in V} + \underbrace{\mathbf{x}}_{\in W}.$$

By uniqueness these must coincide, forcing $\mathbf{x} = \mathbf{0}$. Hence $V \cap W = \{\mathbf{0}\}$, so the sum is direct. ■

Theorem 3.4.4 (Dimension of a direct sum)
Let V, W be finite-dimensional subspaces of U. If $U = V \oplus W$, then $\dim(U) = \dim(V) + \dim(W)$.

Proof.
Let $B_V = \{\mathbf{v}_1, \ldots, \mathbf{v}_m\}$ be a basis of V and $B_W = \{\mathbf{w}_1, \ldots, \mathbf{w}_n\}$ a basis of W. We show $B_V \cup B_W$ is a basis of U.
Spanning: any $\mathbf{u} \in U = V \oplus W$ can be written $\mathbf{u} = \mathbf{v} + \mathbf{w}$ with $\mathbf{v} \in V$, $\mathbf{w} \in W$. Expanding $\mathbf{v} = \sum_i \alpha_i \mathbf{v}_i$ and $\mathbf{w} = \sum_i \beta_i \mathbf{w}_i$ shows $\mathbf{u}$ is a linear combination of $B_V \cup B_W$.

Independence: suppose $\sum_i \alpha_i \mathbf{v}_i + \sum_i \beta_i \mathbf{w}_i = \mathbf{0}$. Then

$$\sum_i \alpha_i \mathbf{v}_i = -\sum_i \beta_i \mathbf{w}_i,$$

where the left side lies in V and the right in W. So this common vector lies in $V \cap W = \{\mathbf{0}\}$, forcing both sides to be $\mathbf{0}$. Independence of B_V then gives every $\alpha_i = 0$, and independence of B_W gives every $\beta_i = 0$.
Therefore $B_V \cup B_W$ is a basis of U with $m+n$ vectors, so $\dim(U) = m+n = \dim(V) + \dim(W)$. ■

3.5 Coordinates and Change of Basis

Once a basis is fixed and *ordered*, every vector acquires a unique list of coordinates. This is what lets an abstract n-dimensional space be computed with as though it were $\mathbb{R}^n$.

Theorem 3.5.1 (Unique representation)
If $B = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ is a basis for V, then every $\mathbf{v} \in V$ can be written $\mathbf{v} = c_1\mathbf{v}_1 + \cdots + c_n\mathbf{v}_n$ in exactly one way.

Proof.
Existence: B spans V, so such scalars exist. *Uniqueness:* if $\mathbf{v} = \sum c_i \mathbf{v}_i = \sum c_i' \mathbf{v}_i$, then $\sum (c_i - c_i')\mathbf{v}_i = \mathbf{0}$; independence of B forces $c_i - c_i' = 0$ for each i, so the representation is unique. ■

Definition 3.5.2 (Coordinate vector)
If $B = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ is an ordered basis and $\mathbf{v} = c_1\mathbf{v}_1 + \cdots + c_n\mathbf{v}_n$, the scalars c_i are the *coordinates* of $\mathbf{v}$ relative to B, and

$$[\mathbf{v}]_B = (c_1, c_2, \ldots, c_n) \in \mathbb{R}^n$$

is the *coordinate vector* of $\mathbf{v}$ relative to B.

Example 3.5.3 (Coordinates in P_2)

For the basis $B = \{1, x, x^2\}$ of P_2 and $p(x) = a_0 + a_1x + a_2x^2$, we have $p = a_0 \cdot 1 + a_1 \cdot x + a_2 \cdot x^2$, so $[p]_B = (a_0, a_1, a_2)$.

Example 3.5.4 (Coordinates in a skewed basis)

For $B' = \{1+x, 2x, 3x^2\}$ and $p(x) = a_0 + a_1x + a_2x^2$, solve $p = c_1(1+x) + c_2(2x) + c_3(3x^2)$. Matching coefficients: $a_0 = c_1$, $a_1 = c_1 + 2c_2$, $a_2 = 3c_3$, so $c_1 = a_0$, $c_2 = \frac{a_1 - a_0}{2}$, $c_3 = \frac{a_2}{3}$, giving $[p]_{B'} = \left(a_0, \frac{a_1-a_0}{2}, \frac{a_2}{3}\right)$.

How are $[\mathbf{v}]_B$ and $[\mathbf{v}]_{B'}$ related when we change basis? The answer is multiplication by a fixed matrix.

Definition 3.5.5 (Change-of-basis matrix)
For ordered bases $B = \{\mathbf{u}_1, \ldots, \mathbf{u}_n\}$ and $B' = \{\mathbf{u}_1', \ldots, \mathbf{u}_n'\}$ of V, the *change-of-basis matrix from B to B'* is

$$P_{B',B} = \big[\, [\mathbf{u}_1]_{B'} \;\; [\mathbf{u}_2]_{B'} \;\; \cdots \;\; [\mathbf{u}_n]_{B'} \,\big],$$

the $n \times n$ matrix whose columns are the coordinate vectors of the *old* basis B relative to the *new* basis B'.

Theorem 3.5.6 (Change-of-basis formula)
For all $\mathbf{v} \in V$, $[\mathbf{v}]_{B'} = P_{B',B}\,[\mathbf{v}]_B$.

Proof.
Write $[\mathbf{v}]_B = (c_1, \ldots, c_n)$, so $\mathbf{v} = \sum_j c_j \mathbf{u}_j$. Taking coordinates relative to B' and using that coordinates respect linear combinations (a consequence of Theorem 3.5.1),

$$[\mathbf{v}]_{B'} = \Big[\sum_j c_j \mathbf{u}_j\Big]_{B'} = \sum_j c_j [\mathbf{u}_j]_{B'} = \sum_j c_j\,(j\text{th column of } P_{B',B}) = P_{B',B}\,[\mathbf{v}]_B,$$

the last equality by the column interpretation of matrix–vector multiplication (Theorem 1.3.10). ■

Theorem 3.5.7 (The change-of-basis matrix is invertible)
$P_{B',B}$ is invertible, with $P_{B',B}^{-1} = P_{B,B'}$.

Proof.
Applying Theorem 3.5.6 twice, for every $\mathbf{v}$,

$$[\mathbf{v}]_B = P_{B,B'}[\mathbf{v}]_{B'} = P_{B,B'}P_{B',B}[\mathbf{v}]_B.$$

Taking $\mathbf{v} = \mathbf{u}_i$ (so $[\mathbf{u}_i]_B = \mathbf{e}_i$) shows $P_{B,B'}P_{B',B}\mathbf{e}_i = \mathbf{e}_i$ for each i; that is, $P_{B,B'}P_{B',B} = I$. By Theorem 1.5.11, $P_{B',B}$ is invertible with inverse $P_{B,B'}$. ■

Remark 3.5.8. When B is a standard basis, $P_{B,B'}$ is read off directly (its columns are the coordinates of the new basis vectors, which in a standard basis are the vectors themselves). Theorem 3.5.7 then yields the reverse matrix by inversion.

3.6 Row Space, Column Space, and Kernel

To each $m \times n$ matrix A we attach four subspaces. Three appear here; the fourth, the cokernel, is the kernel of the transpose.

Definition 3.6.1–3.6.4 (The four fundamental subspaces)
Let A be an $m \times n$ matrix.

- The *row space* $\mathrm{row}(A)$ is the subspace of $\mathbb{R}^n$ spanned by the rows of A.
- The *column space* $\mathrm{col}(A)$ is the subspace of $\mathbb{R}^m$ spanned by the columns of A.
- The *kernel* $\ker(A)$ is the set of all $\mathbf{x} \in \mathbb{R}^n$ with $A\mathbf{x} = \mathbf{0}$.
- The *cokernel* is $\mathrm{coker}(A) = \ker(A^\mathsf{T})$.

Theorem 3.6.5
The kernel of an $m \times n$ matrix is a subspace of $\mathbb{R}^n$.

Proof.
We verify $\ker(A)$ passes the Subspace Test.

Nonempty: $A\mathbf{0} = \mathbf{0}$, so $\mathbf{0} \in \ker(A)$.

Closure under addition: If $\mathbf{x}, \mathbf{y} \in \ker(A)$ then $A(\mathbf{x}+\mathbf{y}) = A\mathbf{x}+A\mathbf{y} = \mathbf{0}+\mathbf{0} = \mathbf{0}$, so $\mathbf{x}+\mathbf{y} \in \ker(A)$.

Closure under scalar multiplication: If $\mathbf{x} \in \ker(A)$ and k is a scalar, then $A(k\mathbf{x}) = k(A\mathbf{x}) = k\mathbf{0} = \mathbf{0}$, so $k\mathbf{x} \in \ker(A)$.

Therefore $\ker(A)$ is a subspace of $\mathbb{R}^n$. ∎

Theorem 3.6.6

Let U be an echelon form of A. If U has r pivots, then $\dim(\mathrm{row}(A)) = r$.

Proof omitted (MTH2025). The essential points are that row operations do not change the row space, and that the nonzero rows of an echelon form are independent; the pivot count r is therefore the dimension of the row space.

Corollary 3.6.7

Every echelon form of A has the same number of pivots.

Proof.

If U, U' are echelon forms of A with r, r' pivots, then $\dim(\mathrm{row}(A)) = r$ and $\dim(\mathrm{row}(A)) = r'$ by Theorem 3.6.6; hence $r = r'$. ∎

Theorem 3.6.8 (Consistency via the column space)

The system $A\mathbf{x} = \mathbf{b}$ is consistent if and only if $\mathbf{b} \in \mathrm{col}(A)$.

Proof.

By Theorem 1.3.10, $A\mathbf{x} = x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n$ is a linear combination of the columns $\mathbf{a}_j$ of A. Thus $A\mathbf{x} = \mathbf{b}$ has a solution $\mathbf{x}$ if and only if $\mathbf{b}$ is expressible as such a combination, i.e. $\mathbf{b} \in \mathrm{span}\{\mathbf{a}_1, \dots, \mathbf{a}_n\} = \mathrm{col}(A)$. ∎

Example 3.6.9 (Is b in the column space?)

Let $A = \left[\begin{smallmatrix} 1 & 3 \\ 4 & 6 \end{smallmatrix}\right]$, $\mathbf{b} = \left[\begin{smallmatrix} 2 \\ 10 \end{smallmatrix}\right]$. Since $\det(A) = 6 - 12 = -6 \neq 0$, A is invertible, so $A\mathbf{x} = \mathbf{b}$ is consistent (Theorem 1.5.13) and $\mathbf{b} \in \mathrm{col}(A)$. Solving, $\mathbf{x} = A^{-1}\mathbf{b} = (1, \frac{1}{3})^\mathsf{T}$, so $\mathbf{b} = 1 \cdot \left[\begin{smallmatrix} 1 \\ 4 \end{smallmatrix}\right] + \frac{1}{3}\left[\begin{smallmatrix} 3 \\ 6 \end{smallmatrix}\right]$.

3.7 Vector Spaces over Fields

The definition of a vector space (Definition 3.1.1) never used any special property of $\mathbb{R}$ beyond the ordinary rules of arithmetic. Any set of scalars obeying those rules — a *field* — will do.

Definition 3.7.1 (Field)

A *field* $\mathbb{F}$ is a non-empty set with two operations, addition and multiplication, assigning to each $a, b \in \mathbb{F}$ unique elements $a + b$ and ab in $\mathbb{F}$, such that for all $a, b, c \in \mathbb{F}$:

- F1 (commutativity): $a + b = b + a$, $ab = ba$.
- F2 (associativity): $a + (b + c) = (a + b) + c$, $a(bc) = (ab)c$.
- F3 (distributivity): $a(b + c) = ab + ac$.
- F4 (identities): distinct $0, 1 \in \mathbb{F}$ with $a + 0 = a$, $1a = a$.
- F5 (additive inverse): each a has $-a$ with $a + (-a) = 0$.
- F6 (multiplicative inverse): each $a \neq 0$ has a^{-1} with $aa^{-1} = 1$.

Example 3.7.2 (Common fields)

The rationals $\mathbb{Q}$, the reals $\mathbb{R}$, the complex numbers $\mathbb{C}$, and the integers modulo a prime p, written $\mathbb{Z}_p = \{0, 1, \ldots, p-1\}$, are all fields. In $\mathbb{Z}_p$ one adds and multiplies as usual, then reduces modulo p.

Example 3.7.3 (Arithmetic in $\mathbb{Z}_3$)

The addition and multiplication tables for $\mathbb{Z}_3 = \{0, 1, 2\}$:

$+$	0	1	2
0	0	1	2
1	1	2	0
2	2	0	1

$\times$	0	1	2
0	0	0	0
1	0	1	2
2	0	2	1

Every nonzero element has a multiplicative inverse ($1^{-1} = 1$, $2^{-1} = 2$ since $2 \cdot 2 = 4 \equiv 1$), so F6 holds and $\mathbb{Z}_3$ is a field.

Remark 3.7.4. The construction of $\mathbb{Z}_p$ makes sense for any $p \in \mathbb{N}$, but $\mathbb{Z}_p$ is a field only when p is prime — otherwise a nonzero element can fail to have a multiplicative inverse (e.g. 2 in $\mathbb{Z}_4$).

Example 3.7.5 ($\mathbb{F}^n$ is a vector space over $\mathbb{F}$)

For a field $\mathbb{F}$ and $n \in \mathbb{N}$, the set $\mathbb{F}^n$ of n-tuples from $\mathbb{F}$, with componentwise addition and scalar multiplication, satisfies A1–A10 and is a vector space over $\mathbb{F}$. Taking $\mathbb{F} = \mathbb{Z}_2$ gives $\mathbb{Z}_2^n$, the setting for binary codes below.

3.8 Application: Coding Theory

Information sent over a noisy channel can be corrupted. *Coding theory* adds structured redundancy so that errors can be detected, and sometimes corrected. We work over $\mathbb{Z}_2$, so codewords are binary strings in $\mathbb{Z}_2^n$ and arithmetic is mod 2.

3.8.1 Error detection

With plain 3-bit words, every string in $\mathbb{Z}_2^3$ is valid, so a single flipped bit turns one valid word into another and goes undetected. Appending a *checksum* bit — the mod-2 sum of the data bits — fixes this: a single error makes the checksum inconsistent, so the received word is not a codeword and the error is *detected*. (A two-bit error can still slip through.)

3.8.2 Error correction: the Hamming $(7, 4)$ code

To *correct* errors we add more redundancy. The Hamming $(7, 4)$ code encodes a 4-bit word $\mathbf{d} = (d_1, d_2, d_3, d_4)$ as a 7-bit word $(d_1, d_2, d_3, d_4, p_1, p_2, p_3)$ with parity bits

$$p_1 = d_2 + d_3 + d_4, \qquad p_2 = d_1 + d_3 + d_4, \qquad p_3 = d_1 + d_2 + d_4 \pmod 2.$$

Treating $\mathbf{d}$ as a row vector, the codeword is $\mathbf{c} = \mathbf{d}\,G$, where the *generator matrix* is

$$G = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{bmatrix}.$$

Lemma 3.8.2

The set of codewords generated by G is $\operatorname{row}(G)$.

Proof.

The codewords are $\{\mathbf{d}\,G : \mathbf{d} \in \mathbb{Z}_2^4\}$. By Theorem 1.3.10 applied to rows (equivalently, $\mathbf{d}\,G = \sum_i d_i(\text{row } i \text{ of } G)$), each $\mathbf{d}\,G$ is a linear combination of the rows of G, and conversely every such combination arises from some $\mathbf{d}$. Hence the codeword set is exactly $\operatorname{span}\{\text{rows of } G\} = \operatorname{row}(G)$. ■

Example 3.8.3 (Encoding)

For $\mathbf{d} = (1,0,0,1)$: $p_1 = 0{+}0{+}1 = 1$, $p_2 = 1{+}0{+}1 = 0$, $p_3 = 1{+}0{+}1 = 0$, so $\mathbf{c} = \mathbf{d}\,G = (1,0,0,1,1,0,0)$.

To test whether a received word $\mathbf{c} \in \mathbb{Z}_2^7$ is a codeword, introduce the *parity-check matrix*

$$H = \begin{bmatrix} 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 & 1 \end{bmatrix}.$$

Theorem 3.8.4

$\operatorname{row}(G) = \ker(H)$; consequently $\mathbf{c}$ is a valid codeword if and only if $H\mathbf{c}^\mathsf{T} = \mathbf{0}$.

Proof.

We work over $\mathbb{Z}_2$, with the Hamming $(7,4)$ code ($n = 7$, $k = 4$).
One inclusion. By construction of the parity-check matrix, each parity bit is defined so the check equations hold on every row of G; that is, $HG^\mathsf{T} = O$. Now take any codeword $\mathbf{c} \in \operatorname{row}(G)$, say $\mathbf{c} = \mathbf{d}\,G$. Then

$$H\mathbf{c}^\mathsf{T} = H(\mathbf{d}\,G)^\mathsf{T} = HG^\mathsf{T}\mathbf{d}^\mathsf{T} = O\mathbf{d}^\mathsf{T} = \mathbf{0},$$

so $\mathbf{c} \in \ker(H)$. Hence $\operatorname{row}(G) \subseteq \ker(H)$.
Equality by dimension count. The rows of G are independent (the generator carries an identity block), so $\dim \operatorname{row}(G) = 4$. For the kernel, the three rows of H are independent, so $\operatorname{rank}(H) = 3$ and the Rank–Nullity Theorem gives $\dim \ker(H) = 7 - 3 = 4$. Since $\operatorname{row}(G) \subseteq \ker(H)$ and both have dimension 4, Theorem 3.3.14(b) forces $\operatorname{row}(G) = \ker(H)$.
Consequence. Chaining the equivalences,

$$\mathbf{c} \text{ is a valid codeword} \iff \mathbf{c} \in \operatorname{row}(G) \iff \mathbf{c} \in \ker(H) \iff H\mathbf{c}^\mathsf{T} = \mathbf{0}.$$

■

Definition 3.8.5 (Linear code)

A subspace L of $\mathbb{Z}_2^n$ is a *linear code*. A matrix G is a *generator matrix* for L if its rows form a basis of L, and H is a *parity-check matrix* for L if $\ker(H) = \operatorname{row}(G) = L$.

Lemma 3.8.7 (One-bit error correction)

A linear code with parity-check matrix H is one-bit error-correcting if and only if the columns of H are all nonzero and pairwise distinct.

Proof.

The key observation is that a single error in position i adds $\mathbf{e}_i$ to the transmitted codeword, so the received word $\mathbf{r}$ has *syndrome*

$$H\mathbf{r}^\mathsf{T} = H\mathbf{e}_i^\mathsf{T} = (i\text{th column of } H).$$

Thus single-bit errors are detectable and correctable precisely when their syndromes (the columns of H) tell them apart.

($\Leftarrow$) Suppose the columns of H are nonzero and pairwise distinct. Then a single error in position i produces a nonzero syndrome equal to column i, and since the columns are distinct, that syndrome identifies i uniquely. The decoder locates position i and flips it back, correcting the error.

($\Rightarrow$) Conversely, suppose the code corrects one-bit errors. If some column i of H were zero, an error in position i would give syndrome $\mathbf{0}$ — indistinguishable from no error, so undetectable. If two columns $i \neq j$ were equal, errors in positions i and j would give the same syndrome — indistinguishable, so uncorrectable. Both contradict one-bit correctability, so the columns must be nonzero and pairwise distinct. ■

Chapter 3 Summary

- A vector space is any set satisfying A1–A10 (Definition 3.1.1); a subset is a subspace iff it is nonempty and closed under $+$ and scalar multiplication (Theorem 3.1.8), and every subspace contains $\mathbf{0}$ (Theorem 3.1.9).
- $\mathrm{span}(S)$ is the smallest subspace containing S (Theorem 3.2.3); a set is dependent iff some vector is a combination of the others (Theorem 3.2.8).
- A *basis* is an independent spanning set. All bases have the same size (Theorem 3.3.9), the *dimension*. In an n-space, any n independent vectors — or any n spanning vectors — form a basis (Theorem 3.3.13). Independent sets extend to bases; spanning sets contain bases (Theorem 3.3.15).
- For $U = V \oplus W$: decompositions are unique (Theorem 3.4.3) and $\dim U = \dim V + \dim W$ (Theorem 3.4.4).
- Coordinates relative to a basis are unique (Theorem 3.5.1); changing basis multiplies by $P_{B',B}$ (Theorem 3.5.6), which is invertible with inverse $P_{B,B'}$ (Theorem 3.5.7).
- The four subspaces of A: row, column, kernel, cokernel. $\ker(A)$ is a subspace (Theorem 3.6.5); $\dim \mathrm{row}(A) =$ pivot count (Theorem 3.6.6); $A\mathbf{x} = \mathbf{b}$ is consistent iff $\mathbf{b} \in \mathrm{col}(A)$ (Theorem 3.6.8).
- Scalars may come from any field $\mathbb{F}$ (Definition 3.7.1); $\mathbb{Z}_2^n$ underlies binary linear codes, where codewords are $\mathrm{row}(G) = \ker(H)$ (Theorem 3.8.4) and one-bit correction holds iff H has distinct nonzero columns (Lemma 3.8.7).

Chapter 3 Exercises

3.1 Prove Theorem 3.1.6(b) and (d) from the axioms: $k\mathbf{0} = \mathbf{0}$, and if $k\mathbf{u} = \mathbf{0}$ then $k = 0$ or $\mathbf{u} = \mathbf{0}$.

3.2 Determine whether $W = \{(x, y, z) \in \mathbb{R}^3 : x + 2y - z = 0\}$ is a subspace of $\mathbb{R}^3$. If so, find a basis and its dimension.

3.3 Are the polynomials $1 + x$, $x + x^2$, $1 + x^2$ a basis for P_2? Justify using a determinant.

3.4 Let V be the subspace of $\mathbb{R}^{2\times 2}$ of symmetric matrices. Find $\dim(V)$ and exhibit a basis.

3.5 For $A = \left[\begin{smallmatrix} 1 & 2 & 1 \\ 2 & 4 & 3 \end{smallmatrix}\right]$, find a basis for $\ker(A)$ and state $\dim \ker(A)$.

3.6 In $\mathbb{Z}_2^7$ with the Hamming code, the word $\mathbf{r} = (1, 0, 1, 1, 0, 0, 1)$ is received. Compute the syndrome $H\mathbf{r}^{\mathsf{T}}$ and, using Lemma 3.8.7, determine the corrected codeword.

Solutions to Chapter 3 Exercises

3.1 (b) $k\mathbf{0} = k(\mathbf{0} + \mathbf{0}) = k\mathbf{0} + k\mathbf{0}$ by A7; adding $-(k\mathbf{0})$ and using A3,A5,A4 gives $\mathbf{0} = k\mathbf{0}$. (d) Suppose $k\mathbf{u} = \mathbf{0}$ and $k \neq 0$. Multiply by k^{-1}: $\mathbf{u} = 1\mathbf{u} = (k^{-1}k)\mathbf{u} = k^{-1}(k\mathbf{u}) = k^{-1}\mathbf{0} = \mathbf{0}$, using A10,A9 and part (b). So $k = 0$ or $\mathbf{u} = \mathbf{0}$. ■

3.2 W is the solution set of one homogeneous equation, hence $\ker[\,1\ 2\ {-1}\,]$, a subspace by Theorem 3.6.5. Free variables y, z: $x = -2y + z$, so $(x, y, z) = y(-2, 1, 0) + z(1, 0, 1)$. A basis is $\{(-2, 1, 0), (1, 0, 1)\}$ and $\dim(W) = 2$.

3.3 Coordinate vectors relative to $\{1, x, x^2\}$ are $(1,1,0), (0,1,1), (1,0,1)$. Form $M = \left[\begin{smallmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{smallmatrix}\right]$ (columns the coordinate vectors); $\det(M) = 1(1) - 0 + 1(1) = 2 \neq 0$, so M is invertible, the three coordinate vectors are independent in $\mathbb{R}^3$, and hence the polynomials are independent. Three independent vectors in the 3-dimensional P_2 form a basis (Theorem 3.3.13(a)). ✓

3.4 A symmetric 2×2 matrix is $\left[\begin{smallmatrix} a & b \\ b & c \end{smallmatrix}\right] = a\left[\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right] + b\left[\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right] + c\left[\begin{smallmatrix} 0 & 0 \\ 0 & 1 \end{smallmatrix}\right]$. These three matrices are independent and span V, so they form a basis and $\dim(V) = 3$.

3.5 Reduce: $\left[\begin{smallmatrix} 1 & 2 & 1 \\ 2 & 4 & 3 \end{smallmatrix}\right] \to \left[\begin{smallmatrix} 1 & 2 & 1 \\ 0 & 0 & 1 \end{smallmatrix}\right] \to \left[\begin{smallmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \end{smallmatrix}\right]$. Pivots in columns $1, 3$; free variable $x_2 = t$. Then $x_3 = 0$, $x_1 = -2t$, so $\ker(A) = \{t(-2,1,0) : t \in \mathbb{R}\}$, a basis is $\{(-2,1,0)\}$, and $\dim \ker(A) = 1$.

3.6 $H\mathbf{r}^\mathsf{T}$ over $\mathbb{Z}_2$: row 1 of H dotted with $\mathbf{r}$ is $r_4+r_5+r_6+r_7 = 1+0+0+1 = 0$; row 2 is $r_2+r_3+r_6+r_7 = 0+1+0+1 = 0$; row 3 is $r_1+r_3+r_5+r_7 = 1+1+0+1 = 1$. The syndrome is $(0,0,1)^\mathsf{T}$, which matches column 1 of H (entries $0,0,1$). So the error is in position 1; flipping it gives the corrected codeword $\mathbf{c} = (0,0,1,1,0,0,1)$.

CHAPTER 4

LINEAR TRANSFORMATIONS

What this chapter is about. A *linear transformation* is a function between vector spaces that respects the two operations — it sends sums to sums and scalar multiples to scalar multiples. Matrices were our first examples; now we treat the idea abstractly, so that differentiation, transposition, and evaluation all become instances of one concept. Two subspaces measure a transformation: its *kernel* (what collapses to zero) and its *range* (what it can reach). The *Rank–Nullity Theorem* ties their dimensions together, and from it flows the theory of *isomorphism* — the precise sense in which every n-dimensional real space is "the same as" $\mathbb{R}^n$. Finally, fixing bases turns any linear transformation back into a matrix.

Reference. A&R §§8.1–8.4 (and §4.9 for matrix representations).

4.1 Functions Between Sets

We first recall vocabulary for functions in general. A *transformation* (or *function*, or *map*) $T : D \to C$ from a set D to a set C assigns to each $x \in D$ a unique element $T(x) \in C$. The set D is the *domain*, C the *codomain*, and $T(x)$ the *image* of x. The *range* is the set of all images,

$$\operatorname{ran}(T) = \{T(x) : x \in D\} \subseteq C.$$

Definition 4.1.1 (Injective, surjective, bijective)
A transformation $T : D \to C$ is *surjective* (*onto*) if $\operatorname{ran}(T) = C$; *injective* (*one-to-one*) if $x \neq y$ implies $T(x) \neq T(y)$; and *bijective* if it is both.

4.2 Linear Transformations

Definition 4.2.1 (Linear transformation)
Let V, W be vector spaces. A function $T : V \to W$ is a *linear transformation* if

$$T(\mathbf{u} + k\mathbf{v}) = T(\mathbf{u}) + kT(\mathbf{v})$$

for all $\mathbf{u}, \mathbf{v} \in V$ and all scalars k. A linear transformation with $W = V$ is a *linear operator*.

Taking $k = 1$ recovers additivity $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$, and taking $\mathbf{u} = \mathbf{0}$ recovers homogeneity $T(k\mathbf{v}) = kT(\mathbf{v})$; the single condition above packages both.

Lemma 4.2.2
If $T : V \to W$ is linear, then $T(\mathbf{0}) = \mathbf{0}$.

Proof.
Using homogeneity with scalar 0 and Theorem 3.1.6(a): $T(\mathbf{0}) = T(0 \cdot \mathbf{0}) = 0 \cdot T(\mathbf{0}) = \mathbf{0}$. ■

Lemma 4.2.2 gives a quick test: any map with $T(\mathbf{0}) \neq \mathbf{0}$ cannot be linear.

Definition 4.2.3 (Matrix transformation)
For an $m \times n$ matrix A, the map $T_A : \mathbb{R}^n \to \mathbb{R}^m$ defined by $T_A(\mathbf{x}) = A\mathbf{x}$ is a *matrix transformation*, and A is its *standard matrix*.

Example 4.2.4 (Matrix transformations are linear)

For $T_A(\mathbf{x}) = A\mathbf{x}$, the distributive and scalar laws of matrix multiplication (Theorem 1.4.1) give $T_A(\mathbf{u} + k\mathbf{v}) = A(\mathbf{u} + k\mathbf{v}) = A\mathbf{u} + kA\mathbf{v} = T_A(\mathbf{u}) + kT_A(\mathbf{v})$. So every matrix transformation is linear.

Example 4.2.5 (Transposition is linear)

Define $T : \mathbb{R}^{n\times m} \to \mathbb{R}^{m\times n}$ by $T(A) = A^{\mathsf{T}}$. By Lemma 1.4.4, $T(A + kB) = (A + kB)^{\mathsf{T}} = A^{\mathsf{T}} + kB^{\mathsf{T}} = T(A) + kT(B)$, so T is linear.

Example 4.2.6 (Differentiation is linear)

Let $V = C^1(-1, 1)$ and $W = F(-1, 1)$ (all real functions), and define $D : V \to W$ by $D[f] = f'$. By the sum and constant-multiple rules of calculus, $D[f + kg] = (f + kg)' = f' + kg' = D[f] + kD[g]$, so D is linear. For $f(x) = x^2$, $D[f](x) = 2x$.

Theorem 4.2.7 (Linear maps preserve combinations)
If $T : V \to W$ is linear, then for all $\mathbf{u}_1, \ldots, \mathbf{u}_r \in V$ and scalars $k_1, \ldots, k_r$,

$$T(k_1\mathbf{u}_1 + k_2\mathbf{u}_2 + \cdots + k_r\mathbf{u}_r) = k_1T(\mathbf{u}_1) + \cdots + k_rT(\mathbf{u}_r).$$

Proof.
By induction on r, applying the defining property repeatedly. The base $r = 1$ is homogeneity. For the step, $T\big(\sum_{i=1}^{r} k_i\mathbf{u}_i\big) = T\big(\sum_{i=1}^{r-1} k_i\mathbf{u}_i + k_r\mathbf{u}_r\big) = T\big(\sum_{i=1}^{r-1} k_i\mathbf{u}_i\big) + k_rT(\mathbf{u}_r)$ by the defining property, then apply the inductive hypothesis to the first term. ■

Remark 4.2.8. Theorem 4.2.7 has a powerful consequence: a linear transformation is *completely determined* by its values on a basis. If S is a basis of V and we know $T(\mathbf{v})$ for each $\mathbf{v} \in S$, then T is known everywhere, since every input is a combination of basis vectors.

Example 4.2.9 (Determining T from its values on a basis)

Let $S = \{\mathbf{v}_1, \mathbf{v}_2\}$ with $\mathbf{v}_1 = (-2, 1)$, $\mathbf{v}_2 = (1, 3)$ be a basis of $\mathbb{R}^2$, and let $T : \mathbb{R}^2 \to \mathbb{R}^3$ be linear with $T(\mathbf{v}_1) = (-1, 2, 0)$ and $T(\mathbf{v}_2) = (0, 3, 5)$. Find $T(-5, 1)$.

Solution. Express $(-5, 1)$ in the basis: solving $(-5, 1) = c_1(-2, 1) + c_2(1, 3)$ gives $c_1 = 2$, $c_2 = -1$. By linearity,

$$T(-5, 1) = 2T(\mathbf{v}_1) - T(\mathbf{v}_2) = 2(-1, 2, 0) - (0, 3, 5) = (-2, 1, -5).$$

Theorem 4.2.10 (Composition of linear maps)
If $T_1 : U \to V$ and $T_2 : V \to W$ are linear, then $T_2 \circ T_1 : U \to W$ is linear.

Proof.
For $\mathbf{u}, \mathbf{u}' \in U$ and scalar k:

$$\begin{aligned}(T_2 \circ T_1)(\mathbf{u} + k\mathbf{u}') &= T_2\big(T_1(\mathbf{u} + k\mathbf{u}')\big) = T_2\big(T_1(\mathbf{u}) + kT_1(\mathbf{u}')\big) && (T_1 \text{ linear})\\ &= T_2(T_1(\mathbf{u})) + kT_2(T_1(\mathbf{u}')) && (T_2 \text{ linear})\\ &= (T_2 \circ T_1)(\mathbf{u}) + k(T_2 \circ T_1)(\mathbf{u}'). && (\text{definition of } \circ)\end{aligned}$$

■

4.3 Kernel and Range

Definition 4.3.1 (Kernel and range)
For a linear transformation $T : V \to W$,

$$\ker(T) = \{\mathbf{x} \in V : T(\mathbf{x}) = \mathbf{0}\}, \qquad \operatorname{ran}(T) = \{\mathbf{y} \in W : \mathbf{y} = T(\mathbf{x}) \text{ for some } \mathbf{x} \in V\}.$$

Example 4.3.2 (For a matrix transformation)

If $T_A : \mathbb{R}^n \to \mathbb{R}^m$ is the matrix transformation of A, then $\ker(T_A) = \ker(A)$ (the solutions of $A\mathbf{x} = \mathbf{0}$) and $\operatorname{ran}(T_A) = \operatorname{col}(A)$ (by Theorem 1.3.10, the images $A\mathbf{x}$ are exactly the combinations of the columns).

Theorem 4.3.3
If $T : V \to W$ is linear, then (a) $\ker(T)$ is a subspace of V, and (b) $\operatorname{ran}(T)$ is a subspace of W.

Proof.

(a) We verify $\ker(T)$ passes the Subspace Test.

Nonempty: $T(\mathbf{0}) = \mathbf{0}$ (Lemma 4.2.2), so $\mathbf{0} \in \ker(T)$.

Closure under addition: If $\mathbf{u}, \mathbf{v} \in \ker(T)$ then $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v}) = \mathbf{0} + \mathbf{0} = \mathbf{0}$, so $\mathbf{u} + \mathbf{v} \in \ker(T)$.

Closure under scalar multiplication: If $\mathbf{u} \in \ker(T)$ and k is a scalar, then $T(k\mathbf{u}) = kT(\mathbf{u}) = k\mathbf{0} = \mathbf{0}$, so $k\mathbf{u} \in \ker(T)$.

Therefore $\ker(T)$ is a subspace of V.

(b) We verify $\operatorname{ran}(T)$ passes the Subspace Test.

Nonempty: $\mathbf{0} = T(\mathbf{0}) \in \operatorname{ran}(T)$.

Closure under addition: If $\mathbf{y}_1 = T(\mathbf{x}_1)$ and $\mathbf{y}_2 = T(\mathbf{x}_2)$ are in $\operatorname{ran}(T)$, then $\mathbf{y}_1 + \mathbf{y}_2 = T(\mathbf{x}_1) + T(\mathbf{x}_2) = T(\mathbf{x}_1 + \mathbf{x}_2) \in \operatorname{ran}(T)$.

Closure under scalar multiplication: If $\mathbf{y} = T(\mathbf{x}) \in \operatorname{ran}(T)$ and k is a scalar, then $k\mathbf{y} = kT(\mathbf{x}) = T(k\mathbf{x}) \in \operatorname{ran}(T)$.

Therefore $\operatorname{ran}(T)$ is a subspace of W. ■

Definition 4.3.4 (Rank and nullity)
If $\operatorname{ran}(T)$ is finite-dimensional, its dimension is the *rank* of T, $\operatorname{rank}(T)$. If $\ker(T)$ is finite-dimensional, its dimension is the *nullity* of T, $\operatorname{nullity}(T)$.

Theorem 4.3.5 (Rank–Nullity Theorem)
If $T : V \to W$ is linear and V is n-dimensional, then

$$\operatorname{rank}(T) + \operatorname{nullity}(T) = n.$$

Proof.
Let $k = \operatorname{nullity}(T)$.
Edge case ($k = n$): then $\ker(T) = V$ (a k-dimensional subspace of the n-dimensional V is all of V, Theorem 3.3.14(b)), so T is the zero map. Thus $\operatorname{ran}(T) = \{\mathbf{0}\}$, $\operatorname{rank}(T) = 0$, and $\operatorname{rank}(T)+\operatorname{nullity}(T) = 0 + n = n$.
Main case ($0 \le k < n$): choose a basis $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ of $\ker(T)$ and extend it to a basis $\{\mathbf{v}_1, \dots, \mathbf{v}_k, \mathbf{v}_{k+1}, \dots, \mathbf{v}_n\}$ of V (Theorem 3.3.15). We claim $\{T(\mathbf{v}_{k+1}), \dots, T(\mathbf{v}_n)\}$ is a basis of $\operatorname{ran}(T)$. Granting this, $\operatorname{rank}(T) = n - k$, so $\operatorname{rank}(T) + \operatorname{nullity}(T) = (n - k) + k = n$.
Spanning. Any $\mathbf{y} \in \operatorname{ran}(T)$ is $\mathbf{y} = T(\mathbf{x})$ for some $\mathbf{x} = \sum_{i=1}^{n} c_i\mathbf{v}_i$. Since $T(\mathbf{v}_i) = \mathbf{0}$ for $i \le k$,

$$\mathbf{y} = \sum_{i=1}^{n} c_i T(\mathbf{v}_i) = \sum_{i=k+1}^{n} c_i T(\mathbf{v}_i),$$

a combination of $T(\mathbf{v}_{k+1}), \dots, T(\mathbf{v}_n)$.
Independence. Suppose $\sum_{i=k+1}^{n} d_i T(\mathbf{v}_i) = \mathbf{0}$. By linearity $T\big(\sum_{i=k+1}^{n} d_i\mathbf{v}_i\big) = \mathbf{0}$, so $\sum_{i=k+1}^{n} d_i\mathbf{v}_i \in \ker(T)$ and is a combination $\sum_{i=1}^{k} e_i\mathbf{v}_i$ of the kernel basis. Then

$$\sum_{i=1}^{k} e_i\mathbf{v}_i - \sum_{i=k+1}^{n} d_i\mathbf{v}_i = \mathbf{0}$$

is a relation among the basis $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ of V, so all its coefficients vanish; in particular every $d_i = 0$. Hence the images are independent, and the claim holds. ■

Theorem 4.3.6 (Injectivity via the kernel)
A linear transformation $T : V \to W$ is one-to-one if and only if $\ker(T) = \{\mathbf{0}\}$.

Proof.
($\Rightarrow$) If T is injective and $\mathbf{x} \in \ker(T)$, then $T(\mathbf{x}) = \mathbf{0} = T(\mathbf{0})$ (Lemma 4.2.2), so $\mathbf{x} = \mathbf{0}$; thus $\ker(T) = \{\mathbf{0}\}$.
($\Leftarrow$) Suppose $\ker(T) = \{\mathbf{0}\}$ and $T(\mathbf{u}) = T(\mathbf{v})$. Then $T(\mathbf{u}-\mathbf{v}) = T(\mathbf{u}) - T(\mathbf{v}) = \mathbf{0}$, so $\mathbf{u} - \mathbf{v} \in \ker(T) = \{\mathbf{0}\}$, giving $\mathbf{u} = \mathbf{v}$. Hence T is injective. ■

Theorem 4.3.7 (For equal dimensions, one-to-one = onto)
Let V, W be n-dimensional and $T : V \to W$ linear. Then T is one-to-one if and only if T is onto.

Proof.
By Rank–Nullity, $\operatorname{rank}(T) = n - \operatorname{nullity}(T)$. Now T is onto $\iff$ $\operatorname{ran}(T) = W$ $\iff$ $\operatorname{rank}(T) = n$ (Theorem 3.3.14(b)) $\iff$ $\operatorname{nullity}(T) = 0$ $\iff$ $\ker(T) = \{\mathbf{0}\}$ $\iff$ T one-to-one (Theorem 4.3.6). ■

4.4 Isomorphism

Definition 4.4.1–4.4.2 (Isomorphism)
A linear transformation $T : V \to W$ that is both one-to-one and onto is an *isomorphism*. If an isomorphism $V \to W$ exists, V and W are *isomorphic*, written $V \cong W$.

Theorem 4.4.3 ($\cong$ is an equivalence relation)
For all vector spaces U, V, W: (i) $U \cong U$; (ii) if $U \cong V$ then $V \cong U$; (iii) if $U \cong V$ and $V \cong W$ then $U \cong W$.

Proof.

(i) The identity map id_U is a linear bijection.

(ii) If $T : U \to V$ is an isomorphism, its inverse $T^{-1} : V \to U$ is a bijection, and is linear: for $\mathbf{y}_1 = T(\mathbf{x}_1), \mathbf{y}_2 = T(\mathbf{x}_2)$, $T^{-1}(\mathbf{y}_1 + k\mathbf{y}_2) = T^{-1}(T(\mathbf{x}_1 + k\mathbf{x}_2)) = \mathbf{x}_1 + k\mathbf{x}_2 = T^{-1}(\mathbf{y}_1) + kT^{-1}(\mathbf{y}_2)$.

(iii) The composite of isomorphisms is an isomorphism: it is linear by Theorem 4.2.10 and a bijection as a composite of bijections. ■

Remark 4.4.4. "Isomorphic" is from the Greek *iso* (same) and *morphe* (form): isomorphic spaces have identical linear-algebraic structure and differ only in the names of their elements.

Example 4.4.5–4.4.6 (Coordinate isomorphisms)

The coordinate map $T : P_2 \to \mathbb{R}^3$, $T(a_0 + a_1x + a_2x^2) = (a_0, a_1, a_2)$, is linear, one-to-one, and onto, so $P_2 \cong \mathbb{R}^3$. Likewise $T : \mathbb{R}^{2\times2} \to \mathbb{R}^4$ sending a matrix to its list of entries is an isomorphism, so $\mathbb{R}^{2\times2} \cong \mathbb{R}^4$. In general, taking coordinates relative to a basis gives an isomorphism of any n-dimensional real space with $\mathbb{R}^n$.

Theorem 4.4.7 (Classification by dimension)
Let U, V be finite-dimensional. If $\dim(U) = \dim(V)$, then $U \cong V$.

Proof.
Let $\{\mathbf{u}_1, \ldots, \mathbf{u}_n\}$ be a basis of U and $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ a basis of V (equal sizes since the dimensions agree). Define $T : U \to V$ on the basis by $T(\mathbf{u}_i) = \mathbf{v}_i$ and extend linearly (Remark 4.2.8). Then $\mathrm{ran}(T) \supseteq \mathrm{span}\{\mathbf{v}_i\} = V$, so T is onto; by Theorem 4.3.7, T is also one-to-one. Hence T is an isomorphism and $U \cong V$. ■

Remark. Combined with $\dim(\mathbb{R}^n) = n$, Theorem 4.4.7 says *every* n-dimensional real vector space is isomorphic to $\mathbb{R}^n$. This is why studying $\mathbb{R}^n$ loses no generality: P_n, $\mathbb{R}^{m\times n}$, and any other finite-dimensional space is a relabelled copy of some $\mathbb{R}^k$.

4.5 Matrix Representations

Once bases are fixed for the domain and codomain, a linear transformation becomes a matrix acting on coordinate vectors — closing the loop back to Chapter 1.

Definition 4.5.1 (Matrix of a transformation)
Let $T : V \to W$ be linear, $B = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ a basis of V, and $B' = \{\mathbf{w}_1, \ldots, \mathbf{w}_m\}$ a basis of W. The *matrix for T relative to B and B'* is the $m \times n$ matrix

$$[T]_{B',B} = \big[\, [T(\mathbf{v}_1)]_{B'} \;\; [T(\mathbf{v}_2)]_{B'} \;\; \cdots \;\; [T(\mathbf{v}_n)]_{B'} \,\big],$$

whose columns are the coordinate vectors of the images of the domain basis.

Theorem 4.5.2 (The representation acts on coordinates)

With notation as above, for every $\mathbf{x} \in V$,

$$[T]_{B',B}\,[\mathbf{x}]_B = [T(\mathbf{x})]_{B'}.$$

Proof.
Write $[\mathbf{x}]_B = (c_1, \ldots, c_n)$, so $\mathbf{x} = \sum_j c_j \mathbf{v}_j$ and, by linearity, $T(\mathbf{x}) = \sum_j c_j T(\mathbf{v}_j)$. Taking B'-coordinates (which respect linear combinations) and using the column interpretation of the product (Theorem 1.3.10),

$$[T(\mathbf{x})]_{B'} = \sum_j c_j [T(\mathbf{v}_j)]_{B'} = \sum_j c_j \,(j\text{th column of } [T]_{B',B}) = [T]_{B',B}\,[\mathbf{x}]_B.$$

■

Algorithm 4.5.3 (Computing $T(\mathbf{x})$ via its matrix)
Given $T : V \to W$ with bases B, B', and $\mathbf{x} \in V$:

(i) compute the coordinate vector $[\mathbf{x}]_B$;

(ii) compute the matrix $[T]_{B',B}$ (columns = coordinates of the $T(\mathbf{v}_j)$);

(iii) form the product $[T]_{B',B}[\mathbf{x}]_B$, which equals $[T(\mathbf{x})]_{B'}$;

(iv) read $T(\mathbf{x})$ off from its coordinate vector $[T(\mathbf{x})]_{B'}$.

Example 4.5.4 (A representation between polynomial spaces)

Let $T : P_2 \to P_1$ be $T(p) = p'$ (differentiation), with bases $B = \{1, x, x^2\}$ for P_2 and $B' = \{1, x\}$ for P_1. Then $T(1) = 0$, $T(x) = 1$, $T(x^2) = 2x$, with B'-coordinates $(0, 0)$, $(1, 0)$, $(0, 2)$. Hence

$$[T]_{B',B} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}.$$

Check on $p(x) = 3 + 4x + 5x^2$: $[p]_B = (3, 4, 5)$, and $[T]_{B',B}[p]_B = (4, 10)$, i.e. $T(p) = 4 + 10x = p'$, as expected.

Chapter 4 Summary

- T is linear iff $T(\mathbf{u} + k\mathbf{v}) = T(\mathbf{u}) + kT(\mathbf{v})$ (Definition 4.2.1); necessarily $T(\mathbf{0}) = \mathbf{0}$ (Lemma 4.2.2), and T is determined by its values on a basis (Remark 4.2.8).
- Kernel and range are subspaces (Theorem 4.3.3); $\ker(T_A) = \ker(A)$ and $\operatorname{ran}(T_A) = \operatorname{col}(A)$.
- **Rank–Nullity:** $\operatorname{rank}(T) + \operatorname{nullity}(T) = \dim(V)$ (Theorem 4.3.5).
- T is one-to-one iff $\ker(T) = \{\mathbf{0}\}$ (Theorem 4.3.6); for equal finite dimensions, one-to-one $\iff$ onto (Theorem 4.3.7).
- An *isomorphism* is a linear bijection; $\cong$ is an equivalence relation (Theorem 4.4.3), and equal dimension implies isomorphic (Theorem 4.4.7). Every n-dimensional real space is isomorphic to $\mathbb{R}^n$.
- Relative to bases B, B', T is represented by $[T]_{B',B}$, and $[T]_{B',B}[\mathbf{x}]_B = [T(\mathbf{x})]_{B'}$ (Theorem 4.5.2).

Chapter 4 Exercises

4.1 Decide whether $T : \mathbb{R}^2 \to \mathbb{R}^2$, $T(x, y) = (x + 1, y)$, is linear. Justify.

4.2 Let $T : \mathbb{R}^3 \to \mathbb{R}^2$ be $T(x,y,z) = (x-y,\, y-z)$. Find bases for $\ker(T)$ and $\operatorname{ran}(T)$, and verify Rank–Nullity.

4.3 Let $T : P_2 \to P_2$ be $T(p) = p + p'$. Show T is an isomorphism (Hint: find $\ker(T)$, then use Theorem 4.3.7).

4.4 Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ rotate by $90°$ counterclockwise. Find $[T]_{B,B}$ relative to the standard basis $B = \{\mathbf{e}_1, \mathbf{e}_2\}$.

4.5 Prove that if $T : V \to W$ is linear and one-to-one, and $\{\mathbf{v}_1, \ldots, \mathbf{v}_r\}$ is independent in V, then $\{T(\mathbf{v}_1), \ldots, T(\mathbf{v}_r)\}$ is independent in W.

4.6 Let $T : P_1 \to P_1$ be $T(a+bx) = (2a+b) + (a+2b)x$, with $B = \{1, x\}$. Find $[T]_{B,B}$ and use it to compute $T(3-x)$ via Algorithm 4.5.3.

Solutions to Chapter 4 Exercises

4.1 Not linear: $T(0,0) = (1,0) \neq (0,0)$, violating Lemma 4.2.2.

4.2 $\ker(T)$: $x - y = 0$ and $y - z = 0$ give $x = y = z$, so $\ker(T) = \{t(1,1,1)\}$, basis $\{(1,1,1)\}$, nullity $= 1$. $\operatorname{ran}(T)$: the standard matrix is $\left[\begin{smallmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \end{smallmatrix}\right]$, whose columns span $\mathbb{R}^2$ (the first two are independent), so $\operatorname{ran}(T) = \mathbb{R}^2$, basis $\{(1,0),(0,1)\}$, rank $= 2$. Check: $2 + 1 = 3 = \dim(\mathbb{R}^3)$. ✓

4.3 $T(p) = p + p'$. If $T(p) = 0$ with $p = a + bx + cx^2$, then $p' = b + 2cx$ and $p + p' = (a+b) + (b+2c)x + cx^2 = 0$ forces $c = 0$, $b + 2c = 0 \Rightarrow b = 0$, $a + b = 0 \Rightarrow a = 0$. So $\ker(T) = \{0\}$, and T is one-to-one (Theorem 4.3.6); since $\dim P_2 = \dim P_2$, T is onto (Theorem 4.3.7), hence an isomorphism. ∎

4.4 Rotation by $90°$ sends $\mathbf{e}_1 = (1,0) \mapsto (0,1)$ and $\mathbf{e}_2 = (0,1) \mapsto (-1,0)$. The columns are these images: $[T]_{B,B} = \left[\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right]$.

4.5 Suppose $\sum_i c_i T(\mathbf{v}_i) = \mathbf{0}$. By linearity $T\big(\sum_i c_i \mathbf{v}_i\big) = \mathbf{0}$, so $\sum_i c_i \mathbf{v}_i \in \ker(T) = \{\mathbf{0}\}$ (using Theorem 4.3.6). Thus $\sum_i c_i \mathbf{v}_i = \mathbf{0}$, and independence of $\{\mathbf{v}_i\}$ forces all $c_i = 0$. Hence $\{T(\mathbf{v}_i)\}$ is independent. ∎

4.6 $T(1) = 2 + x$, $T(x) = 1 + 2x$, with B-coordinates $(2,1)$ and $(1,2)$, so $[T]_{B,B} = \left[\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right]$. For $3 - x$: $[3-x]_B = (3,-1)$, and $[T]_{B,B}(3,-1)^{\mathsf{T}} = (6-1,\, 3-2)^{\mathsf{T}} = (5,1)$, so $T(3-x) = 5 + x$. (Direct check: $T(3-x) = (2\cdot 3 + (-1)) + (3 + 2(-1))x = 5 + x$.) ✓

CHAPTER 5

INNER PRODUCT SPACES

What this chapter is about. Vector spaces have no built-in notion of length or angle. An *inner product* supplies one: a way to multiply two vectors into a scalar, generalising the dot product. From it we recover lengths, distances, the Cauchy–Schwarz inequality, angles, and above all *orthogonality*. Orthonormal bases make computation trivial, and the *Gram–Schmidt* process manufactures them from any basis. We then prove the *Fundamental Theorem of Linear Algebra*, relating the four subspaces of a matrix, and apply orthogonal projection to solve the *least-squares problem* — the best approximate solution of an inconsistent system.

Reference. A&R §§6.1–6.4 and §7.7.

5.1 Inner Products

Definition 5.1.1 (Inner product)
An *inner product* on a real vector space V is a function $\langle \cdot, \cdot \rangle$ assigning a real number $\langle \mathbf{u}, \mathbf{v} \rangle$ to each pair of vectors, such that for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$ and all $k \in \mathbb{R}$:

1. *Symmetry:* $\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle$;
2. *Linearity:* $\langle \mathbf{u}, \mathbf{v} + k\mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{v} \rangle + k\langle \mathbf{u}, \mathbf{w} \rangle$;
3. *Positive definiteness:* $\langle \mathbf{v}, \mathbf{v} \rangle \geq 0$, with $\langle \mathbf{v}, \mathbf{v} \rangle = 0$ if and only if $\mathbf{v} = \mathbf{0}$.

Definition 5.1.2 (Inner product space)
A real vector space with an inner product is a *real inner product space*.

Linearity is stated in the second slot; symmetry transfers it to the first.

Lemma 5.1.3 (Linearity in the first argument)
$\langle \mathbf{u} + k\mathbf{w}, \mathbf{v} \rangle = \langle \mathbf{u}, \mathbf{v} \rangle + k\langle \mathbf{w}, \mathbf{v} \rangle$.

Proof.

$$\begin{aligned}
\langle \mathbf{u} + k\mathbf{w}, \mathbf{v} \rangle &= \langle \mathbf{v}, \mathbf{u} + k\mathbf{w} \rangle && \text{(symmetry)} \\
&= \langle \mathbf{v}, \mathbf{u} \rangle + k\langle \mathbf{v}, \mathbf{w} \rangle && \text{(linearity in slot 2)} \\
&= \langle \mathbf{u}, \mathbf{v} \rangle + k\langle \mathbf{w}, \mathbf{v} \rangle. && \text{(symmetry, twice)}
\end{aligned}$$

∎

Example 5.1.4 (Euclidean inner product)

The dot product $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u} \cdot \mathbf{v} = \sum_{i=1}^{n} u_i v_i$ on $\mathbb{R}^n$ satisfies all three axioms. With it, $\mathbb{R}^n$ is *Euclidean n-space.*

Example 5.1.5 (Weighted inner product)

For positive reals $w_1, \dots, w_n$, the formula $\langle \mathbf{u}, \mathbf{v} \rangle = \sum_{i=1}^{n} w_i u_i v_i$ is an inner product on $\mathbb{R}^n$. Positive definiteness uses $w_i > 0$: $\langle \mathbf{v}, \mathbf{v} \rangle = \sum w_i v_i^2 \geq 0$, vanishing only when every $v_i = 0$.

Example 5.1.6 (Matrix inner product)

For an invertible $n \times n$ matrix A, the formula $\langle \mathbf{u}, \mathbf{v} \rangle = (A\mathbf{u}) \cdot (A\mathbf{v})$ is an inner product on $\mathbb{R}^n$. Positive definiteness holds because $\langle \mathbf{v}, \mathbf{v} \rangle = \|A\mathbf{v}\|^2 \geq 0$, and equals 0 only if $A\mathbf{v} = \mathbf{0}$, i.e. $\mathbf{v} = \mathbf{0}$ since A is invertible (Theorem 1.5.7).

Remark 5.1.7. The Euclidean inner product is the case $A = I$ of the matrix inner product; the weighted inner product is the case where $A = \operatorname{diag}(\sqrt{w_1}, \dots, \sqrt{w_n})$.

Example 5.1.8 (Function-space inner product)

On $C[a, b]$, the formula

$$\langle f, g \rangle = \int_a^b f(x) g(x)\, dx$$

is an inner product. Symmetry and linearity are clear; positive definiteness, $\langle f, f \rangle = \int_a^b f(x)^2\, dx \geq 0$ with equality iff $f \equiv 0$, uses continuity of f.

Remark 5.1.9. The integral inner product is the continuous analogue of the dot product: summation over coordinates becomes integration over the domain.

Definition 5.1.10 (Norm and distance)

In a real inner product space, the *length* (or *norm*) of $\mathbf{v}$ is $\|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}$, and the *distance* between $\mathbf{u}, \mathbf{v}$ is

$$\operatorname{dist}(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\| = \sqrt{\langle \mathbf{u} - \mathbf{v}, \mathbf{u} - \mathbf{v} \rangle}.$$

Theorem 5.1.11 (Properties of norm and distance)

For $\mathbf{u}, \mathbf{v}, \mathbf{w}$ in a real inner product space and $k \in \mathbb{R}$:

(a) $\|\mathbf{v}\| \geq 0$, with $\|\mathbf{v}\| = 0 \iff \mathbf{v} = \mathbf{0}$;

(b) $\|k\mathbf{v}\| = |k|\, \|\mathbf{v}\|$;

(c) $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$ (triangle inequality);

(d) $\operatorname{dist}(\mathbf{u}, \mathbf{v}) = \operatorname{dist}(\mathbf{v}, \mathbf{u})$;

(e) $\operatorname{dist}(\mathbf{u}, \mathbf{v}) \geq 0$, with equality $\iff \mathbf{u} = \mathbf{v}$;

(f) $\operatorname{dist}(\mathbf{u}, \mathbf{v}) \leq \operatorname{dist}(\mathbf{u}, \mathbf{w}) + \operatorname{dist}(\mathbf{w}, \mathbf{v})$.

Proof.

(a) is positive definiteness restated.

(b) $\|k\mathbf{v}\| = \sqrt{\langle k\mathbf{v}, k\mathbf{v} \rangle} = \sqrt{k^2 \langle \mathbf{v}, \mathbf{v} \rangle} = |k| \|\mathbf{v}\|$.

(c) is proved below using Cauchy–Schwarz (Theorem 5.2.1). Parts (d)–(f) follow from (a)–(c) applied to difference vectors: (d) from $\|-\mathbf{x}\| = \|\mathbf{x}\|$, (e) from (a), and (f) (the triangle inequality for distance) from (c) with $\mathbf{u} - \mathbf{v} = (\mathbf{u} - \mathbf{w}) + (\mathbf{w} - \mathbf{v})$. ∎

Example 5.1.12 (A non-standard inner product on $\mathbb{R}^2$)

Suppose $\mathbb{R}^2$ carries $\langle \mathbf{u}, \mathbf{v} \rangle = u_1 v_1 + 3u_2 v_2$ (a weighted inner product). Then $\|(1,1)\| = \sqrt{1+3} = 2$, not $\sqrt{2}$ — length depends on the inner product, not just the coordinates.

Example 5.1.13 (Norms of $\sin$ and $\cos$)

On $C[0, 2\pi]$ with $\langle f, g \rangle = \int_0^{2\pi} fg\,dx$,

$$\|\sin x\|^2 = \int_0^{2\pi} \sin^2 x\,dx = \pi, \qquad \|\cos x\|^2 = \int_0^{2\pi} \cos^2 x\,dx = \pi,$$

so $\|\sin x\| = \|\cos x\| = \sqrt{\pi}$.

5.2 Angle and Orthogonality

Theorem 5.2.1 (Cauchy–Schwarz inequality)
In a real inner product space, $|\langle \mathbf{u}, \mathbf{v} \rangle| \le \|\mathbf{u}\|\,\|\mathbf{v}\|$ for all $\mathbf{u}, \mathbf{v}$.

Proof.

Case $\mathbf{v} = \mathbf{0}$ Then $\langle \mathbf{u}, \mathbf{0} \rangle = 0$ by linearity, and $\|\mathbf{v}\| = 0$, so both sides are zero and the inequality holds with equality.

Case $\mathbf{u}, \mathbf{v} \ne \mathbf{0}$ By positive definiteness, $\langle \mathbf{u} + \beta\mathbf{v}, \mathbf{u} + \beta\mathbf{v} \rangle \ge 0$ for every $\beta \in \mathbb{R}$. Expanding this with linearity and symmetry,

$$\begin{aligned}
&\langle \mathbf{u} + \beta\mathbf{v}, \mathbf{u} \rangle + \beta\langle \mathbf{u} + \beta\mathbf{v}, \mathbf{v} \rangle \ge 0 && \text{(linearity in 2nd argument)}\\
&\langle \mathbf{u}, \mathbf{u} \rangle + \beta\langle \mathbf{v}, \mathbf{u} \rangle + \beta\langle \mathbf{u}, \mathbf{v} \rangle + \beta^2\langle \mathbf{v}, \mathbf{v} \rangle \ge 0 && \text{(linearity in 1st argument)}\\
&\langle \mathbf{u}, \mathbf{u} \rangle + 2\beta\langle \mathbf{u}, \mathbf{v} \rangle + \beta^2\langle \mathbf{v}, \mathbf{v} \rangle \ge 0 && \text{(symmetry)}
\end{aligned}$$

Rearranging gives, for every $\beta \in \mathbb{R}$,

$$-2\beta\langle \mathbf{u}, \mathbf{v} \rangle \le \langle \mathbf{u}, \mathbf{u} \rangle + \beta^2\langle \mathbf{v}, \mathbf{v} \rangle. \tag{$*$}$$

The right-hand side is nonnegative, so $(*)$ is only informative for one sign of β. We pick that sign in each case so that the left-hand side becomes $2\lambda|\langle \mathbf{u}, \mathbf{v} \rangle|$ for some $\lambda > 0$.

- If $\langle \mathbf{u}, \mathbf{v} \rangle \ge 0$, set $\beta = -\lambda$ with $\lambda > 0$. Then $-2\beta\langle \mathbf{u}, \mathbf{v} \rangle = 2\lambda\langle \mathbf{u}, \mathbf{v} \rangle = 2\lambda|\langle \mathbf{u}, \mathbf{v} \rangle|$, and $(*)$ becomes $2\lambda|\langle \mathbf{u}, \mathbf{v} \rangle| \le \langle \mathbf{u}, \mathbf{u} \rangle + \lambda^2\langle \mathbf{v}, \mathbf{v} \rangle$.
- If $\langle \mathbf{u}, \mathbf{v} \rangle < 0$, set $\beta = \lambda$ with $\lambda > 0$. Then $-2\beta\langle \mathbf{u}, \mathbf{v} \rangle = -2\lambda\langle \mathbf{u}, \mathbf{v} \rangle = 2\lambda|\langle \mathbf{u}, \mathbf{v} \rangle|$, giving the same inequality.

In either case, dividing by $\lambda > 0$ and writing $\langle \mathbf{u}, \mathbf{u} \rangle = \|\mathbf{u}\|^2$, $\langle \mathbf{v}, \mathbf{v} \rangle = \|\mathbf{v}\|^2$,

$$2|\langle \mathbf{u}, \mathbf{v} \rangle| \le \tfrac{1}{\lambda}\|\mathbf{u}\|^2 + \lambda\|\mathbf{v}\|^2 \qquad \text{for all } \lambda > 0.$$

This holds for every $\lambda > 0$, so we are free to choose the value that makes the bound tightest. Take $\lambda = \|\mathbf{u}\|/\|\mathbf{v}\|$ (valid since $\mathbf{v} \ne \mathbf{0}$):

$$\tfrac{1}{\lambda}\|\mathbf{u}\|^2 + \lambda\|\mathbf{v}\|^2 = \tfrac{\|\mathbf{v}\|}{\|\mathbf{u}\|}\|\mathbf{u}\|^2 + \tfrac{\|\mathbf{u}\|}{\|\mathbf{v}\|}\|\mathbf{v}\|^2 = \|\mathbf{u}\|\|\mathbf{v}\| + \|\mathbf{u}\|\|\mathbf{v}\| = 2\|\mathbf{u}\|\|\mathbf{v}\|.$$

Therefore $2|\langle \mathbf{u}, \mathbf{v} \rangle| \le 2\|\mathbf{u}\|\|\mathbf{v}\|$, that is, $|\langle \mathbf{u}, \mathbf{v} \rangle| \le \|\mathbf{u}\|\|\mathbf{v}\|$. ∎

Proof of the triangle inequality (Theorem 5.1.11(c)).
Using Cauchy–Schwarz, $\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + 2\langle \mathbf{u}, \mathbf{v} \rangle + \|\mathbf{v}\|^2 \le \|\mathbf{u}\|^2 + 2\|\mathbf{u}\|\|\mathbf{v}\| + \|\mathbf{v}\|^2 = (\|\mathbf{u}\| + \|\mathbf{v}\|)^2$;

take square roots. ■

Definition 5.2.2 (Angle)

The *angle* between nonzero $\mathbf{u}, \mathbf{v}$ is $\theta = \arccos\left(\frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\| \|\mathbf{v}\|}\right)$. Cauchy–Schwarz guarantees the argument lies in $[-1, 1]$, so θ is well-defined.

Example 5.2.3 (Angle between sin and cos)

On $C[0, 2\pi]$, $\langle \sin x, \cos x \rangle = \int_0^{2\pi} \sin x \cos x \, dx = 0$, so $\theta = \arccos 0 = \frac{\pi}{2}$: sin and cos are perpendicular as functions.

Definition 5.2.4 (Orthogonality)

Vectors $\mathbf{u}, \mathbf{v}$ are *orthogonal* if $\langle \mathbf{u}, \mathbf{v} \rangle = 0$. (Example 5.2.3 shows sin and cos are orthogonal in $C[0, 2\pi]$.)

Example 5.2.5

For $\mathbf{u} = (1, 1)$ and $\mathbf{v} = (1, -1)$ in Euclidean $\mathbb{R}^2$, $\langle \mathbf{u}, \mathbf{v} \rangle = 1 - 1 = 0$, so they are orthogonal.

Theorem 5.2.6 (Generalised Pythagorean theorem)

If $\mathbf{u}, \mathbf{v}$ are orthogonal in a real inner product space, then $\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$.

Proof.

$\|\mathbf{u} + \mathbf{v}\|^2 = \langle \mathbf{u} + \mathbf{v}, \mathbf{u} + \mathbf{v} \rangle = \|\mathbf{u}\|^2 + 2\langle \mathbf{u}, \mathbf{v} \rangle + \|\mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$, since $\langle \mathbf{u}, \mathbf{v} \rangle = 0$. ■

5.3 Orthogonal Complements

Definition 5.3.1 (Orthogonal subspaces)

Subspaces U, W of an inner product space V are *orthogonal*, written $U \perp W$, if every vector of U is orthogonal to every vector of W.

Lemma 5.3.2

If $U \perp W$, then $U \cap W = \{\mathbf{0}\}$.

Proof.

If $\mathbf{x} \in U \cap W$, then $\mathbf{x}$ is orthogonal to itself: $\langle \mathbf{x}, \mathbf{x} \rangle = 0$. By positive definiteness, $\mathbf{x} = \mathbf{0}$. ■

Corollary 5.3.3

If U, W are orthogonal subspaces of V, then $U + W = U \oplus W$.

Proof.

By Lemma 5.3.2, $U \cap W = \{\mathbf{0}\}$, which is exactly the condition for the sum to be direct (Definition 3.4.1). ■

Definition 5.3.4 (Orthogonal complement)
The *orthogonal complement* of a subspace $U \subseteq V$ is

$$U^\perp = \{\mathbf{v} \in V : \langle \mathbf{v}, \mathbf{u} \rangle = 0 \text{ for all } \mathbf{u} \in U\}.$$

Lemma 5.3.5
$U^\perp$ is a subspace of V.

Proof.
We verify $U^\perp$ passes the Subspace Test.

Nonempty: $\langle \mathbf{0}, \mathbf{u} \rangle = 0$ for all $\mathbf{u} \in U$, so $\mathbf{0} \in U^\perp$.

Closure under addition: If $\mathbf{x}, \mathbf{y} \in U^\perp$ then for every $\mathbf{u} \in U$, $\langle \mathbf{x}+\mathbf{y}, \mathbf{u} \rangle = \langle \mathbf{x}, \mathbf{u} \rangle + \langle \mathbf{y}, \mathbf{u} \rangle = 0+0 = 0$ (Lemma 5.1.3), so $\mathbf{x} + \mathbf{y} \in U^\perp$.

Closure under scalar multiplication: If $\mathbf{x} \in U^\perp$ and k is a scalar, then for every $\mathbf{u} \in U$, $\langle k\mathbf{x}, \mathbf{u} \rangle = k\langle \mathbf{x}, \mathbf{u} \rangle = k \cdot 0 = 0$, so $k\mathbf{x} \in U^\perp$.

Therefore $U^\perp$ is a subspace. ∎

5.4 Gram–Schmidt and Orthonormal Bases

Definition 5.4.1 (Orthogonal and orthonormal sets)
A set of two or more vectors is *orthogonal* if every pair is orthogonal, and *orthonormal* if in addition every vector has length 1. For nonzero $\mathbf{v}$, the vector $\mathbf{v}/\|\mathbf{v}\|$ is a *unit vector*; replacing $\mathbf{v}$ by $\mathbf{v}/\|\mathbf{v}\|$ is *normalisation*.

Example 5.4.2

$S = \{(1, 1, 1), (0, 1, -1), (2, -1, -1)\}$ is orthogonal in Euclidean $\mathbb{R}^3$: each pairwise dot product is zero (e.g. $(1, 1, 1) \cdot (0, 1, -1) = 0$). Normalising each vector turns S into an orthonormal set.

Theorem 5.4.3 (Orthogonal sets are independent)
If $S = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ is an orthogonal set of *nonzero* vectors, then S is linearly independent.

Proof.
Suppose $\sum_i c_i\mathbf{v}_i = \mathbf{0}$. Take the inner product with $\mathbf{v}_j$; by orthogonality all terms vanish except $i = j$:

$$\begin{aligned} 0 &= \langle \textstyle\sum_i c_i\mathbf{v}_i, \mathbf{v}_j \rangle = \sum_i c_i\langle \mathbf{v}_i, \mathbf{v}_j \rangle && \text{(linearity)} \\ &= c_j\langle \mathbf{v}_j, \mathbf{v}_j \rangle = c_j\|\mathbf{v}_j\|^2. && \text{(orthogonality kills } i \neq j) \end{aligned}$$

Since $\mathbf{v}_j \neq \mathbf{0}$, $\|\mathbf{v}_j\|^2 > 0$, so $c_j = 0$. As j was arbitrary, every $c_j = 0$ and S is independent. ∎

Corollary 5.4.4
An orthonormal set is linearly independent.

Proof.
Orthonormal vectors are nonzero (they have length 1) and orthogonal, so Theorem 5.4.3 applies. ∎

Theorem 5.4.5 (Inner products via orthonormal coordinates)
If B is an orthonormal basis for an n-dimensional inner product space V, then $\langle \mathbf{u}, \mathbf{v} \rangle = [\mathbf{u}]_B \cdot [\mathbf{v}]_B$ for all $\mathbf{u}, \mathbf{v} \in V$.

Proof.
Write $\mathbf{u} = \sum_i a_i \mathbf{v}_i$, $\mathbf{v} = \sum_j b_j \mathbf{v}_j$ with $B = \{\mathbf{v}_i\}$ orthonormal. Then by bilinearity and $\langle \mathbf{v}_i, \mathbf{v}_j \rangle = \delta_{ij}$,

$$\langle \mathbf{u}, \mathbf{v} \rangle = \sum_{i,j} a_i b_j \langle \mathbf{v}_i, \mathbf{v}_j \rangle = \sum_i a_i b_i = [\mathbf{u}]_B \cdot [\mathbf{v}]_B.$$

■

Theorem 5.4.6 (Fourier expansion)
If $B = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ is an orthonormal basis for V, then every $\mathbf{u} \in V$ satisfies

$$\mathbf{u} = \langle \mathbf{u}, \mathbf{v}_1 \rangle \mathbf{v}_1 + \langle \mathbf{u}, \mathbf{v}_2 \rangle \mathbf{v}_2 + \cdots + \langle \mathbf{u}, \mathbf{v}_n \rangle \mathbf{v}_n.$$

Proof.
Since B is a basis, $\mathbf{u} = \sum_i c_i \mathbf{v}_i$ for unique scalars c_i. Taking the inner product with $\mathbf{v}_j$ and using orthonormality, $\langle \mathbf{u}, \mathbf{v}_j \rangle = \sum_i c_i \langle \mathbf{v}_i, \mathbf{v}_j \rangle = c_j$. So $c_j = \langle \mathbf{u}, \mathbf{v}_j \rangle$, as claimed. ■

The coefficients $\langle \mathbf{u}, \mathbf{v}_j \rangle$ are the orthogonal projections of $\mathbf{u}$ onto the basis directions, which leads to projection onto a subspace.

Theorem 5.4.7 (Projection Theorem)
Let W be a finite-dimensional subspace of an inner product space V. Then every $\mathbf{u} \in V$ can be written uniquely as

$$\mathbf{u} = \mathbf{w} + \mathbf{w}^\perp, \qquad \mathbf{w} \in W,\ \mathbf{w}^\perp \in W^\perp.$$

Proof.
Existence. Take an orthonormal basis $\{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ of W (one exists by Theorem 5.4.10), and set

$$\mathbf{w} = \sum_{j=1}^{k} \langle \mathbf{u}, \mathbf{v}_j \rangle \mathbf{v}_j \in W, \qquad \mathbf{w}^\perp = \mathbf{u} - \mathbf{w}.$$

Then $\mathbf{u} = \mathbf{w} + \mathbf{w}^\perp$ with $\mathbf{w} \in W$. To see $\mathbf{w}^\perp \in W^\perp$, it suffices to check $\mathbf{w}^\perp$ is orthogonal to each basis vector $\mathbf{v}_i$:

$$\langle \mathbf{w}^\perp, \mathbf{v}_i \rangle = \langle \mathbf{u}, \mathbf{v}_i \rangle - \sum_{j=1}^{k} \langle \mathbf{u}, \mathbf{v}_j \rangle \langle \mathbf{v}_j, \mathbf{v}_i \rangle = \langle \mathbf{u}, \mathbf{v}_i \rangle - \langle \mathbf{u}, \mathbf{v}_i \rangle = 0,$$

where the middle sum collapses because $\langle \mathbf{v}_j, \mathbf{v}_i \rangle = \delta_{ij}$ (orthonormality).
Uniqueness. Suppose $\mathbf{u} = \mathbf{w}_1 + \mathbf{w}_1^\perp = \mathbf{w}_2 + \mathbf{w}_2^\perp$ are two such decompositions. Then

$$\mathbf{w}_1 - \mathbf{w}_2 = \mathbf{w}_2^\perp - \mathbf{w}_1^\perp \in W \cap W^\perp = \{\mathbf{0}\}$$

by Lemma 5.3.2, so $\mathbf{w}_1 = \mathbf{w}_2$ and $\mathbf{w}_1^\perp = \mathbf{w}_2^\perp$. ■

Definition 5.4.8 (Orthogonal projection)
The vector $\mathbf{w}$ in Theorem 5.4.7 is the *orthogonal projection* of $\mathbf{u}$ on W, written $\mathbf{w} = \operatorname{proj}_W \mathbf{u}$.

Remark 5.4.9. If $\{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ is merely orthogonal (not normalised), then

$$\operatorname{proj}_W \mathbf{u} = \sum_{j=1}^{k} \frac{\langle \mathbf{u}, \mathbf{v}_j \rangle}{\|\mathbf{v}_j\|^2} \mathbf{v}_j,$$

each term dividing by $\|\mathbf{v}_j\|^2$ to compensate for unnormalised lengths.

Theorem 5.4.10 (Existence of orthonormal bases)
Every nonzero finite-dimensional inner product space has an orthonormal basis.

Proof.
Let $\{\mathbf{u}_1, \ldots, \mathbf{u}_n\}$ be any basis of the space. We build an orthogonal basis $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ step by step, then normalise.
Construction. Set $\mathbf{v}_1 = \mathbf{u}_1$. Having built $\mathbf{v}_1, \ldots, \mathbf{v}_{k-1}$, let $W_{k-1} = \operatorname{span}\{\mathbf{v}_1, \ldots, \mathbf{v}_{k-1}\}$ and define

$$\mathbf{v}_k = \mathbf{u}_k - \operatorname{proj}_{W_{k-1}} \mathbf{u}_k = \mathbf{u}_k - \sum_{j=1}^{k-1} \frac{\langle \mathbf{u}_k, \mathbf{v}_j \rangle}{\langle \mathbf{v}_j, \mathbf{v}_j \rangle} \mathbf{v}_j.$$

By the Projection Theorem (Theorem 5.4.7), $\mathbf{v}_k = \mathbf{u}_k - \operatorname{proj}_{W_{k-1}} \mathbf{u}_k \in W_{k-1}^{\perp}$, so $\mathbf{v}_k$ is orthogonal to every earlier $\mathbf{v}_j$.
Each $\mathbf{v}_k$ is nonzero. Expanding the projection and substituting the earlier $\mathbf{v}_j$ (each itself a combination of $\mathbf{u}_1, \ldots, \mathbf{u}_j$), we may write $\mathbf{v}_k$ as a linear combination of $\mathbf{u}_1, \ldots, \mathbf{u}_k$ in which the coefficient of $\mathbf{u}_k$ is 1. This is a *nontrivial* combination of the basis vectors, so by linear independence of $\{\mathbf{u}_1, \ldots, \mathbf{u}_n\}$ it cannot equal $\mathbf{0}$.
Conclusion. After n steps we have an orthogonal set $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ of nonzero vectors, hence linearly independent (Theorem 5.4.3) and therefore a basis. Normalising each, $\mathbf{w}_j = \mathbf{v}_j / \|\mathbf{v}_j\|$, yields an orthonormal basis $\{\mathbf{w}_1, \ldots, \mathbf{w}_n\}$. ∎

Algorithm 5.4.11 (Gram–Schmidt)
To convert a basis $\{\mathbf{u}_1, \ldots, \mathbf{u}_n\}$ into an orthogonal basis $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ spanning the same space, set

$$\begin{aligned}
\mathbf{v}_1 &= \mathbf{u}_1, \\
\mathbf{v}_2 &= \mathbf{u}_2 - \frac{\langle \mathbf{u}_2, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|^2} \mathbf{v}_1, \\
\mathbf{v}_3 &= \mathbf{u}_3 - \frac{\langle \mathbf{u}_3, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|^2} \mathbf{v}_1 - \frac{\langle \mathbf{u}_3, \mathbf{v}_2 \rangle}{\|\mathbf{v}_2\|^2} \mathbf{v}_2, \\
&\ \vdots \\
\mathbf{v}_k &= \mathbf{u}_k - \sum_{j=1}^{k-1} \frac{\langle \mathbf{u}_k, \mathbf{v}_j \rangle}{\|\mathbf{v}_j\|^2} \mathbf{v}_j.
\end{aligned}$$

At each step, $\mathbf{v}_k = \mathbf{u}_k - \operatorname{proj}_{W_{k-1}} \mathbf{u}_k$ where $W_{k-1} = \operatorname{span}\{\mathbf{v}_1, \ldots, \mathbf{v}_{k-1}\}$. Normalise ($\mathbf{e}_j = \mathbf{v}_j / \|\mathbf{v}_j\|$) to obtain an orthonormal basis.

Example 5.4.12 (Gram–Schmidt in $\mathbb{R}^2$)

Apply Gram–Schmidt to $\mathbf{u}_1 = (1, 1)$, $\mathbf{u}_2 = (0, 1)$ in Euclidean $\mathbb{R}^2$. First $\mathbf{v}_1 = (1, 1)$. Then

$$\mathbf{v}_2 = (0, 1) - \frac{\langle (0, 1), (1, 1) \rangle}{\|(1, 1)\|^2} (1, 1) = (0, 1) - \frac{1}{2}(1, 1) = \left(-\tfrac{1}{2}, \tfrac{1}{2}\right).$$

Check: $\mathbf{v}_1 \cdot \mathbf{v}_2 = -\frac{1}{2} + \frac{1}{2} = 0$. Normalising, $\mathbf{e}_1 = \frac{1}{\sqrt{2}}(1,1)$ and $\mathbf{e}_2 = \frac{1}{\sqrt{2}}(-1,1)$.

Theorem 5.4.13
If V is a finite-dimensional inner product space and W is a subspace, then $V = W \oplus W^\perp$, and $\dim V = \dim W + \dim W^\perp$.

Proof.
By the Projection Theorem (Theorem 5.4.7), every $\mathbf{u} \in V$ decomposes uniquely as $\mathbf{w} + \mathbf{w}^\perp$ with $\mathbf{w} \in W$, $\mathbf{w}^\perp \in W^\perp$, so $V = W + W^\perp$ with the sum direct (Corollary 5.3.3). The dimension formula is Theorem 3.4.4. ■

5.5 The Fundamental Theorem of Linear Algebra

The four subspaces of a matrix are not independent of each other: each is the orthogonal complement of another, and their dimensions are governed by a single number, the rank.

Theorem 5.5.1 (Row space and kernel are complementary)
If A is an $m \times n$ matrix, then $\operatorname{row}(A)^\perp = \ker(A)$ in $\mathbb{R}^n$.

Proof.
Let $\mathbf{r}_1, \ldots, \mathbf{r}_m$ be the rows of A. Then $\mathbf{x} \in \ker(A) \iff A\mathbf{x} = \mathbf{0} \iff \mathbf{r}_i \cdot \mathbf{x} = 0$ for every i. By linearity of the dot product, $\mathbf{r}_i \cdot \mathbf{x} = 0$ for all i if and only if $\mathbf{w} \cdot \mathbf{x} = 0$ for every $\mathbf{w} \in \operatorname{span}\{\mathbf{r}_i\} = \operatorname{row}(A)$. That last condition is exactly $\mathbf{x} \in \operatorname{row}(A)^\perp$. Hence $\ker(A) = \operatorname{row}(A)^\perp$. ■

Corollary 5.5.2 (Orthogonal decompositions)
For any $m \times n$ matrix A: (a) $\mathbb{R}^n = \operatorname{row}(A) \oplus \ker(A)$; (b) $\mathbb{R}^m = \operatorname{col}(A) \oplus \ker(A^\mathsf{T})$.

Proof.
(a) By Theorem 5.5.1, $\ker(A) = \operatorname{row}(A)^\perp$, and Theorem 5.4.13 gives $\mathbb{R}^n = \operatorname{row}(A) \oplus \operatorname{row}(A)^\perp$.
(b) Apply (a) to A^T: since $\operatorname{row}(A^\mathsf{T}) = \operatorname{col}(A)$ and $\ker(A^\mathsf{T}) = \operatorname{coker}(A)$, we get $\mathbb{R}^m = \operatorname{col}(A) \oplus \ker(A^\mathsf{T})$. ■

Theorem 5.5.6 (Fundamental Theorem of Linear Algebra)
Let A be an $m \times n$ matrix with r pivots. Then

(i) $\dim(\operatorname{col}(A)) = r = \operatorname{rank}(A)$;
(ii) $\dim(\operatorname{row}(A)) = r = \operatorname{rank}(A)$;
(iii) $\dim(\ker(A)) = n - r = \operatorname{nullity}(A)$;
(iv) $\dim(\operatorname{coker}(A)) = m - r = \operatorname{nullity}(A^\mathsf{T})$.

Proof.
(ii) is Theorem 3.6.6: the pivot count equals $\dim \operatorname{row}(A)$. (iii): $\ker(A)$ is the solution space of $A\mathbf{x} = \mathbf{0}$, with one basis vector per free variable; there are $n - r$ free variables, so $\dim \ker(A) = n - r$, which by Corollary 5.5.2(a) and Theorem 3.4.4 equals $n - \dim \operatorname{row}(A) = n - r$ consistently. (i): from $\mathbb{R}^n = \operatorname{row}(A) \oplus \ker(A)$ (Corollary 5.5.2(a)), $\dim \operatorname{row}(A) = n - (n - r) = r$; and applying the Rank–Nullity Theorem to T_A (whose range is $\operatorname{col}(A)$, Example 4.3.2) gives $\dim \operatorname{col}(A) = n - \dim \ker(A) = r$. So row rank $=$ column rank $= r$. (iv): apply (iii) to A^T, an $n \times m$ matrix with the same pivot count r, giving $\dim \ker(A^\mathsf{T}) = m - r$. ■

Remark. The equality of (i) and (ii) — row rank equals column rank — is remarkable: the number of independent rows of any matrix equals the number of independent columns, though rows and columns live in different spaces.

For a square matrix, the Fundamental Theorem adds many further equivalent conditions to the invertibility list.

Theorem 5.5.7 (Equivalent statements (the master list))
For an $n \times n$ matrix A, the following are equivalent:

(a) A is invertible;

(b) $A\mathbf{x} = \mathbf{0}$ has only the trivial solution;

(c) $\mathrm{RREF}(A) = I_n$;

(d) A is a product of elementary matrices;

(e) $A\mathbf{x} = \mathbf{b}$ is consistent for every $\mathbf{b}$;

(f) $A\mathbf{x} = \mathbf{b}$ has a unique solution for every $\mathbf{b}$;

(g) $\det(A) \neq 0$;

(h) the columns of A are linearly independent;

(i) the rows of A are linearly independent;

(j) the columns of A span $\mathbb{R}^n$;

(k) the rows of A span $\mathbb{R}^n$;

(l) the columns of A form a basis of $\mathbb{R}^n$;

(m) the rows of A form a basis of $\mathbb{R}^n$;

(n) $\mathrm{rank}(A) = n$;

(o) $\mathrm{nullity}(A) = 0$;

(p) $\mathrm{ran}(T_A) = \mathbb{R}^n$.

Proof.
Statements (a)–(g) are exactly Theorem 2.2.12, so they are equivalent to one another. It remains to fold in (h)–(p). We show each is equivalent to "A has rank n" (condition (n)), which is equivalent to (c) — the reduced echelon form being I_n, i.e. having n pivots — by Theorem 5.5.6.

(n) $\iff$ (o) By the Fundamental Theorem (Theorem 5.5.6), $\mathrm{rank}(A) + \mathrm{nullity}(A) = n$. Hence $\mathrm{rank}(A) = n$ if and only if $\mathrm{nullity}(A) = 0$.

(h), (j), (l), (n) These concern the n columns of A, which live in $\mathbb{R}^n$. By Theorem 3.3.13, n vectors in an n-dimensional space are linearly independent $\iff$ they span $\iff$ they form a basis; and each of these holds $\iff$ $\dim(\mathrm{col}(A)) = n$ $\iff$ $\mathrm{rank}(A) = n$. So (h), (j), (l) are each equivalent to (n).

(i), (k), (m) These are the same three statements for the *rows* of A. They are equivalent to their column counterparts because $\mathrm{rank}(A) = \mathrm{rank}(A^\mathsf{T})$ (row rank equals column rank, Theorem 5.5.6); equivalently, $\det(A) = \det(A^\mathsf{T})$ (Theorem 2.2.2) ties the row conditions to (g).

(p) The range of T_A is the column space: $\mathrm{ran}(T_A) = \mathrm{col}(A)$. So $\mathrm{ran}(T_A) = \mathbb{R}^n \iff \mathrm{col}(A) = \mathbb{R}^n$, which is condition (j).

Every one of (h)–(p) reduces to $\mathrm{rank}(A) = n$, and hence to (a). This completes the chain of equivalences. ■

5.6 Least Squares

When $A\mathbf{x} = \mathbf{b}$ is inconsistent — as overdetermined systems from real data usually are — we cannot solve it exactly, but we can find the $\mathbf{x}$ making $A\mathbf{x}$ as close to $\mathbf{b}$ as possible. The geometry of projection makes this precise.

Theorem 5.6.1 (Closest Point)
If W is a finite-dimensional subspace of an inner product space V and $\mathbf{b} \in V$, then the point of W closest to $\mathbf{b}$ is $\mathrm{proj}_W \mathbf{b}$: that is, $\mathrm{dist}(\mathbf{b}, \mathrm{proj}_W \mathbf{b}) < \mathrm{dist}(\mathbf{b}, \mathbf{w})$ for every $\mathbf{w} \in W$ with $\mathbf{w} \neq \mathrm{proj}_W \mathbf{b}$.

Proof.

Write $\mathbf{b} = \operatorname{proj}_W \mathbf{b} + \mathbf{z}$ with $\mathbf{z} = \mathbf{b} - \operatorname{proj}_W \mathbf{b} \in W^\perp$ (Theorem 5.4.7). For any $\mathbf{w} \in W$, the vector $\operatorname{proj}_W \mathbf{b} - \mathbf{w} \in W$ is orthogonal to $\mathbf{z}$, so by Pythagoras (Theorem 5.2.6),

$$\operatorname{dist}(\mathbf{b}, \mathbf{w})^2 = \|\mathbf{z} + (\operatorname{proj}_W \mathbf{b} - \mathbf{w})\|^2 = \|\mathbf{z}\|^2 + \|\operatorname{proj}_W \mathbf{b} - \mathbf{w}\|^2 \geq \|\mathbf{z}\|^2 = \operatorname{dist}(\mathbf{b}, \operatorname{proj}_W \mathbf{b})^2,$$

with equality only when $\mathbf{w} = \operatorname{proj}_W \mathbf{b}$. Taking square roots gives the strict inequality for $\mathbf{w} \neq \operatorname{proj}_W \mathbf{b}$. ∎

Definition 5.6.2 (Least-squares problem)

Given $A\mathbf{x} = \mathbf{b}$ (m equations, n unknowns), a *least-squares solution* is a vector $\mathbf{x}$ minimising $\|\mathbf{b} - A\mathbf{x}\|$ in the Euclidean inner product on $\mathbb{R}^m$. The vector $\mathbf{b} - A\mathbf{x}$ is the *error vector* and $\|\mathbf{b} - A\mathbf{x}\|$ the *least-squares error.*

Remark 5.6.3. The name reflects that $\|\mathbf{b} - A\mathbf{x}\|^2 = \sum_i (b_i - (A\mathbf{x})_i)^2$ is a sum of squares, which least-squares solutions minimise.

Theorem 5.6.4

Let $W = \operatorname{col}(A)$. Then $\mathbf{x}$ is a least-squares solution of $A\mathbf{x} = \mathbf{b}$ if and only if $A\mathbf{x} = \operatorname{proj}_W \mathbf{b}$.

Proof.

The achievable vectors $A\mathbf{x}$ are exactly the elements of $W = \operatorname{col}(A)$ (Theorem 1.3.10). Minimising $\|\mathbf{b} - A\mathbf{x}\|$ over all $\mathbf{x}$ is therefore minimising $\|\mathbf{b} - \mathbf{w}\|$ over $\mathbf{w} \in W$, whose unique minimiser is $\mathbf{w} = \operatorname{proj}_W \mathbf{b}$ (Theorem 5.6.1). So $\mathbf{x}$ is least-squares $\iff$ $A\mathbf{x} = \operatorname{proj}_W \mathbf{b}$. ∎

Theorem 5.6.5 (Normal equations)

A vector $\mathbf{x}$ is a least-squares solution of $A\mathbf{x} = \mathbf{b}$ if and only if it solves the *normal system*

$$A^\mathsf{T} A\,\mathbf{x} = A^\mathsf{T}\mathbf{b}.$$

Moreover, the normal system is always consistent.

Proof.

Let $W = \operatorname{col}(A)$. By Theorem 5.6.4, $\mathbf{x}$ is a least-squares solution if and only if $A\mathbf{x} = \operatorname{proj}_W \mathbf{b}$, which holds if and only if $\mathbf{b} - A\mathbf{x} \in W^\perp$.
Now identify $W^\perp$. Since $\operatorname{col}(A) = \operatorname{row}(A^\mathsf{T})$, applying Theorem 5.5.1 to A^T gives

$$W^\perp = \operatorname{col}(A)^\perp = \operatorname{row}(A^\mathsf{T})^\perp = \ker(A^\mathsf{T}).$$

So the condition $\mathbf{b} - A\mathbf{x} \in W^\perp$ becomes $A^\mathsf{T}(\mathbf{b} - A\mathbf{x}) = \mathbf{0}$, that is,

$$A^\mathsf{T} A\,\mathbf{x} = A^\mathsf{T}\mathbf{b}.$$

This is exactly the normal system, proving the equivalence.
Consistency. Since $\operatorname{proj}_W \mathbf{b} \in W = \operatorname{col}(A)$, there is some $\mathbf{x}$ with $A\mathbf{x} = \operatorname{proj}_W \mathbf{b}$; by the equivalence just proved, that $\mathbf{x}$ solves the normal system. So the normal system is always consistent. ∎

Example 5.6.6 (Least-squares solutions)

Find all least-squares solutions of $A\mathbf{x} = \mathbf{b}$ with $A = \begin{bmatrix} 1 & 3 \\ 2 & 6 \\ 3 & 9 \end{bmatrix}$, $\mathbf{b} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$.

Solution. Form the normal system $A^\mathsf{T}A\mathbf{x} = A^\mathsf{T}\mathbf{b}$:

$$A^\mathsf{T}A = \begin{bmatrix} 14 & 42 \\ 42 & 126 \end{bmatrix}, \qquad A^\mathsf{T}\mathbf{b} = \begin{bmatrix} 1+0+3 \\ 3+0+9 \end{bmatrix} = \begin{bmatrix} 4 \\ 12 \end{bmatrix}.$$

Row-reducing $[A^\mathsf{T}A \mid A^\mathsf{T}\mathbf{b}]$: $R_2 \mapsto R_2 - 3R_1$ gives $\left[\begin{array}{cc|c} 14 & 42 & 4 \\ 0 & 0 & 0 \end{array}\right]$, so $14x_1 + 42x_2 = 4$, i.e. $x_1 = \frac{2}{7} - 3x_2$. The least-squares solutions form a line:

$$\mathbf{x} = \left(\tfrac{2}{7} - 3t,\ t\right), \qquad t \in \mathbb{R}.$$

(There are infinitely many because the columns of A are dependent — the second is 3 times the first.)

Definition 5.6.7 (Gram matrix)
For an $m \times n$ matrix A, the $n \times n$ matrix $A^\mathsf{T}A$ is the *Gram matrix* of A.

Theorem 5.6.8 (Properties of the Gram matrix)
For any $m \times n$ matrix A: (a) $\ker(A^\mathsf{T}A) = \ker(A)$; (b) $\operatorname{rank}(A^\mathsf{T}A) = \operatorname{rank}(A)$; (c) $A^\mathsf{T}A$ is invertible if and only if A has linearly independent columns.

Proof.

(a) Clearly $\ker(A) \subseteq \ker(A^\mathsf{T}A)$. Conversely, if $A^\mathsf{T}A\mathbf{x} = \mathbf{0}$, then $\|A\mathbf{x}\|^2 = (A\mathbf{x}) \cdot (A\mathbf{x}) = \mathbf{x}^\mathsf{T}A^\mathsf{T}A\mathbf{x} = \mathbf{x}^\mathsf{T}\mathbf{0} = 0$, so $A\mathbf{x} = \mathbf{0}$; thus $\ker(A^\mathsf{T}A) \subseteq \ker(A)$.

(b) By Rank–Nullity applied to the n-column matrices A and $A^\mathsf{T}A$, equal kernels (part (a)) give equal nullities, hence equal ranks.

(c) $A^\mathsf{T}A$ is $n \times n$, so it is invertible $\iff$ $\operatorname{rank}(A^\mathsf{T}A) = n$ (Theorem 5.5.7) $\iff$ $\operatorname{rank}(A) = n$ (part (b)) $\iff$ the n columns of A are independent (Theorem 5.5.7). ∎

Remark 5.6.9. When A has independent columns, $A^\mathsf{T}A$ is invertible and the least-squares solution is *unique*:

$$\mathbf{x} = (A^\mathsf{T}A)^{-1}A^\mathsf{T}\mathbf{b}.$$

This is the workhorse formula for line- and curve-fitting: to fit data (x_i, y_i) by $y = c_0 + c_1x$, take A with rows $(1, x_i)$ and $\mathbf{b} = (y_i)$; provided the x_i are not all equal, the columns are independent and the best-fit line is unique.

Chapter 5 Summary

- An inner product is symmetric, linear, and positive-definite (Definition 5.1.1); it defines norm $\|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}$ and distance (Definition 5.1.10). Cauchy–Schwarz $|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\|\|\mathbf{v}\|$ (Theorem 5.2.1) underlies the triangle inequality and the definition of angle.
- Orthogonal sets of nonzero vectors are independent (Theorem 5.4.3); in an orthonormal basis, coordinates are inner products (Fourier expansion, Theorem 5.4.6).
- **Gram–Schmidt** (Algorithm 5.4.11) turns any basis into an orthogonal one by subtracting projections; every finite-dimensional inner product space has an orthonormal basis (Theorem 5.4.10).
- Projection onto W gives the closest point (Theorem 5.6.1); $V = W \oplus W^\perp$ (Theorem 5.4.13).
- **Fundamental Theorem (Theorem 5.5.6):** row rank = column rank = r, $\dim\ker(A) = n - r$, $\dim\operatorname{coker}(A) = m - r$; with $\operatorname{row}(A)^\perp = \ker(A)$ (Theorem 5.5.1). The square-matrix master list is Theorem 5.5.7.

- **Least squares:** the best approximate solution of $A\mathbf{x} = \mathbf{b}$ solves the normal equations $A^\top A\mathbf{x} = A^\top\mathbf{b}$ (Theorem 5.6.5), always consistent; unique as $(A^\top A)^{-1}A^\top\mathbf{b}$ when A has independent columns (Theorem 5.6.8).

Chapter 5 Exercises

5.1 On $\mathbb{R}^2$ with $\langle \mathbf{u}, \mathbf{v}\rangle = 2u_1v_1 + u_2v_2$, compute $\|(1, 2)\|$ and the angle between $(1, 0)$ and $(1, 1)$.

5.2 Apply Gram–Schmidt to $\mathbf{u}_1 = (1, 1, 0)$, $\mathbf{u}_2 = (1, 0, 1)$, $\mathbf{u}_3 = (0, 1, 1)$ in Euclidean $\mathbb{R}^3$ to produce an orthogonal basis.

5.3 Let $W = \operatorname{span}\{(1, 1, 0), (0, 1, 1)\} \subseteq \mathbb{R}^3$. Find $\operatorname{proj}_W(2, 0, 0)$ and the distance from $(2, 0, 0)$ to W.

5.4 For $A = \left[\begin{smallmatrix}1&0\\1&1\\0&1\end{smallmatrix}\right]$, $\mathbf{b} = (1, 2, 3)^\top$, find the unique least-squares solution of $A\mathbf{x} = \mathbf{b}$.

5.5 Prove that for any $\mathbf{u}, \mathbf{v}$ in a real inner product space, $\|\mathbf{u} + \mathbf{v}\|^2 + \|\mathbf{u} - \mathbf{v}\|^2 = 2\|\mathbf{u}\|^2 + 2\|\mathbf{v}\|^2$ (the parallelogram law).

5.6 Let A be $m \times n$ with independent columns. Prove that $P = A(A^\top A)^{-1}A^\top$ satisfies $P^2 = P$ and $P^\top = P$, and explain why $P\mathbf{b} = \operatorname{proj}_{\operatorname{col}(A)}\mathbf{b}$.

Solutions to Chapter 5 Exercises

5.1 $\|(1, 2)\| = \sqrt{2(1)^2 + (2)^2} = \sqrt{6}$. For the angle: $\langle(1, 0), (1, 1)\rangle = 2(1)(1) + 0 = 2$, $\|(1, 0)\| = \sqrt{2}$, $\|(1, 1)\| = \sqrt{2 + 1} = \sqrt{3}$, so $\cos\theta = \frac{2}{\sqrt{2}\sqrt{3}} = \frac{2}{\sqrt{6}}$, giving $\theta = \arccos(2/\sqrt{6}) \approx 35.3°$.

5.2 $\mathbf{v}_1 = (1, 1, 0)$. $\mathbf{v}_2 = (1, 0, 1) - \frac{1}{2}(1, 1, 0) = (\frac{1}{2}, -\frac{1}{2}, 1)$. For $\mathbf{v}_3$: $\langle\mathbf{u}_3, \mathbf{v}_1\rangle = 1$, $\langle\mathbf{u}_3, \mathbf{v}_2\rangle = -\frac{1}{2} + 1 = \frac{1}{2}$, $\|\mathbf{v}_2\|^2 = \frac{3}{2}$, so $\mathbf{v}_3 = (0, 1, 1) - \frac{1}{2}(1, 1, 0) - \frac{1/2}{3/2}(\frac{1}{2}, -\frac{1}{2}, 1) = (0, 1, 1) - (\frac{1}{2}, \frac{1}{2}, 0) - (\frac{1}{6}, -\frac{1}{6}, \frac{1}{3}) = (-\frac{2}{3}, \frac{2}{3}, \frac{2}{3})$. Orthogonal basis: $\{(1, 1, 0), (\frac{1}{2}, -\frac{1}{2}, 1), (-\frac{2}{3}, \frac{2}{3}, \frac{2}{3})\}$.

5.3 The spanning vectors $\mathbf{v}_1 = (1, 1, 0)$, $\mathbf{v}_2 = (0, 1, 1)$ are not orthogonal ($\mathbf{v}_1 \cdot \mathbf{v}_2 = 1$). Orthogonalise: $\mathbf{v}_2' = (0, 1, 1) - \frac{1}{2}(1, 1, 0) = (-\frac{1}{2}, \frac{1}{2}, 1)$. Then with $\mathbf{u} = (2, 0, 0)$: $\operatorname{proj}_W \mathbf{u} = \frac{\langle\mathbf{u},\mathbf{v}_1\rangle}{\|\mathbf{v}_1\|^2}\mathbf{v}_1 + \frac{\langle\mathbf{u},\mathbf{v}_2'\rangle}{\|\mathbf{v}_2'\|^2}\mathbf{v}_2' = \frac{2}{2}(1, 1, 0) + \frac{-1}{3/2}(-\frac{1}{2}, \frac{1}{2}, 1) = (1, 1, 0) + (\frac{1}{3}, -\frac{1}{3}, -\frac{2}{3}) = (\frac{4}{3}, \frac{2}{3}, -\frac{2}{3})$. Distance $= \|\mathbf{u} - \operatorname{proj}_W \mathbf{u}\| = \|(\frac{2}{3}, -\frac{2}{3}, \frac{2}{3})\| = \frac{2}{\sqrt{3}}$.

5.4 $A^\top A = \left[\begin{smallmatrix}2&1\\1&2\end{smallmatrix}\right]$, $A^\top\mathbf{b} = \left[\begin{smallmatrix}1+2\\2+3\end{smallmatrix}\right] = \left[\begin{smallmatrix}3\\5\end{smallmatrix}\right]$. Columns of A are independent, so the solution is unique: $\mathbf{x} = (A^\top A)^{-1}A^\top\mathbf{b} = \frac{1}{3}\left[\begin{smallmatrix}2&-1\\-1&2\end{smallmatrix}\right]\left[\begin{smallmatrix}3\\5\end{smallmatrix}\right] = \frac{1}{3}(1, 7) = (\frac{1}{3}, \frac{7}{3})$.

5.5 Expand both norms via the inner product: $\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + 2\langle\mathbf{u}, \mathbf{v}\rangle + \|\mathbf{v}\|^2$ and $\|\mathbf{u} - \mathbf{v}\|^2 = \|\mathbf{u}\|^2 - 2\langle\mathbf{u}, \mathbf{v}\rangle + \|\mathbf{v}\|^2$. Adding, the cross terms cancel: $\|\mathbf{u} + \mathbf{v}\|^2 + \|\mathbf{u} - \mathbf{v}\|^2 = 2\|\mathbf{u}\|^2 + 2\|\mathbf{v}\|^2$. ∎

5.6 Independent columns make $A^\top A$ invertible (Theorem 5.6.8). *Idempotent:*

$$P^2 = A(A^\top A)^{-1}\underbrace{A^\top A(A^\top A)^{-1}}_{=I}A^\top = A(A^\top A)^{-1}A^\top = P.$$

Symmetric: $P^\top = \big(A(A^\top A)^{-1}A^\top\big)^\top = A\big((A^\top A)^{-1}\big)^\top A^\top = A(A^\top A)^{-1}A^\top = P$, since $A^\top A$ is symmetric. For any $\mathbf{b}$, $P\mathbf{b} \in \operatorname{col}(A)$ and $\mathbf{b} - P\mathbf{b} \in \ker(A^\top) = \operatorname{col}(A)^\perp$ (check $A^\top(\mathbf{b} - P\mathbf{b}) = A^\top\mathbf{b} - A^\top\mathbf{b} = \mathbf{0}$), so by the uniqueness in Theorem 5.4.7, $P\mathbf{b} = \operatorname{proj}_{\operatorname{col}(A)}\mathbf{b}$. ∎

CHAPTER 6

EIGENVALUES AND EIGENVECTORS

What this chapter is about. For most vectors, multiplying by a matrix A changes both length and direction. But some special nonzero vectors are merely *scaled* by A — these are its *eigenvectors*, and the scaling factors are its *eigenvalues*. They expose the intrinsic action of A: along each eigenvector, A acts as simple multiplication. We find eigenvalues as roots of the *characteristic polynomial* $\det(\lambda I - A)$, find eigenvectors by solving $(\lambda I - A)\mathbf{x} = \mathbf{0}$, and ask when a matrix is *diagonalizable* — similar to a diagonal matrix via a basis of eigenvectors. Diagonalization turns hard computations (like high powers A^k) into trivial ones, and sets up the spectral theory of Chapter 7.

Reference. A&R §§5.1–5.2.

Proof convention. As in the seminar notes, Theorem 6.2.6 (eigenvectors for distinct eigenvalues are independent) is stated without proof, belonging to MTH2025; all other results are proved in full.

6.1 Eigenvalues and Eigenvectors

Definition 6.1.1 (Eigenvector and eigenvalue)

If A is an $n \times n$ matrix, a *nonzero* vector $\mathbf{x} \in \mathbb{R}^n$ is an *eigenvector* of A (or of the operator T_A) if

$$A\mathbf{x} = \lambda\mathbf{x}$$

for some scalar $\lambda \in \mathbb{R}$. The scalar λ is the corresponding *eigenvalue*, and $\mathbf{x}$ is *an eigenvector corresponding to* λ.

Remark 6.1.2. "Eigenvalue" blends the German *eigen* ("own", "characteristic") with "value": the eigenvalues are the characteristic scaling factors belonging to A.

Remark 6.1.3. If $\mathbf{x}$ is an eigenvector for λ, so is $c\mathbf{x}$ for any $c \neq 0$: $A(c\mathbf{x}) = cA\mathbf{x} = c\lambda\mathbf{x} = \lambda(c\mathbf{x})$. Eigenvectors are never unique — only their *directions* are determined.

Example 6.1.4 (Verifying an eigenvector)

Let $A = \left[\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right]$ and $\mathbf{x} = (1, 1)$. Then $A\mathbf{x} = (2 + 1,\ 1 + 2) = (3, 3) = 3(1, 1) = 3\mathbf{x}$, so $\mathbf{x}$ is an eigenvector with eigenvalue $\lambda = 3$.

How do we *find* eigenvalues without guessing? Rewrite $A\mathbf{x} = \lambda\mathbf{x}$ as $(\lambda I - A)\mathbf{x} = \mathbf{0}$. A nonzero solution $\mathbf{x}$ exists exactly when $\lambda I - A$ is singular.

Theorem 6.1.5 (Characteristic equation)
$\lambda \in \mathbb{R}$ is an eigenvalue of the $n \times n$ matrix A if and only if

$$\det(\lambda I - A) = 0.$$

Proof.
λ is an eigenvalue $\iff$ there is a nonzero $\mathbf{x}$ with $A\mathbf{x} = \lambda\mathbf{x}$ $\iff$ $(\lambda I - A)\mathbf{x} = \mathbf{0}$ has a nonzero solution $\iff$ $(\lambda I - A)$ is not invertible (Theorem 1.5.7) $\iff$ $\det(\lambda I - A) = 0$ (Theorem 2.2.9). ∎

Definition 6.1.6 (Characteristic equation)
The equation $\det(\lambda I - A) = 0$ is the *characteristic equation* of A.

Example 6.1.7 (Finding eigenvalues)

For $A = \left[\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right]$ of Example 6.1.4,

$$\det(\lambda I - A) = \det \begin{bmatrix} \lambda - 2 & -1 \\ -1 & \lambda - 2 \end{bmatrix} = (\lambda - 2)^2 - 1 = \lambda^2 - 4\lambda + 3 = (\lambda - 1)(\lambda - 3).$$

The eigenvalues are $\lambda = 1$ and $\lambda = 3$ (the latter matching Example 6.1.4).

Lemma 6.1.8 (Characteristic polynomial)
For any $n \times n$ matrix A, the quantity $\det(\lambda I - A)$ is a monic polynomial in λ of degree n, called the *characteristic polynomial* of A.

Proof.
Expand $\det(\lambda I - A)$ by cofactors. The product of the diagonal entries,

$$\prod_{i=1}^{n} (\lambda - a_{ii}),$$

contributes the top-degree term: multiplying out, the highest power is λ^n with coefficient 1, so the polynomial is monic of degree n.
Every other term in the cofactor expansion omits at least two of the diagonal factors $(\lambda - a_{ii})$ — because choosing an off-diagonal entry in some row removes both that row's and that column's diagonal factor — and so contributes a polynomial of degree at most $n - 2$ in λ. These lower-degree terms cannot affect the leading behaviour. Hence $\det(\lambda I - A)$ is a monic polynomial of degree exactly n. ∎

Example 6.1.9 (The 2×2 case)

For $A = \left[\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right]$,

$$\det(\lambda I - A) = (\lambda - a)(\lambda - d) - bc = \lambda^2 - (a + d)\lambda + (ad - bc) = \lambda^2 - \operatorname{tr}(A)\lambda + \det(A),$$

a monic quadratic, confirming Lemma 6.1.8 for $n = 2$. Notice the coefficients are $\operatorname{tr}(A)$ and $\det(A)$.

Theorem 6.1.10 (Eigenvalues of a triangular matrix)
If A is triangular, its eigenvalues are exactly its diagonal entries.

Proof.
If A is triangular, so is $\lambda I - A$, with diagonal entries $\lambda - a_{ii}$. By Lemma 2.2.1, $\det(\lambda I - A) = \prod_i (\lambda - a_{ii})$, whose roots are the a_{ii}. ■

6.1.1 Eigenvectors and eigenspaces

Once an eigenvalue λ is known, its eigenvectors are the nonzero solutions of $(\lambda I - A)\mathbf{x} = \mathbf{0}$.

Example 6.1.11 (Finding eigenvectors)

For $A = \left[\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right]$ (eigenvalues $1, 3$): for $\lambda = 3$, solve $(3I - A)\mathbf{x} = \left[\begin{smallmatrix} 1 & -1 \\ -1 & 1 \end{smallmatrix}\right]\mathbf{x} = \mathbf{0}$, giving $x_1 = x_2$, so eigenvectors are multiples of $(1, 1)$. For $\lambda = 1$, $(I - A)\mathbf{x} = \left[\begin{smallmatrix} -1 & -1 \\ -1 & -1 \end{smallmatrix}\right]\mathbf{x} = \mathbf{0}$ gives $x_1 = -x_2$, eigenvectors multiples of $(1, -1)$.

Definition 6.1.12 (Eigenspace)
The *eigenspace* of A for the eigenvalue λ is $\ker(\lambda I - A)$, the solution space of $(\lambda I - A)\mathbf{x} = \mathbf{0}$.

Each eigenspace is a subspace of $\mathbb{R}^n$, being the kernel of the matrix $\lambda I - A$ (every kernel is a subspace, Theorem 3.6.5). In particular it always contains $\mathbf{0}$, even though $\mathbf{0}$ is never itself an eigenvector.

Example 6.1.13 (Eigenspaces)

For $A = \left[\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right]$: the eigenspace for $\lambda = 3$ is $\operatorname{span}\{(1, 1)\}$ and for $\lambda = 1$ is $\operatorname{span}\{(1, -1)\}$, each of dimension 1.

6.1.2 Powers and invertibility

Theorem 6.1.14 (Eigenvalues of powers)
If k is a positive integer, λ an eigenvalue of A, and $\mathbf{x}$ a corresponding eigenvector, then λ^k is an eigenvalue of A^k with the same eigenvector $\mathbf{x}$.

Proof.
By induction on k. The base $k = 1$ is the hypothesis. Assuming $A^k\mathbf{x} = \lambda^k\mathbf{x}$,

$$A^{k+1}\mathbf{x} = A(A^k\mathbf{x}) = A(\lambda^k\mathbf{x}) = \lambda^k(A\mathbf{x}) = \lambda^k(\lambda\mathbf{x}) = \lambda^{k+1}\mathbf{x}.$$

■

Example 6.1.15 (Eigenvalues of A^2)

Let $M = A^2$ where $A = \left[\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right]$, so $M = \left[\begin{smallmatrix} 5 & 4 \\ 4 & 5 \end{smallmatrix}\right]$. By Theorem 6.1.14 and the eigenvalues $1, 3$ of A, the eigenvalues of M are $1^2 = 1$ and $3^2 = 9$.

Theorem 6.1.16 (Invertibility via eigenvalues)
A square matrix A is invertible if and only if $\lambda = 0$ is *not* an eigenvalue.

Proof.
$\lambda = 0$ is an eigenvalue $\iff \det(0 \cdot I - A) = 0 \iff \det(-A) = 0 \iff \det(A) = 0$ (Theorem 2.2.7) $\iff A$ is not invertible (Theorem 2.2.9). Negating, A invertible $\iff$ 0 is not an eigenvalue. ■

6.1.3 Multiplicities

Definition 6.1.17 (Algebraic and geometric multiplicity)
Let α be an eigenvalue of A. The *algebraic multiplicity* of α is the largest k such that $(\lambda - \alpha)^k$ divides the characteristic polynomial. The *geometric multiplicity* of α is the dimension of its eigenspace.

Theorem 6.1.18
For every eigenvalue of a square matrix, the geometric multiplicity is at most the algebraic multiplicity.

Proof.
Let α have geometric multiplicity $g = \dim\ker(\alpha I - A)$. Choose a basis $\{\mathbf{p}_1, \ldots, \mathbf{p}_g\}$ of the eigenspace and extend it to a basis of $\mathbb{R}^n$ (Theorem 3.3.15(b)); let P be the invertible matrix with these basis vectors as its columns.
Since $A\mathbf{p}_i = \alpha\mathbf{p}_i$ for $i \le g$, the matrix $B = P^{-1}AP$ sends $\mathbf{e}_i \mapsto \alpha\mathbf{e}_i$ for $i \le g$, so it has the block form

$$B = \begin{bmatrix} \alpha I_g & * \\ O & * \end{bmatrix}.$$

Computing the characteristic polynomial of this block-triangular matrix,

$$\det(\lambda I - B) = \det\big((\lambda - \alpha)I_g\big) \cdot \det(\cdots) = (\lambda - \alpha)^g\, q(\lambda)$$

for some polynomial q. So $(\lambda - \alpha)^g$ divides the characteristic polynomial of B.
Finally, A and $B = P^{-1}AP$ are similar, so they have the same characteristic polynomial (Theorem 6.2.3). Hence $(\lambda - \alpha)^g$ divides the characteristic polynomial of A, which means α has algebraic multiplicity at least g. That is, geometric multiplicity $\le$ algebraic multiplicity. ∎

Example 6.1.19 (A defective matrix)

Let $A = \left[\begin{smallmatrix} 4 & -4 \\ 1 & 0 \end{smallmatrix}\right]$. The characteristic polynomial is $\lambda^2 - 4\lambda + 4 = (\lambda - 2)^2$, so $\lambda = 2$ has algebraic multiplicity 2. The eigenspace solves $(2I - A)\mathbf{x} = \left[\begin{smallmatrix} -2 & 4 \\ -1 & 2 \end{smallmatrix}\right]\mathbf{x} = \mathbf{0}$, i.e. $x_1 = 2x_2$, spanned by $(2, 1)$ — geometric multiplicity 1. Here geometric $(1) <$ algebraic (2): the matrix is *defective* and, as we will see, not diagonalizable.

6.2 Similarity and Diagonalization

Diagonal matrices are the easiest to work with — powers, determinants, and eigenvalues are read off instantly. The central question of this chapter is when a given matrix can be transformed into a diagonal one by a change of basis.

Definition 6.2.2 (Similar matrices)
A square matrix B is *similar* to A, written $B \sim A$, if there is an invertible matrix P with $B = P^{-1}AP$.

Theorem 6.2.3 (Similar matrices share invariants)
Suppose $A \sim B$. Then (a) $\det(A) = \det(B)$; (b) A and B have the same characteristic polynomial; (c) corresponding eigenvalues have the same geometric multiplicities.

Proof.
Write $B = P^{-1}AP$.
(a) $\det(B) = \det(P^{-1})\det(A)\det(P) = \frac{1}{\det P}\det(A)\det(P) = \det(A)$ by Theorem 2.2.10 and Theorem 2.2.11.

(b) Using $\lambda I - B = P^{-1}(\lambda I - A)P$ (since $P^{-1}(\lambda I)P = \lambda I$),

$$\det(\lambda I - B) = \det\bigl(P^{-1}(\lambda I - A)P\bigr) = \det(\lambda I - A)$$

by the same multiplicative computation as (a).

(c) Fix an eigenvalue λ. We show P^{-1} maps the λ-eigenspace of A bijectively onto that of B. If $A\mathbf{x} = \lambda\mathbf{x}$, then

$$B(P^{-1}\mathbf{x}) = P^{-1}APP^{-1}\mathbf{x} = P^{-1}A\mathbf{x} = P^{-1}(\lambda\mathbf{x}) = \lambda(P^{-1}\mathbf{x}),$$

so $P^{-1}\mathbf{x}$ is a λ-eigenvector of B. Thus $\mathbf{x} \mapsto P^{-1}\mathbf{x}$ sends the λ-eigenspace of A into that of B; it is linear and injective (as P^{-1} is invertible), and the same argument with P in place of P^{-1} gives the inverse map. So the two eigenspaces are isomorphic and have equal dimension — i.e. the geometric multiplicity of λ is the same for A and B. ■

Definition 6.2.4 (Diagonalizable)
A square matrix is *diagonalizable* if it is similar to a diagonal matrix.

Theorem 6.2.5 (Diagonalizability criteria)
For an $n \times n$ matrix A, the following are equivalent:

(a) A is diagonalizable;

(b) A has n linearly independent eigenvectors;

(c) for each eigenvalue the geometric multiplicity equals the algebraic multiplicity, and the algebraic multiplicities sum to n.

Proof.

(a)⇒(b) Suppose $P^{-1}AP = D$ is diagonal with diagonal entries λ_i, so $AP = PD$. Reading column i of $AP = PD$ gives $A\mathbf{p}_i = \lambda_i\mathbf{p}_i$, so each column $\mathbf{p}_i$ of P is an eigenvector. The columns are linearly independent because P is invertible (Theorem 5.5.7). So A has n independent eigenvectors.

(b)⇒(a) Suppose $\mathbf{p}_1, \dots, \mathbf{p}_n$ are independent eigenvectors with eigenvalues λ_i. Set $P = [\mathbf{p}_1 \ \cdots \ \mathbf{p}_n]$, which is invertible (its columns are independent, Theorem 5.5.7). Then

$$AP = [\lambda_1\mathbf{p}_1 \ \cdots \ \lambda_n\mathbf{p}_n] = PD, \qquad D = \mathrm{diag}(\lambda_1, \dots, \lambda_n),$$

so $P^{-1}AP = D$ and A is diagonalizable.

(b) $\iff$ (c) Eigenvectors from distinct eigenvalues are independent (Theorem 6.2.6), so collecting a basis of each eigenspace yields $\sum_j g_j$ independent eigenvectors, where g_j is the geometric multiplicity of the jth distinct eigenvalue — and one can do no better. So A has n independent eigenvectors iff $\sum_j g_j = n$. Writing a_j for the algebraic multiplicities, we always have $g_j \le a_j$ (Theorem 6.1.18) and $\sum_j a_j \le n$ (the characteristic polynomial has degree n). Hence $\sum_j g_j = n$ forces $\sum_j a_j = n$ with $g_j = a_j$ for every j; conversely those equalities give $\sum_j g_j = n$. This is exactly condition (c). ■

Theorem 6.2.6 (Eigenvectors for distinct eigenvalues)
If $\mathbf{v}_1, \dots, \mathbf{v}_r$ are eigenvectors of A corresponding to *distinct* eigenvalues, then $\{\mathbf{v}_1, \dots, \mathbf{v}_r\}$ is linearly independent.

Proof omitted (MTH2025). The idea: a shortest nontrivial dependence among eigenvectors for distinct eigenvalues can be shortened further by applying A and subtracting a multiple, a contradiction.

Corollary 6.2.7 (n distinct eigenvalues $\Rightarrow$ diagonalizable)
If an $n \times n$ matrix has n distinct eigenvalues, then it is diagonalizable.

Proof.
Choosing one eigenvector per eigenvalue gives n eigenvectors for n distinct eigenvalues, independent by Theorem 6.2.6. By Theorem 6.2.5(b), A is diagonalizable. ∎

Algorithm 6.2.8 (Diagonalization)
Let A be $n \times n$.

(i) Find the eigenvalues, and a basis for each eigenspace. Let E be the union of all these basis vectors. If E has fewer than n vectors, A is *not* diagonalizable.

(ii) If $E = \{\mathbf{p}_1, \dots, \mathbf{p}_n\}$, form $P = [\mathbf{p}_1\ \mathbf{p}_2\ \cdots\ \mathbf{p}_n]$.

(iii) Then $D = P^{-1}AP$ is diagonal, with diagonal entries the eigenvalues $\lambda_1, \dots, \lambda_n$ corresponding (in order) to $\mathbf{p}_1, \dots, \mathbf{p}_n$.

Example 6.2.9 (Diagonalize and compute a power)

Diagonalize $A = \left[\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right]$ and compute A^{10}.

Solution. From Examples 6.1.7 and 6.1.11, the eigenvalues are $3, 1$ with eigenvectors $(1,1)$, $(1,-1)$. Set

$$P = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \qquad D = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}, \qquad P^{-1} = \tfrac{1}{2}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

Then $A = PDP^{-1}$, and since $A^k = PD^kP^{-1}$,

$$A^{10} = P\begin{bmatrix} 3^{10} & 0 \\ 0 & 1 \end{bmatrix}P^{-1} = \tfrac{1}{2}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} 3^{10} & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \tfrac{1}{2}\begin{bmatrix} 3^{10}+1 & 3^{10}-1 \\ 3^{10}-1 & 3^{10}+1 \end{bmatrix}.$$

With $3^{10} = 59049$, $A^{10} = \left[\begin{smallmatrix} 29525 & 29524 \\ 29524 & 29525 \end{smallmatrix}\right]$ — obtained without ten matrix multiplications.

Chapter 6 Summary

- An eigenvector satisfies $A\mathbf{x} = \lambda\mathbf{x}$ with $\mathbf{x} \neq \mathbf{0}$ (Definition 6.1.1); λ is an eigenvalue iff $\det(\lambda I - A) = 0$ (Theorem 6.1.5), the characteristic equation.
- $\det(\lambda I - A)$ is monic of degree n (Lemma 6.1.8); for triangular A the eigenvalues are the diagonal entries (Theorem 6.1.10).
- The eigenspace for λ is $\ker(\lambda I - A)$, a subspace (Definition 6.1.12). Geometric multiplicity $\leq$ algebraic multiplicity (Theorem 6.1.18).
- A^k has eigenvalues λ^k (Theorem 6.1.14); A is invertible iff 0 is not an eigenvalue (Theorem 6.1.16).
- Similar matrices ($B = P^{-1}AP$) share determinant, characteristic polynomial, and multiplicities (Theorem 6.2.3).
- A is *diagonalizable* iff it has n independent eigenvectors iff every geometric multiplicity matches its algebraic one (Theorem 6.2.5); n distinct eigenvalues suffice (Corollary 6.2.7). Diagonalization gives $A^k = PD^kP^{-1}$ cheaply (Algorithm 6.2.8).

Chapter 6 Exercises

6.1 Find the eigenvalues and a basis for each eigenspace of $A = \left[\begin{smallmatrix} 3 & 2 \\ 0 & 1 \end{smallmatrix}\right]$.

6.2 Show that $A = \left[\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right]$ has no real eigenvalues, and explain geometrically why.

6.3 Determine whether $A = \left[\begin{smallmatrix} 5 & -3 \\ 6 & -4 \end{smallmatrix}\right]$ is diagonalizable; if so, find P and D with $P^{-1}AP = D$.

6.4 Let A be a 3×3 matrix with eigenvalues $2, 2, 5$. State the possible values of the geometric multiplicity of $\lambda = 2$, and say in each case whether A is diagonalizable.

6.5 Prove that if λ is an eigenvalue of an invertible matrix A, then λ^{-1} is an eigenvalue of A^{-1} (with the same eigenvector).

6.6 Use diagonalization to compute A^k in closed form for $A = \left[\begin{smallmatrix} 1 & 2 \\ 0 & 3 \end{smallmatrix}\right]$.

Solutions to Chapter 6 Exercises

6.1 Triangular, so eigenvalues are the diagonal entries $3, 1$ (Theorem 6.1.10). For $\lambda = 3$: $(3I - A)\mathbf{x} = \left[\begin{smallmatrix} 0 & -2 \\ 0 & 2 \end{smallmatrix}\right]\mathbf{x} = \mathbf{0}$ gives $x_2 = 0$, basis $\{(1, 0)\}$. For $\lambda = 1$: $\left[\begin{smallmatrix} -2 & -2 \\ 0 & 0 \end{smallmatrix}\right]\mathbf{x} = \mathbf{0}$ gives $x_1 = -x_2$, basis $\{(1, -1)\}$.

6.2 Characteristic polynomial $\lambda^2 + 1$, with no real roots, so no real eigenvalues. Geometrically A is rotation by $90°$, which sends no nonzero real vector to a scalar multiple of itself — every direction is turned.

6.3 Characteristic polynomial $\lambda^2 - \operatorname{tr}\lambda + \det = \lambda^2 - 1 \cdot \lambda + (-20 + 18) = \lambda^2 - \lambda - 2 = (\lambda - 2)(\lambda + 1)$. Two distinct eigenvalues $2, -1$, so diagonalizable (Corollary 6.2.7). For $\lambda = 2$: $\left[\begin{smallmatrix} -3 & 3 \\ -6 & 6 \end{smallmatrix}\right]\mathbf{x} = \mathbf{0} \Rightarrow x_1 = x_2$, eigenvector $(1, 1)$. For $\lambda = -1$: $\left[\begin{smallmatrix} -6 & 3 \\ -6 & 3 \end{smallmatrix}\right]\mathbf{x} = \mathbf{0} \Rightarrow x_2 = 2x_1$, eigenvector $(1, 2)$. So $P = \left[\begin{smallmatrix} 1 & 1 \\ 1 & 2 \end{smallmatrix}\right]$, $D = \left[\begin{smallmatrix} 2 & 0 \\ 0 & -1 \end{smallmatrix}\right]$.

6.4 The algebraic multiplicity of $\lambda = 2$ is 2, so its geometric multiplicity is 1 or 2 (Theorem 6.1.18, and ≥ 1 since it is an eigenvalue). If 2: geometric $=$ algebraic for both eigenvalues, total $= 3$, so A is diagonalizable. If 1: geometric $<$ algebraic for $\lambda = 2$, so A is *not* diagonalizable (Theorem 6.2.5).

6.5 Suppose $A\mathbf{x} = \lambda\mathbf{x}$ with $\mathbf{x} \neq \mathbf{0}$. Then $\lambda \neq 0$ (Theorem 6.1.16, A invertible). Apply A^{-1}: $\mathbf{x} = A^{-1}(\lambda\mathbf{x}) = \lambda A^{-1}\mathbf{x}$, so $A^{-1}\mathbf{x} = \lambda^{-1}\mathbf{x}$. Thus λ^{-1} is an eigenvalue of A^{-1} with eigenvector $\mathbf{x}$. ∎

6.6 Triangular eigenvalues $1, 3$. For $\lambda = 1$: $\left[\begin{smallmatrix} 0 & -2 \\ 0 & -2 \end{smallmatrix}\right]\mathbf{x} = \mathbf{0} \Rightarrow x_2 = 0$, eigenvector $(1, 0)$. For $\lambda = 3$: $\left[\begin{smallmatrix} 2 & -2 \\ 0 & 0 \end{smallmatrix}\right]\mathbf{x} = \mathbf{0} \Rightarrow x_1 = x_2$, eigenvector $(1, 1)$. With $P = \left[\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right]$, $P^{-1} = \left[\begin{smallmatrix} 1 & -1 \\ 0 & 1 \end{smallmatrix}\right]$, $D = \operatorname{diag}(1, 3)$:

$$A^k = PD^kP^{-1} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & 3^k \end{bmatrix}\begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 3^k - 1 \\ 0 & 3^k \end{bmatrix}.$$

CHAPTER 7

ORTHOGONAL MATRICES, THE SPECTRAL THEOREM, AND THE SVD

What this chapter is about. We combine the orthogonality of Chapter 5 with the diagonalization of Chapter 6. An *orthogonal matrix* is one whose columns form an orthonormal basis; equivalently $A^{-1} = A^{\mathsf{T}}$. Such matrices preserve lengths and angles. The *Spectral Theorem* is the crowning result for symmetric matrices: a matrix is orthogonally diagonalizable if and only if it is symmetric. Finally, the *Singular Value Decomposition* extends diagonalization to *every* matrix — rectangular, singular, anything — factoring $A = U\Sigma V^{\mathsf{T}}$ with U, V orthogonal and Σ diagonal. The SVD yields the *pseudoinverse*, which solves least-squares problems and inverts what cannot truly be inverted.

Reference. A&R §§7.1–7.4.
Proof convention. Following the seminar notes, the existence of the SVD (Theorem 7.3.8) and the hard direction of the Spectral Theorem (Theorem 7.2.2, (c)⇒(a)) are stated with proofs deferred to MTH2025; all other structural results are proved in full.

7.1 Orthogonal Matrices

Definition 7.1.1 (Orthogonal matrix)
A square matrix A is *orthogonal* if $A^{-1} = A^{\mathsf{T}}$, equivalently $A^{\mathsf{T}}A = AA^{\mathsf{T}} = I$.

Example 7.1.2

The matrix $A = \frac{1}{3}\begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{bmatrix}$ is orthogonal: each column has length $\frac{1}{3}\sqrt{1+4+4} = 1$, and distinct columns are orthogonal, so $A^{\mathsf{T}}A = I$.

Example 7.1.3 (Rotation matrices)

For any θ, the rotation $R_\theta = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$ is orthogonal: $R_\theta^{\mathsf{T}} R_\theta = I$ by the identity $\cos^2\theta + \sin^2\theta = 1$.

Theorem 7.1.4 (Characterisation by orthonormal rows/columns)
For an $n \times n$ matrix A, the following are equivalent: (a) A is orthogonal; (b) the rows of A form an orthonormal basis of $\mathbb{R}^n$; (c) the columns of A form an orthonormal basis of $\mathbb{R}^n$.

Proof.

The key identity is that the (i, j) entry of $A^\mathsf{T}A$ is the dot product of column i and column j of A.

(a) $\iff$ (c) A is orthogonal means $A^\mathsf{T}A = I$, i.e. $(A^\mathsf{T}A)_{ij} = \delta_{ij}$ for all i, j. By the key identity, this says the columns of A are orthonormal. For n vectors in $\mathbb{R}^n$, being orthonormal makes them independent (Theorem 5.4.3) and hence a basis (Theorem 3.3.13). So A orthogonal $\iff$ the columns form an orthonormal basis.

(a) $\iff$ (b) Apply the same argument to $AA^\mathsf{T} = I$, whose (i, j) entry is the dot product of row i and row j of A. So A orthogonal $\iff$ the rows form an orthonormal basis. ∎

Lemma 7.1.5 (Product of orthogonal matrices)

If A, B are orthogonal $n \times n$ matrices, then AB is orthogonal.

Proof.

$(AB)^\mathsf{T}(AB) = B^\mathsf{T}A^\mathsf{T}AB = B^\mathsf{T}IB = B^\mathsf{T}B = I$, using $A^\mathsf{T}A = I$ and $B^\mathsf{T}B = I$. ∎

Lemma 7.1.6 (Determinant of an orthogonal matrix)

If A is orthogonal, then $\det(A) = \pm 1$.

Proof.

From $A^\mathsf{T}A = I$ and Theorem 2.2.10, Theorem 2.2.2, $\det(A^\mathsf{T})\det(A) = \det(A)^2 = \det(I) = 1$, so $\det(A) = \pm 1$. ∎

Theorem 7.1.7 (Orthogonal matrices preserve geometry)

For an $n \times n$ matrix A, the following are equivalent: (a) A is orthogonal; (b) $\|A\mathbf{x}\| = \|\mathbf{x}\|$ for all $\mathbf{x} \in \mathbb{R}^n$; (c) $A\mathbf{x} \cdot A\mathbf{y} = \mathbf{x} \cdot \mathbf{y}$ for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$.

Proof.

We prove (a)⇒(c)⇒(b)⇒(a).

(a)⇒(c) If A is orthogonal then $A^\mathsf{T}A = I$, so for all $\mathbf{x}, \mathbf{y}$,

$$A\mathbf{x} \cdot A\mathbf{y} = (A\mathbf{x})^\mathsf{T}(A\mathbf{y}) = \mathbf{x}^\mathsf{T}A^\mathsf{T}A\mathbf{y} = \mathbf{x}^\mathsf{T}\mathbf{y} = \mathbf{x} \cdot \mathbf{y}.$$

(c)⇒(b) Take $\mathbf{y} = \mathbf{x}$ in (c): $\|A\mathbf{x}\|^2 = A\mathbf{x} \cdot A\mathbf{x} = \mathbf{x} \cdot \mathbf{x} = \|\mathbf{x}\|^2$, so $\|A\mathbf{x}\| = \|\mathbf{x}\|$.

(b)⇒(a) This direction needs the polarisation identity. Suppose $\|A\mathbf{x}\| = \|\mathbf{x}\|$ for all $\mathbf{x}$. Then for any $\mathbf{x}, \mathbf{y}$, expanding $\|A(\mathbf{x} + \mathbf{y})\|^2 = \|\mathbf{x} + \mathbf{y}\|^2$ and cancelling the $\|A\mathbf{x}\|^2 = \|\mathbf{x}\|^2$ and $\|A\mathbf{y}\|^2 = \|\mathbf{y}\|^2$ terms gives $A\mathbf{x} \cdot A\mathbf{y} = \mathbf{x} \cdot \mathbf{y}$ — that is, condition (c). Rewriting (c) as $\mathbf{x}^\mathsf{T}(A^\mathsf{T}A - I)\mathbf{y} = 0$ for all $\mathbf{x}, \mathbf{y}$ and taking $\mathbf{x} = \mathbf{e}_i$, $\mathbf{y} = \mathbf{e}_j$ shows every entry of $A^\mathsf{T}A - I$ is zero. Hence $A^\mathsf{T}A = I$ and A is orthogonal. ∎

Example 7.1.8

The rotation R_θ of Example 7.1.3 satisfies $\|R_\theta\mathbf{x}\| = \|\mathbf{x}\|$: rotation preserves length, as direct multiplication confirms.

Theorem 7.1.9 (Change between orthonormal bases)

Let V be a finite-dimensional inner product space. If P is the change-of-basis matrix from one orthonormal basis to another, then P is orthogonal.

Proof.
Let $B = \{\mathbf{u}_1, \ldots, \mathbf{u}_n\}$ and $B' = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ be the two orthonormal bases, and write $P = P_{B',B}$ for the change-of-basis matrix.
Since B is orthonormal, Theorem 5.4.5 gives $\langle \mathbf{x}, \mathbf{x} \rangle = [\mathbf{x}]_B \cdot [\mathbf{x}]_B$ for every $\mathbf{x} \in V$; since B' is orthonormal, likewise $\langle \mathbf{x}, \mathbf{x} \rangle = [\mathbf{x}]_{B'} \cdot [\mathbf{x}]_{B'}$. Equating the two,

$$[\mathbf{x}]_B \cdot [\mathbf{x}]_B = [\mathbf{x}]_{B'} \cdot [\mathbf{x}]_{B'} \qquad \text{for all } \mathbf{x} \in V.$$

Now $[\mathbf{x}]_{B'} = P[\mathbf{x}]_B$, so this reads $[\mathbf{x}]_B \cdot [\mathbf{x}]_B = (P[\mathbf{x}]_B) \cdot (P[\mathbf{x}]_B)$. As $\mathbf{x}$ ranges over V, the coordinate vector $[\mathbf{x}]_B$ ranges over all of $\mathbb{R}^n$ (the coordinate map is an isomorphism, Theorem 4.4.8), so

$$\mathbf{y} \cdot \mathbf{y} = (P\mathbf{y}) \cdot (P\mathbf{y}) \qquad \text{for all } \mathbf{y} \in \mathbb{R}^n,$$

that is, $\|P\mathbf{y}\| = \|\mathbf{y}\|$ for all $\mathbf{y}$. By Theorem 7.1.7, P is orthogonal. ■

7.2 Orthogonal Diagonalization and the Spectral Theorem

Definition 7.2.1 (Orthogonally diagonalizable)
Square matrices A, B are *orthogonally similar* if $P^\mathsf{T} AP = B$ for some orthogonal P. If A is orthogonally similar to a diagonal matrix, A is *orthogonally diagonalizable*.

Note that for orthogonal P, $P^\mathsf{T} = P^{-1}$, so orthogonal similarity is a special case of ordinary similarity — but with the bonus that the change of basis preserves all lengths and angles.

Theorem 7.2.2 (Spectral Theorem)
For an $n \times n$ matrix A, the following are equivalent: (a) A is orthogonally diagonalizable; (b) A has an orthonormal set of n eigenvectors; (c) A is symmetric.

Proof.

(a) $\iff$ (b) If $P^\mathsf{T} AP = D$ with P orthogonal and D diagonal, then $AP = PD$; reading column i shows each column of P is an eigenvector, and orthogonality of P makes these columns orthonormal (Theorem 7.1.4). So A has n orthonormal eigenvectors. Conversely, n orthonormal eigenvectors assembled as the columns of P give an orthogonal matrix with $P^\mathsf{T} AP$ diagonal. So A is orthogonally diagonalizable iff it has an orthonormal set of n eigenvectors.

(a)⇒(c) If $A = PDP^\mathsf{T}$ with D diagonal and P orthogonal, then

$$A^\mathsf{T} = (PDP^\mathsf{T})^\mathsf{T} = PD^\mathsf{T} P^\mathsf{T} = PDP^\mathsf{T} = A$$

(using $D^\mathsf{T} = D$ for a diagonal matrix), so A is symmetric.

(c)⇒(a) This is the substantive direction. A symmetric matrix has real eigenvalues, eigenvectors from distinct eigenvalues are orthogonal (Theorem 7.2.3), and each eigenspace has an orthonormal basis (Theorem 5.4.10); assembling these yields an orthonormal eigenbasis, giving orthogonal diagonalizability. The full argument — in particular that the eigenspaces span all of $\mathbb{R}^n$ — is developed in MTH2025. ■

Theorem 7.2.3 (Symmetric matrices: distinct eigenvalues give orthogonal eigenvectors)
If A is symmetric, then eigenvectors corresponding to distinct eigenvalues are orthogonal.

Proof.

Let $A\mathbf{u} = \lambda\mathbf{u}$ and $A\mathbf{v} = \mu\mathbf{v}$ with $\lambda \neq \mu$. Using $A^\mathsf{T} = A$,

$$\lambda(\mathbf{u}\cdot\mathbf{v}) = (A\mathbf{u})\cdot\mathbf{v} = (A\mathbf{u})^\mathsf{T}\mathbf{v} = \mathbf{u}^\mathsf{T}A^\mathsf{T}\mathbf{v} = \mathbf{u}^\mathsf{T}A\mathbf{v} = \mathbf{u}\cdot(A\mathbf{v}) = \mu(\mathbf{u}\cdot\mathbf{v}).$$

Thus $(\lambda-\mu)(\mathbf{u}\cdot\mathbf{v}) = 0$; since $\lambda \neq \mu$, $\mathbf{u}\cdot\mathbf{v} = 0$. ■

Algorithm 7.2.4 (Orthogonal diagonalization)
Let A be symmetric.

(i) Find a basis for each eigenspace of A.

(ii) Apply Gram–Schmidt within each eigenspace to obtain an orthonormal basis of it (eigenvectors from different eigenspaces are already orthogonal by Theorem 7.2.3).

(iii) Form P with columns the union of all these orthonormal vectors. Then P is orthogonal and $D = P^\mathsf{T}AP$ is diagonal, with eigenvalues ordered to match the columns of P.

Example 7.2.5 (Orthogonally diagonalize a symmetric matrix)

Let $A = \left[\begin{smallmatrix}3&1\\1&3\end{smallmatrix}\right]$ (symmetric, hence orthogonally diagonalizable). The characteristic polynomial is $(\lambda-3)^2-1 = (\lambda-2)(\lambda-4)$, so eigenvalues are $2, 4$. For $\lambda = 4$: $(4I-A)\mathbf{x} = \left[\begin{smallmatrix}1&-1\\-1&1\end{smallmatrix}\right]\mathbf{x} = \mathbf{0} \Rightarrow$ eigenvector $(1,1)$. For $\lambda = 2$: eigenvector $(1,-1)$. These are orthogonal (Theorem 7.2.3); normalising,

$$P = \frac{1}{\sqrt{2}}\begin{bmatrix}1&1\\1&-1\end{bmatrix}, \qquad D = \begin{bmatrix}4&0\\0&2\end{bmatrix}, \qquad P^\mathsf{T}AP = D.$$

Theorem 7.2.6 (Spectral decomposition)
Let A be symmetric $n\times n$, with orthonormal eigenbasis $\mathbf{u}_1,\dots,\mathbf{u}_n$ and corresponding eigenvalues $\lambda_1,\dots,\lambda_n$. Then

$$A = \sum_{i=1}^{n}\lambda_i\,\mathbf{u}_i\mathbf{u}_i^\mathsf{T}.$$

Proof.
Since A is symmetric, the Spectral Theorem (Theorem 7.2.2) gives $A = PDP^\mathsf{T}$ with $P = [\mathbf{u}_1 \ \cdots \ \mathbf{u}_n]$ orthogonal and $D = \operatorname{diag}(\lambda_1,\dots,\lambda_n)$. Expanding this product column-by-column: $PD = [\lambda_1\mathbf{u}_1 \ \cdots \ \lambda_n\mathbf{u}_n]$, and multiplying on the right by P^T (whose rows are $\mathbf{u}_1^\mathsf{T},\dots,\mathbf{u}_n^\mathsf{T}$) gives

$$A = PDP^\mathsf{T} = \sum_{i=1}^{n}(\lambda_i\mathbf{u}_i)\mathbf{u}_i^\mathsf{T} = \sum_{i=1}^{n}\lambda_i\,\mathbf{u}_i\mathbf{u}_i^\mathsf{T}.$$

Geometrically, each $\mathbf{u}_i\mathbf{u}_i^\mathsf{T}$ is the rank-one orthogonal projection onto the eigenline $\operatorname{span}\{\mathbf{u}_i\}$, so the formula expresses A as acting by the scalar λ_i along each eigendirection. ■

Example 7.2.7 (Building a symmetric matrix from a spectrum)

Find a 2×2 symmetric matrix with eigenvalues $\lambda_1 = 3$, $\lambda_2 = -2$ and corresponding orthonormal eigenvectors $\mathbf{u}_1 = \frac{1}{\sqrt{2}}(1,1)$, $\mathbf{u}_2 = \frac{1}{\sqrt{2}}(1,-1)$. By spectral decomposition,

$$A = 3\,\mathbf{u}_1\mathbf{u}_1^\mathsf{T} - 2\,\mathbf{u}_2\mathbf{u}_2^\mathsf{T} = \tfrac{3}{2}\begin{bmatrix}1&1\\1&1\end{bmatrix} - 1\begin{bmatrix}1&-1\\-1&1\end{bmatrix} = \begin{bmatrix}\frac{1}{2}&\frac{5}{2}\\\frac{5}{2}&\frac{1}{2}\end{bmatrix}.$$

7.3 The Singular Value Decomposition

Only symmetric matrices are orthogonally diagonalizable, and only square matrices have eigenvalues at all. The SVD removes both restrictions: *every* matrix, of any shape, factors through orthogonal matrices and a diagonal one. The key is that while A itself may be unmanageable, $A^\mathsf{T}A$ is always symmetric and well-behaved.

Theorem 7.3.1 ($A^\mathsf{T}A$ is well-behaved)
For any $m \times n$ matrix A: (a) $A^\mathsf{T}A$ is orthogonally diagonalizable; and (b) the eigenvalues of $A^\mathsf{T}A$ are nonnegative.

Proof.

(a) $A^\mathsf{T}A$ is symmetric, since $(A^\mathsf{T}A)^\mathsf{T} = A^\mathsf{T}(A^\mathsf{T})^\mathsf{T} = A^\mathsf{T}A$; by the Spectral Theorem (Theorem 7.2.2) it is orthogonally diagonalizable.

(b) If $A^\mathsf{T}A\mathbf{x} = \lambda\mathbf{x}$ with $\mathbf{x} \neq \mathbf{0}$, then $\|A\mathbf{x}\|^2 = (A\mathbf{x})^\mathsf{T}(A\mathbf{x}) = \mathbf{x}^\mathsf{T}A^\mathsf{T}A\mathbf{x} = \lambda\mathbf{x}^\mathsf{T}\mathbf{x} = \lambda\|\mathbf{x}\|^2$. The left side is ≥ 0 and $\|\mathbf{x}\|^2 > 0$, so $\lambda \geq 0$. ∎

Definition 7.3.2 (Singular values)
If $\lambda_1, \ldots, \lambda_n$ are the eigenvalues of $A^\mathsf{T}A$, the *singular values* of A are $\sigma_i = \sqrt{\lambda_i}$ (real and nonnegative by Theorem 7.3.1(b)).

Theorem 7.3.3 (Number of positive singular values)
If A has rank r, then A has exactly r positive singular values.

Proof.
The number of positive singular values is the number of nonzero eigenvalues of $A^\mathsf{T}A$, i.e. $n - \dim\ker(A^\mathsf{T}A)$. By Theorem 5.6.8(a), $\ker(A^\mathsf{T}A) = \ker(A)$, so this is $n - \mathrm{nullity}(A) = \mathrm{rank}(A) = r$ by Rank–Nullity. ∎

Example 7.3.4 (Singular values)

For $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$, $A^\mathsf{T}A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ with eigenvalues $3, 1$. The singular values are $\sigma_1 = \sqrt{3}$, $\sigma_2 = 1$. Since $\mathrm{rank}(A) = 2$, both are positive, matching Theorem 7.3.3.

Definition 7.3.5 (Singular Value Decomposition)
A *Singular Value Decomposition* (SVD) of an $m \times n$ matrix A is a factorisation $A = U\Sigma V^\mathsf{T}$, where U is $m \times m$ orthogonal, V is $n \times n$ orthogonal, and Σ is an $m \times n$ “diagonal” matrix (entries off the main diagonal are zero) with nonnegative entries.

Theorem 7.3.6 (Structure of an SVD)
Let A be $m \times n$ of rank $r > 0$, with SVD $A = U\Sigma V^\mathsf{T}$. Then:

(a) the nonzero entries are $(\Sigma)_{ii} = \sigma_i$ for $1 \leq i \leq r$, where $\sigma_1 \geq \cdots \geq \sigma_r > 0$ are the nonzero singular values;

(b) the ith column $\mathbf{v}_i$ of V is an eigenvector of $A^\mathsf{T}A$ with eigenvalue σ_i^2;

(c) for $1 \leq i \leq r$, the ith column of U is $\mathbf{u}_i = \frac{1}{\sigma_i}A\mathbf{v}_i$.

Proof.
The whole proof rests on computing $A^\top A$. Using $A = U\Sigma V^\top$ and $U^\top U = I$ (orthogonality of U),

$$A^\top A = V\Sigma^\top U^\top U\Sigma V^\top = V(\Sigma^\top\Sigma)V^\top.$$

Here $\Sigma^\top\Sigma$ is the $n \times n$ diagonal matrix with diagonal entries σ_i^2. So this equation exhibits V as orthogonally diagonalizing $A^\top A$.

(a), (b) Reading off the diagonalization: the columns $\mathbf{v}_i$ of V are eigenvectors of $A^\top A$ with eigenvalues σ_i^2, which is (b). Since the singular values are by definition the square roots of these eigenvalues, the nonzero σ_i are exactly the nonzero diagonal entries of Σ, giving (a).

(c) Multiply $A = U\Sigma V^\top$ on the right by V to get $AV = U\Sigma$. Comparing column i for $i \le r$ gives $A\mathbf{v}_i = \sigma_i\mathbf{u}_i$, hence $\mathbf{u}_i = \frac{1}{\sigma_i}A\mathbf{v}_i$. ■

Remark 7.3.7. A consequence of Theorem 7.3.6 is that the singular values, and the singular value σ_i together with the directions $\mathbf{v}_i, \mathbf{u}_i$, are determined by A (up to the usual choices within eigenspaces); the SVD packages the action of A as: rotate by $V^\top$, scale the axes by the σ_i, rotate by U.

Theorem 7.3.8 (Existence of the SVD)
Every matrix has a singular value decomposition.

Proof omitted (MTH2025). The construction is exactly Algorithm 7.3.10: orthonormal eigenvectors of $A^\top A$ furnish V, the singular values furnish Σ, and the normalised images $A\mathbf{v}_i/\sigma_i$ (extended to an orthonormal basis) furnish U.

Theorem 7.3.9 (The SVD as a matrix representation)
Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be $T(\mathbf{x}) = A\mathbf{x}$, and let $A = U\Sigma V^\top$. If B is the basis of columns of V and B' the basis of columns of U, then $[T]_{B',B} = \Sigma$.

Proof.
The ith column of $[T]_{B',B}$ is the B'-coordinate vector of $T(\mathbf{v}_i)$, where $\mathbf{v}_i$ is the ith column of V. By Theorem 7.3.6(c), for $i \le r$ we have $T(\mathbf{v}_i) = A\mathbf{v}_i = \sigma_i\mathbf{u}_i$, and for $i > r$ we have $A\mathbf{v}_i = \mathbf{0}$. Since B' is the basis of columns of U, the B'-coordinate vector of $\mathbf{u}_i$ is $\mathbf{e}_i$. Therefore the ith column of $[T]_{B',B}$ is

$$[\sigma_i\mathbf{u}_i]_{B'} = \sigma_i\mathbf{e}_i \quad (i \le r), \qquad [\mathbf{0}]_{B'} = \mathbf{0} \quad (i > r),$$

which is exactly column i of the diagonal matrix Σ. Hence $[T]_{B',B} = \Sigma$: relative to the singular bases, T is pure coordinate scaling. ■

Algorithm 7.3.10 (Computing an SVD)
For an $m \times n$ matrix A:

(i) find the eigenvalues λ_i and an orthonormal eigenbasis $\mathbf{v}_i$ of $A^\top A$, ordered so $\lambda_1 \ge \cdots \ge \lambda_r > 0$ and $\lambda_i = 0$ for $i > r$;

(ii) set $V = [\mathbf{v}_1 \ \cdots \ \mathbf{v}_n]$;

(iii) the nonzero diagonal entries of Σ are $\sigma_i = \sqrt{\lambda_i}$ for $i \le r$ (rest zero);

(iv) for $i \le r$ set $\mathbf{u}_i = \frac{1}{\sigma_i}A\mathbf{v}_i$; extend $\{\mathbf{u}_1, \dots, \mathbf{u}_r\}$ to an orthonormal basis $\mathbf{u}_1, \dots, \mathbf{u}_m$ of $\mathbb{R}^m$ and set $U = [\mathbf{u}_1 \ \cdots \ \mathbf{u}_m]$.

Then $A = U\Sigma V^\top$.

Example 7.3.11ex (A worked SVD)

Find an SVD of $A = \left[\begin{smallmatrix}1&1\\0&1\\1&0\end{smallmatrix}\right]$ (from Example 7.3.4).

Solution. We found $A^\mathsf{T}A = \left[\begin{smallmatrix}2&1\\1&2\end{smallmatrix}\right]$ with eigenvalues $\lambda_1 = 3, \lambda_2 = 1$ and orthonormal eigenvectors $\mathbf{v}_1 = \frac{1}{\sqrt{2}}(1,1)$, $\mathbf{v}_2 = \frac{1}{\sqrt{2}}(1,-1)$. So

$$V = \frac{1}{\sqrt{2}}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix}, \qquad \sigma_1 = \sqrt{3},\ \sigma_2 = 1, \qquad \Sigma = \begin{bmatrix}\sqrt{3} & 0\\ 0 & 1\\ 0 & 0\end{bmatrix}.$$

Then $\mathbf{u}_1 = \frac{1}{\sqrt{3}}A\mathbf{v}_1 = \frac{1}{\sqrt{6}}(2,1,1)$ and $\mathbf{u}_2 = \frac{1}{1}A\mathbf{v}_2 = \frac{1}{\sqrt{2}}(0,-1,1)$. Extending to an orthonormal basis with $\mathbf{u}_3 = \frac{1}{\sqrt{3}}(1,-1,-1)$ (a unit vector orthogonal to $\mathbf{u}_1, \mathbf{u}_2$),

$$U = \begin{bmatrix}2/\sqrt{6} & 0 & 1/\sqrt{3}\\ 1/\sqrt{6} & -1/\sqrt{2} & -1/\sqrt{3}\\ 1/\sqrt{6} & 1/\sqrt{2} & -1/\sqrt{3}\end{bmatrix},$$

and $A = U\Sigma V^\mathsf{T}$.

Definition 7.3.11 (Singular vectors)
An eigenvector of $A^\mathsf{T}A$ is a *right singular vector* of A; an eigenvector of AA^T is a *left singular vector*.

Lemma 7.3.12
If $A = U\Sigma V^\mathsf{T}$, then the columns of U are left singular vectors and the columns of V are right singular vectors of A.

Proof.
Theorem 7.3.6(b) shows the columns of V are eigenvectors of $A^\mathsf{T}A$ (right singular vectors). Symmetrically, $AA^\mathsf{T} = U\Sigma V^\mathsf{T}V\Sigma^\mathsf{T}U^\mathsf{T} = U(\Sigma\Sigma^\mathsf{T})U^\mathsf{T}$, so the columns of U are eigenvectors of AA^T (left singular vectors). ■

Theorem 7.3.14 (Singular value expansion)
Let A be $m \times n$ of rank $r > 0$ with SVD $A = U\Sigma V^\mathsf{T}$, columns $\mathbf{u}_i$ of U and $\mathbf{v}_i$ of V. Then

$$A = \sigma_1\mathbf{u}_1\mathbf{v}_1^\mathsf{T} + \sigma_2\mathbf{u}_2\mathbf{v}_2^\mathsf{T} + \cdots + \sigma_r\mathbf{u}_r\mathbf{v}_r^\mathsf{T}. \tag{7.1}$$

Proof.
We compare entries. Fix $1 \le i \le m$ and $1 \le j \le n$. Expanding the product,

$$(A)_{ij} = (U\Sigma V^\mathsf{T})_{ij} = \sum_{k=1}^{m}\sum_{l=1}^{n}(U)_{ik}(\Sigma)_{kl}(V)_{jl}.$$

Since Σ is diagonal with only its first r diagonal entries nonzero, the only surviving terms have $k = l \le r$ with $(\Sigma)_{kk} = \sigma_k$:

$$(A)_{ij} = \sum_{k=1}^{r}(U)_{ik}\,\sigma_k\,(V)_{jk} = \sum_{k=1}^{r}\sigma_k\,(\mathbf{u}_k)_i\,(\mathbf{v}_k)_j,$$

using that $(U)_{ik} = (\mathbf{u}_k)_i$ and $(V)_{jk} = (\mathbf{v}_k)_j$. The product $\mathbf{u}_k\mathbf{v}_k^\mathsf{T}$ has (i,j) entry $(\mathbf{u}_k)_i(\mathbf{v}_k)_j$, so

$$(A)_{ij} = \sum_{k=1}^{r}\sigma_k\,(\mathbf{u}_k\mathbf{v}_k^\mathsf{T})_{ij} = \Big(\sum_{k=1}^{r}\sigma_k\,\mathbf{u}_k\mathbf{v}_k^\mathsf{T}\Big)_{ij}.$$

As this holds for every entry, $A = \sum_{k=1}^{r} \sigma_k \mathbf{u}_k \mathbf{v}_k^\top$. ■

Definition 7.3.15 (Singular value expansion)

Equation (7.1) is the *singular value expansion* of A; its matrix form $A = \tilde{U}\tilde{\Sigma}\tilde{V}^\top$ (keeping only the first r columns/rows) is the *reduced SVD*. Truncating the sum after $k < r$ terms gives the best rank-k approximation to A.

7.4 The Pseudoinverse

A non-square or singular matrix has no inverse, yet we often need to "undo" it as far as possible — for instance to solve least-squares problems. The SVD provides the answer.

Definition 7.4.1 (Pseudoinverse)

Let Σ be an $m \times n$ diagonal matrix with $(\Sigma)_{ii} = \sigma_i > 0$ for $1 \le i \le r$. Its *pseudoinverse* Σ^+ is the $n \times m$ diagonal matrix with $(\Sigma^+)_{ii} = 1/\sigma_i$ for $1 \le i \le r$ (and zeros elsewhere). For any $m \times n$ matrix A with SVD $A = U\Sigma V^\top$, the *pseudoinverse* is

$$A^+ = V\Sigma^+ U^\top.$$

Lemma 7.4.2

Every matrix has a unique pseudoinverse.

Proof omitted (Problem Set). Existence follows from existence of the SVD; uniqueness is shown by verifying A^+ satisfies the defining Moore–Penrose conditions, which pin it down independently of the chosen SVD.

Lemma 7.4.3 (Pseudoinverse from the reduced SVD)

If $A = \tilde{U}\tilde{\Sigma}\tilde{V}^\top$ is a reduced SVD, then $A^+ = \tilde{V}\tilde{\Sigma}^{-1}\tilde{U}^\top$.

Proof omitted (Problem Set). It follows by restricting the definition to the nonzero block, where $\tilde{\Sigma}$ is genuinely invertible.

Lemma 7.4.4

For any matrix A, $\operatorname{rank}(A) = \operatorname{rank}(A^+)$.

Proof.

The rank of A equals the number r of nonzero singular values (Theorem 7.3.3). Since Σ^+ has the same number r of nonzero entries, and pre/post-multiplying by the invertible orthogonal $V, U^\top$ preserves rank, $\operatorname{rank}(A^+) = \operatorname{rank}(\Sigma^+) = r = \operatorname{rank}(A)$. ■

Lemma 7.4.5 (Pseudoinverse generalises the inverse)

If A is invertible, then $A^+ = A^{-1}$.

Proof.

If A is invertible (square, full rank n), then Σ is square invertible with $\Sigma^+ = \Sigma^{-1}$. Hence $A^+ = V\Sigma^{-1}U^\top = (U\Sigma V^\top)^{-1} = A^{-1}$, using that U, V are orthogonal so $(V^\top)^{-1} = V$ and $U^{-1} = U^\top$. ■

Example 7.4.6 (Computing a pseudoinverse)

Let $A = \left[\begin{smallmatrix}1 & 0\\ 1 & 0\end{smallmatrix}\right]$ (rank 1). Then $A^\mathsf{T}A = \left[\begin{smallmatrix}2 & 0\\ 0 & 0\end{smallmatrix}\right]$ with eigenvalues $2, 0$, so $\sigma_1 = \sqrt{2}$, and $\mathbf{v}_1 = (1,0)$, $\mathbf{v}_2 = (0,1)$. Here $\mathbf{u}_1 = \frac{1}{\sqrt{2}}A\mathbf{v}_1 = \frac{1}{\sqrt{2}}(1,1)$, $\mathbf{u}_2 = \frac{1}{\sqrt{2}}(1,-1)$. The reduced SVD gives

$$A^+ = \tilde{V}\tilde{\Sigma}^{-1}\tilde{U}^\mathsf{T} = \begin{bmatrix}1\\0\end{bmatrix} \cdot \tfrac{1}{\sqrt{2}} \cdot \tfrac{1}{\sqrt{2}}\begin{bmatrix}1 & 1\end{bmatrix} = \tfrac{1}{2}\begin{bmatrix}1 & 1\\ 0 & 0\end{bmatrix}.$$

Check: A^+ has rank $1 = \operatorname{rank}(A)$, consistent with Lemma 7.4.4.

Theorem 7.4.7 (The four subspaces under the pseudoinverse)

For any $m \times n$ matrix A: (a) $\operatorname{col}(A^+) = \operatorname{row}(A)$; (b) $\operatorname{row}(A^+) = \operatorname{col}(A)$; (c) $\ker(A^+) = \operatorname{coker}(A)$; (d) $\operatorname{coker}(A^+) = \ker(A)$.

Proof.

Let $A = \hat{U}\hat{\Sigma}\hat{V}^\mathsf{T}$ be a reduced SVD, so $A^+ = \hat{V}\hat{\Sigma}^{-1}\hat{U}^\mathsf{T}$, where $\hat{V} = [\mathbf{v}_1 \ \cdots \ \mathbf{v}_r]$ and $\hat{U} = [\mathbf{u}_1 \ \cdots \ \mathbf{u}_r]$ collect the first r singular vectors.

(a) Take any $\mathbf{x} \in \operatorname{col}(A^+)$, say $\mathbf{x} = A^+\mathbf{b}$. Then

$$\mathbf{x} = \hat{V}\hat{\Sigma}^{-1}\hat{U}^\mathsf{T}\mathbf{b} = \hat{V}\mathbf{c}, \qquad \mathbf{c} = \hat{\Sigma}^{-1}\hat{U}^\mathsf{T}\mathbf{b} \in \mathbb{R}^r,$$

so $\mathbf{x} = \sum_{i=1}^r c_i\mathbf{v}_i$ is a combination of $\mathbf{v}_1, \ldots, \mathbf{v}_r$. These span $\operatorname{row}(A)$ (Theorem 7.3.6), so $\mathbf{x} \in \operatorname{row}(A)$, giving $\operatorname{col}(A^+) \subseteq \operatorname{row}(A)$. Moreover $\operatorname{rank}(A^+) = \operatorname{rank}(A)$ (Lemma 7.4.4), so $\dim \operatorname{col}(A^+) = \dim \operatorname{row}(A)$; an inclusion of subspaces of equal dimension is an equality (Theorem 3.3.14(b)), giving $\operatorname{col}(A^+) = \operatorname{row}(A)$.

(b) Apply (a) to A^+ in place of A, using $(A^+)^+ = A$: this gives $\operatorname{col}(A) = \operatorname{row}(A^+)$, i.e. $\operatorname{row}(A^+) = \operatorname{col}(A)$.

(c) Using Theorem 5.5.1 (applied to A^+) and part (b),

$$\ker(A^+) = \operatorname{row}(A^+)^\perp = \operatorname{col}(A)^\perp = \operatorname{coker}(A).$$

(d) Likewise, $\operatorname{coker}(A^+) = \operatorname{col}(A^+)^\perp = \operatorname{row}(A)^\perp = \ker(A)$, using part (a) and Theorem 5.5.1. ■

Theorem 7.4.8 (The pseudoinverse inverts A on the row space)

Define $T_A : \operatorname{row}(A) \to \operatorname{col}(A)$ by $T_A(\mathbf{x}) = A\mathbf{x}$, and $T_{A^+} : \operatorname{col}(A) \to \operatorname{row}(A)$ by $T_{A^+}(\mathbf{x}) = A^+\mathbf{x}$. Then T_A is an isomorphism with $T_A^{-1} = T_{A^+}$.

Proof omitted (Problem Set). Restricted to the row space, A acts as scaling by the positive σ_i along the right singular directions, an invertible map; A^+ scales back by $1/\sigma_i$.

Theorem 7.4.9 (Minimal-length least-squares solution)

The system $A\mathbf{x} = \mathbf{b}$ has a unique least-squares solution of minimal length, given by

$$\mathbf{x}^* = A^+\mathbf{b}.$$

Proof.

The least-squares solutions of $A\mathbf{x} = \mathbf{b}$ are exactly the solutions of the normal system $A^\mathsf{T}A\mathbf{x} = A^\mathsf{T}\mathbf{b}$ (Theorem 5.6.5); any two differ by an element of $\ker(A)$. We show $\mathbf{x}^* = A^+\mathbf{b}$ is one of them, and is the shortest.

$\mathbf{x}^*$ *is a least-squares solution.* Let $A = \hat{U}\hat{\Sigma}\hat{V}^\mathsf{T}$ be a reduced SVD, so $A^+ = \hat{V}\hat{\Sigma}^{-1}\hat{U}^\mathsf{T}$. Using $\hat{U}^\mathsf{T}\hat{U} = I$ and $\hat{V}^\mathsf{T}\hat{V} = I$,

$$A^\mathsf{T}A(A^+\mathbf{b}) = (\hat{U}\hat{\Sigma}\hat{V}^\mathsf{T})^\mathsf{T}(\hat{U}\hat{\Sigma}\hat{V}^\mathsf{T})(\hat{V}\hat{\Sigma}^{-1}\hat{U}^\mathsf{T})\mathbf{b} = \hat{V}\hat{\Sigma}(\hat{U}^\mathsf{T}\hat{U})\hat{\Sigma}(\hat{V}^\mathsf{T}\hat{V})\hat{\Sigma}^{-1}\hat{U}^\mathsf{T}\mathbf{b} = \hat{V}\hat{\Sigma}\hat{U}^\mathsf{T}\mathbf{b} = A^\mathsf{T}\mathbf{b},$$

so $\mathbf{x}^*$ satisfies the normal equations.
$\mathbf{x}^*$ *has minimal length.* By Theorem 7.4.7(a), $\mathbf{x}^* = A^+\mathbf{b} \in \mathrm{col}(A^+) = \mathrm{row}(A) = \ker(A)^\perp$ (using Theorem 5.5.1). Any other least-squares solution has the form $\mathbf{x} = \mathbf{x}^* + \mathbf{z}$ with $\mathbf{z} \in \ker(A)$. Since $\mathbf{x}^* \perp \mathbf{z}$, Pythagoras (Theorem 5.2.6) gives

$$\|\mathbf{x}\|^2 = \|\mathbf{x}^*\|^2 + \|\mathbf{z}\|^2 \geq \|\mathbf{x}^*\|^2,$$

with equality only when $\mathbf{z} = \mathbf{0}$. So $\mathbf{x}^* = A^+\mathbf{b}$ is the unique least-squares solution of minimal length. ■

Chapter 7 Summary

- An *orthogonal matrix* has $A^{-1} = A^\mathsf{T}$ (Definition 7.1.1), equivalently orthonormal rows/columns (Theorem 7.1.4); it preserves lengths and dot products (Theorem 7.1.7) and has $\det = \pm 1$ (Lemma 7.1.6).
- **Spectral Theorem (Theorem 7.2.2):** A is orthogonally diagonalizable $\iff$ A is symmetric. For symmetric A, eigenvectors from distinct eigenvalues are orthogonal (Theorem 7.2.3); orthogonally diagonalize via Algorithm 7.2.4, and $A = \sum_i \lambda_i \mathbf{u}_i\mathbf{u}_i^\mathsf{T}$ (Theorem 7.2.6).
- For *any* matrix, $A^\mathsf{T}A$ is symmetric with nonnegative eigenvalues (Theorem 7.3.1); the *singular values* are $\sigma_i = \sqrt{\lambda_i}$ (Definition 7.3.2), with exactly $\mathrm{rank}(A)$ of them positive (Theorem 7.3.3).
- **SVD (Theorem 7.3.8):** every $A = U\Sigma V^\mathsf{T}$ with U, V orthogonal and Σ diagonal; columns of V are eigenvectors of $A^\mathsf{T}A$, $\mathbf{u}_i = A\mathbf{v}_i/\sigma_i$ (Theorem 7.3.6); singular value expansion $A = \sum_{i=1}^r \sigma_i \mathbf{u}_i\mathbf{v}_i^\mathsf{T}$ (Theorem 7.3.14).
- The *pseudoinverse* $A^+ = V\Sigma^+U^\mathsf{T}$ (Definition 7.4.1) generalises the inverse (Lemma 7.4.5), has the same rank (Lemma 7.4.4), swaps the four subspaces (Theorem 7.4.7), and delivers the minimal-length least-squares solution $\mathbf{x}^* = A^+\mathbf{b}$ (Theorem 7.4.9).

Chapter 7 Exercises

7.1 Verify that $A = \frac{1}{5}\begin{bmatrix} 3 & -4 \\ 4 & 3 \end{bmatrix}$ is orthogonal, and find $\det(A)$.

7.2 Orthogonally diagonalize $A = \begin{bmatrix} 2 & -2 \\ -2 & 5 \end{bmatrix}$: find an orthogonal P and diagonal D with $P^\mathsf{T}AP = D$.

7.3 Find the singular values of $A = \begin{bmatrix} 2 & 0 \\ 0 & -3 \\ 0 & 0 \end{bmatrix}$.

7.4 Compute a full SVD of $A = \begin{bmatrix} 3 & 0 \\ 0 & -2 \end{bmatrix}$. (Hint: it is already almost diagonal.)

7.5 Prove that the eigenvalues of a symmetric matrix A satisfy: A is invertible iff all singular values are positive.

7.6 For $A = \begin{bmatrix} 1 & 1 \end{bmatrix}$ (a 1×2 matrix), compute A^+ and verify that $\mathbf{x}^* = A^+b$ solves the minimal-length least-squares problem for $A\mathbf{x} = b$ (take $b = 2$).

Solutions to Chapter 7 Exercises

7.1 Columns $\frac{1}{5}(3, 4)$ and $\frac{1}{5}(-4, 3)$ each have length $\frac{1}{5}\sqrt{9 + 16} = 1$ and dot product $\frac{1}{25}(-12 + 12) = 0$, so $A^\mathsf{T}A = I$ and A is orthogonal (Theorem 7.1.4). $\det(A) = \frac{1}{25}(9+16) = 1$ (consistent with Lemma 7.1.6).

7.2 Characteristic polynomial $(\lambda - 2)(\lambda - 5) - 4 = \lambda^2 - 7\lambda + 6 = (\lambda - 1)(\lambda - 6)$, eigenvalues $1, 6$. For $\lambda = 1$: $\begin{bmatrix} -1 & -2 \\ -2 & -4 \end{bmatrix}\mathbf{x} = \mathbf{0} \Rightarrow x_1 = -2x_2$, eigenvector $(2, -1)$ (note: solving $-x_1 - 2x_2 = 0$ gives $x_1 = -2x_2$, so $(-2,1)$; take $(2,-1)$). For $\lambda = 6$: $\left[\begin{smallmatrix} 4 & -2 \\ -2 & 1 \end{smallmatrix}\right] \mathbf{x} = \mathbf{0} \Rightarrow x_2 = 2x_1$, eigenvector $(1,2)$. Orthogonal (Theorem 7.2.3); normalise: $P = \frac{1}{\sqrt{5}} \left[\begin{smallmatrix} 2 & 1 \\ -1 & 2 \end{smallmatrix}\right]$, $D = \left[\begin{smallmatrix} 1 & 0 \\ 0 & 6 \end{smallmatrix}\right]$.

7.3 $A^\top A = \left[\begin{smallmatrix} 4 & 0 \\ 0 & 9 \end{smallmatrix}\right]$, eigenvalues $4, 9$, so singular values $\sigma_1 = 3$, $\sigma_2 = 2$. (The middle row of zeros does not affect $A^\top A$.)

7.4 $A^\top A = \left[\begin{smallmatrix} 9 & 0 \\ 0 & 4 \end{smallmatrix}\right]$, eigenvalues $9, 4$, ordered: $\sigma_1 = 3$ (eigenvector $\mathbf{e}_1$), $\sigma_2 = 2$ (eigenvector $\mathbf{e}_2$). So $V = I$, $\Sigma = \left[\begin{smallmatrix} 3 & 0 \\ 0 & 2 \end{smallmatrix}\right]$. Then $\mathbf{u}_1 = \frac{1}{3} A\mathbf{e}_1 = (1,0)$, $\mathbf{u}_2 = \frac{1}{2} A\mathbf{e}_2 = (0,-1)$, so $U = \left[\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right]$. Check: $U\Sigma V^\top = \left[\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right] \left[\begin{smallmatrix} 3 & 0 \\ 0 & 2 \end{smallmatrix}\right] = \left[\begin{smallmatrix} 3 & 0 \\ 0 & -2 \end{smallmatrix}\right] = A$. ✓

7.5 For symmetric A, the singular values are $\sigma_i = \sqrt{\lambda_i(A^\top A)} = \sqrt{\lambda_i(A^2)} = |\lambda_i(A)|$ (since $A^\top A = A^2$ and eigenvalues of A^2 are squares of those of A, Theorem 6.1.14). All $\sigma_i > 0 \iff$ no eigenvalue of A is $0 \iff A$ invertible (Theorem 6.1.16). ■

7.6 $A^\top A = \left[\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right]$, eigenvalues $2, 0$; $\sigma_1 = \sqrt{2}$, $\mathbf{v}_1 = \frac{1}{\sqrt{2}}(1,1)$. Then $\mathbf{u}_1 = \frac{1}{\sqrt{2}} A\mathbf{v}_1 = \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}}(1+1) = 1$ (the 1×1 matrix $[1]$). Reduced SVD: $A = \tilde{U}\tilde{\Sigma}\tilde{V}^\top$ with $\tilde{U} = [1]$, $\tilde{\Sigma} = [\sqrt{2}]$, $\tilde{V} = \frac{1}{\sqrt{2}}(1,1)^\top$. So $A^+ = \tilde{V}\tilde{\Sigma}^{-1}\tilde{U}^\top = \frac{1}{\sqrt{2}}(1,1)^\top \cdot \frac{1}{\sqrt{2}} = \frac{1}{2}(1,1)^\top$. For $b = 2$: $\mathbf{x}^* = A^+ b = (1,1)^\top$. Check: $A\mathbf{x}^* = 1 + 1 = 2 = b$ exactly (so error is 0), and $\mathbf{x}^* \in \operatorname{row}(A) = \operatorname{span}\{(1,1)\}$, confirming minimal length. ✓